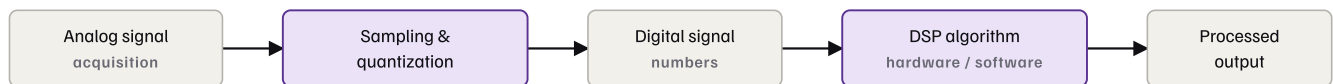


SIX UNITS · BUILT FROM YOUR LECTURE MODULES AND THE CLASS NOTEBOOK

Digital Signal Processing, from the ground up

Exactly what your teacher has taught so far, in the order it was taught — six lecture modules and five tutorials — rewritten in plain language for someone starting from zero. Nothing extra, nothing skipped.

UNITS	LECTURE MODULES	TUTORIALS WORKED	COURSE
6	11	All 5	EC303, SVNIT Surat



UNIT 1
how a signal becomes numbers

UNITS 2–5
what we do with the numbers: systems, convolution, z-transform, spectra, DFT, FFT

The DSP pipeline, as your Module 01 draws it. An analog signal is turned into numbers by sampling and quantization; a DSP algorithm then works on those numbers. Unit 1 is the left half of this picture. Everything after is the right half.

HOW TO READ THIS COURSE

- Every section opens with **one italic line** saying what it is about in plain words.
- The **grey badge** under it names the lecture module (or tutorial) it comes from, so you can read the slides alongside.
- **Blue boxes** are the things to remember. **Orange boxes** are the mistakes people actually make.
- **Worked** boxes are your teacher's own examples, done step by step. **Check yourself** boxes (gold) are your tutorial questions, answered. Both are closed until you click.

UNIT 1 · MODULE 01 · M01L003 · M01L001 · TUTORIALS 1, 2

Signals: from analog to digital

What a signal is, how we sort signals into types, what "frequency" means once time becomes a list of numbers, how an analog signal is turned into that list, and the basic discrete-time signals we will use for the rest of the course.

1.1

Signals and systems

A signal carries information. A system takes a signal in and gives another one out.

Module 01 · Review of Signal & System, §3–4

DEFINITIONS

A **signal** is a mathematical description of a physical happening that carries information. It is written as a function of one or more independent variables.

A **system** is a physical device or a mathematical operation that takes an input signal, processes it, and produces an output signal: $y(t) = T\{x(t)\}$.

- **One variable (time):** $s(t) = 5t$, or a speech waveform.
- **Two variables (space):** an image, $I(x, y)$ — the value is brightness at a point.
- Real signals like speech, ECG, EEG or a seismic recording are *measured*, not written as a formula.

Speech is a good example. Over a whole sentence it keeps changing, but over any short frame of 10–30 ms it barely changes, so each frame can be modelled as a sum of sinusoids:

$$s(t) = \sum_{i=1}^N A_i(t) \sin(2\pi F_i(t)t + \theta_i(t))$$

A fresh set of $\{A_i, F_i, \theta_i\}$ is found for every frame. This is how every speech coder works.

A signal comes from a source — a physical stimulus plus the thing that responds to it. Vocal cords make speech; heart muscle makes an ECG. Measuring the signal tells you about the source, which is the whole point.

A system acts on it. The teacher's example: speech with noise on it, $x(t) = s(t) + n(t)$, goes through a filter designed so that the output is $y(t) \approx s(t)$. The filter suppresses the noise and keeps the useful information. Filters, amplifiers, compressors and modulators are all systems in this sense.

1.2

What DSP is, and its basic elements

Turn the signal into numbers, then do arithmetic on the numbers.

Module 01 · §5-7

DEFINITION

Digital Signal Processing is the analysis and manipulation of signals using digital computers or dedicated digital processors, by carrying out mathematical operations on a digitised signal.

FROM THE CLASS NOTEBOOK

The one-line version your teacher opened with: **a DSP is just a MAC unit with memory** — a multiply-accumulate. The general LTI equation $y(n) = \sum_k a_k x(n-k) + \sum_k b_k y(n-k)$ is nothing but multiplies, adds and delays, and that is all the processor is built to do fast. Keep it in mind through Units 2 and 6: every filter in this course is that one equation with different coefficients.

A DSP system is not tied to a box of hardware. It can be software on an ordinary computer, dedicated hardware (a DSP chip, an FPGA, an ASIC), or a mix of both. The recipe it runs — a finite sequence of mathematical operations — is called an **algorithm**. Filtering, the FFT, compression, noise reduction and feature extraction are all algorithms.



Analog systems process $x_a(t)$ directly with circuits. Digital systems digitise, process, then reconstruct.

The three basic elements. Most real signals are analog — they vary continuously in time and in value. To process them digitally they must first be converted to numbers, and often converted back afterwards.

A/D converter (ADC)

Turns the analog signal into digital samples. Three steps inside: sample, quantize, code (§1.9–1.10).

DSP processor

Executes the signal-processing algorithm.

D/A converter (DAC)
Rebuilds an analog signal from the processed numbers.

1.3

Advantages and limitations of DSP

Digital wins on flexibility, accuracy and reliability — analog still wins on raw speed.

Module 01 · §8-9

Advantages

- 1. **Programmable** — change the algorithm in software; no hardware redesign.
- 2. **High accuracy** — better precision and repeatability than analog components allow.
- 3. **Reliable storage** — digital signals can be stored, copied and transmitted with no real loss.
- 4. **Advanced algorithms** — FFT, adaptive filtering, image compression are all practical.
- 5. **Cost-effective** — modern DSP processors give high performance for little money.

FEATURE	ANALOG PROCESSING	DIGITAL PROCESSING
Implementation	Analog circuits	Hardware + software
Flexibility	Low	High (programmable)
Accuracy	Limited by component tolerance	High numerical precision
Storage	Difficult	Easy and reliable
Noise immunity	Poor	Excellent
Complex algorithms	Limited	Readily implemented
Cost	Higher for complex systems	Lower for large volumes

Limitations

- High-speed A/D and D/A converters are needed.
- Very wideband signals need very powerful DSP hardware.
- Quantization brings finite-precision errors that cannot be avoided.
- Real-time work needs enough computing power and memory to finish each sample before the next arrives.
- Power consumption can be large in high-performance systems.

THE SENTENCE TO END THE ANSWER WITH

For extremely high-frequency or ultra-wideband applications, **analog processing may still be preferred**, because the converters and the processor cannot keep up. Writing this shows you understand the trade-off rather than just listing benefits.

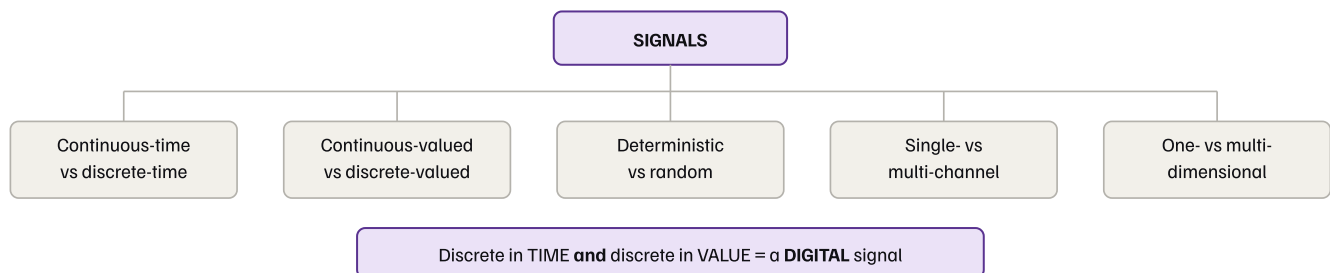
1.4

Classification of signals

We sort signals into types because the method that works on one type often fails on another.

Module 01 · §11-15

The way we process a signal, or analyse how a system responds to it, depends heavily on the signal's characteristics. Some techniques only apply to particular families. So classifying is the first step, not decoration.



Five independent questions. Every signal has an answer to each one.

Continuous-time vs discrete-time

Continuous-time: defined at every instant, $-\infty < t < \infty$. Written $x(t)$. Voltage, temperature, speech.

Discrete-time: defined only at particular instants, $n = 0, \pm 1, \pm 2, \dots$. Written $x(n)$. Obtained by sampling. Digital audio, sensor readings.

Continuous-valued vs discrete-valued

Continuous-valued: the amplitude can be any value in a range — infinitely many possible levels. A microphone output.

Discrete-valued: the amplitude is limited to a fixed set of levels. Produced by quantization. Binary data, an ADC output.

THE DEFINITION THAT GETS EXAMINED

A **digital signal** is discrete in time **and** discrete in value. Sampling gives the first; quantization gives the second. Discrete-time alone is not digital.

Deterministic vs random

Deterministic: completely described by a formula or a table, so every future value can be predicted exactly. No uncertainty. A sine wave, a square wave, an exponential.

Random: values cannot be predicted; each observation is one possible realisation. Described with probability and statistics. Noise, real speech, thermal fluctuation.

Multichannel and multidimensional

A **multichannel** signal is several synchronised signals from the same event, written as a vector. The teacher's example: an earthquake sensor records ground acceleration in three directions, $S(t) = [s_1(t), s_2(t), s_3(t)]^T$. A 12-lead ECG has 12 channels.

A **multidimensional** signal depends on two or more independent variables. An image $I(x, y)$ is 2-D; a video $I(x, y, t)$ is 3-D.

SIGNAL	CHANNELS	DIMENSIONS
Speech	1	1-D
Stereo audio	2	1-D
ECG (12-lead)	12	1-D
Grey image	1	2-D
RGB image	3	2-D
Colour video	3	3-D

A colour TV picture is therefore a **three-channel, three-dimensional** signal. More channels or dimensions means a more expensive algorithm.

1.5

Frequency in continuous time

Frequency means how fast something repeats — and for an analog sinusoid there is no limit.

Module 01 §16-20 · Teacher module M01L003

Two facts make sinusoids the most important signals in the course:

1. Any signal can be written as a combination of harmonics — a periodic one from harmonics at separate frequencies (Fourier series), an aperiodic one from harmonics at all frequencies (Fourier transform).
2. The response of a linear system to a harmonic input is a harmonic at the *same* frequency. That is what makes frequency-domain analysis possible.

$$x_a(t) = A \cos(\Omega t + \theta) = A \cos(2\pi F t + \theta), -\infty < F < \infty$$

A

amplitude

$$\Omega = 2\pi F$$

frequency in radians per second; F in Hz (cycles/second)

 θ

phase in radians

$$T_p = 1/F$$

the period. Raising F shortens the period — more oscillations in the same time.

THREE PROPERTIES OF THE ANALOG SINUSOID

1. It is **periodic for every F**: $x_a(t + T_p) = x_a(t)$.
2. Sinusoids with different F are **always different**.
3. Raising F raises the rate of oscillation **without limit**.

Keep these three in mind. In §1.6 all of them break.

As a complex exponential

$$A \cos(\Omega t + \theta) = \left(\frac{A}{2}\right) e^{j(\Omega t + \theta)} + \left(\frac{A}{2}\right) e^{-j(\Omega t + \theta)}$$

Two phasors: the $+\Omega$ one rotates anticlockwise, the $-\Omega$ one clockwise, and their sum is the real cosine. This is why **negative frequency** appears in every spectrum — it is bookkeeping that keeps the answer real.

Phase delay

$$A \cos(\Omega_0 t + \theta) = A \cos[\Omega_0(t - t_{\text{pha}})], t_{\text{pha}} = -\left(\frac{\theta}{\Omega_0}\right)$$

A phase angle is just a time shift. The waveform is unchanged; it only slides along the axis.

A combination of sinusoids**WHEN IS A SUM OF SINUSOIDS PERIODIC?**

Only when the ratio of every pair of periods (or frequencies) is a **rational fraction**. Then the common period is the **LCM** of the individual periods, and the fundamental frequency is the **GCD** of the individual frequencies. Power simply adds: $P_y = P_{x1} + P_{x2} + \dots$

If a ratio is irrational — $2\cos(\pi t) + 4\sin(3t)$ — there is no common period. Such a signal is called **almost periodic**.

WORKED Common period of a sum (teacher's example)

$$x(t) = 2\sin(3t) + 4\cos(\tfrac{1}{2}t) + 4\cos(\tfrac{1}{3}t - \pi/5)$$

$$\text{periods : } T_1 = \frac{2\pi}{3}, T_2 = 4\pi, T_3 = 6\pi$$

$$\text{common period } T = \text{LCM} = 12\pi \text{ seconds}$$

$$\text{fundamental } \Omega_0 = \frac{2\pi}{T} = \frac{1}{6} \text{ rad/s} = \text{GCD} \left(3, \left(\frac{1}{2} \right), \left(\frac{1}{3} \right) \right)$$

Check: $3 \div (1/6) = 18$, $\tfrac{1}{2} \div (1/6) = 3$, $\tfrac{1}{3} \div (1/6) = 2$ — all whole numbers, so every part completes whole cycles in 12π s ✓

1.6

Frequency in discrete time

Once time is a list of whole numbers, frequency behaves strangely — all three analog facts fail.

Module 01 §21-22 · Teacher module M01L003 §6-7

Sample the analog sinusoid every T_s seconds, so $t = nT_s$:

$$x(n) = A \cos(2\pi F n T_s + \theta) = A \cos \left(2\pi \left(\frac{F}{F_s} \right) n + \theta \right) = A \cos(2\pi f n + \theta) = A \cos(\omega n + \theta)$$

THE DIGITAL FREQUENCIES

$$f = \left(\frac{F}{F_s} \right)$$

cycles per sample – the normalised frequency

$$\omega = 2\pi f = \left(\frac{\Omega}{F_s} \right) = \Omega T_s$$

radians per sample

Digital frequency is the analog frequency divided by the sampling rate.

	CONTINUOUS TIME	DISCRETE TIME
Angular	Ω rad/s	$\omega = \Omega T_s$ rad/sample
Cyclic	F Hz	$f = F/F_s$ cycles/sample
Range	$-\infty < \Omega < \infty$	$-\pi \leq \omega \leq \pi$
Range	$-\infty < F < \infty$	$-\frac{1}{2} \leq f \leq \frac{1}{2}$

Property 1 — periodic only if f is a fraction

For the sequence to repeat after N samples, $2\pi fN$ must be a whole number of turns, $2\pi k$:

$$f = \left(\frac{k}{N} \right)$$

a ratio of whole numbers, common factors cancelled

The period N is the denominator. And k means: it takes k full cycles of the analog wave to make one period of the sampled sequence. A combination of DT sinusoids has period equal to the LCM of the individual periods.

NOT EVERY DT SINUSOID IS PERIODIC

If f is irrational (e.g. $f = \sqrt{3}$, or $\sin(3n)$ where $f = 3/2\pi$) the sequence **never repeats**. It still has a periodic-looking envelope, but no true period.

Property 2 — frequencies 2π apart are the same signal

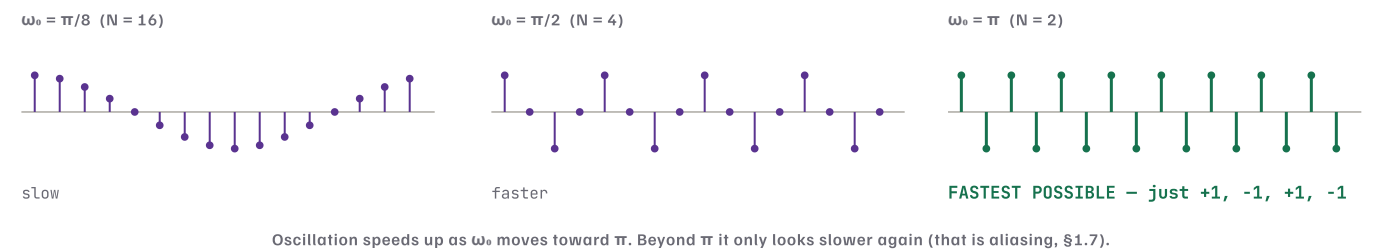
$$\cos[(\omega_0 + 2\pi)n + \theta] = \cos(\omega_0 n + 2\pi n + \theta) = \cos(\omega_0 n + \theta)$$

because $2\pi n$ is always a whole number of turns. So ω_0 and $\omega_0 + 2\pi n$ are indistinguishable. A DT sinusoid is **always periodic in frequency** with period 2π (or 1 in f), and it is **unique only in the principal range** $-\frac{1}{2} \leq f \leq \frac{1}{2}$. If $f_0 > \frac{1}{2}$, the unique frequency is $f_0 - M$ for the whole number M that brings it into range.

Property 3 — the fastest wiggle is at $\omega = \pi$

The teacher's table, sweeping ω_0 from 0 to π :

Ω_0	θ	$\pi/8$	$\pi/4$	$\pi/2$	π
f	0	1/16	1/8	1/4	1/2
N	∞	16	8	4	2



Module 01, Figure 4 redrawn. At $\omega = \pi$ the samples simply alternate sign. Nothing can wiggle faster than that with one sample per step.

CHECK YOURSELF

Tutorial 1, Q3 and Q4 — periodic or not, and the period

Method: write as $\cos(2\pi f n)$, read f , decide if it is a fraction.

Q3	F	VERDICT
$\cos(0.01\pi n)$	$0.005 = 1/200$	periodic, N = 200
$\cos(\pi \cdot 30n/105)$	$15/105 = 1/7$	periodic, N = 7
$\cos(3\pi n)$	$3/2$	periodic, N = 2
$\sin(3n)$	$3/2\pi$ — irrational	not periodic
$\sin(\pi \cdot 62n/10)$	$31/10$	periodic, N = 10

Q4(a) $\cos(n/8) \cdot \cos(\pi n/8)$: the first factor has $f = 1/16\pi$, irrational $\rightarrow$ the product is **not periodic**.

Q4(b) $\cos(\pi n/2) - \sin(\pi n/8) + 3\cos(\pi n/4 + \pi/3)$: periods 4, 16, 8 $\rightarrow$ **N = LCM = 16**. The phase $\pi/3$ plays no part.

1.7

Aliasing

Two different frequencies that give exactly the same samples.

Module 01 §23, §30 · Teacher module M01L003 §8

The teacher's argument. Let $\pi < \omega_0 < 2\pi$ and put $\omega_2 = 2\pi - \omega_0$. Then

$$A \cos(\omega_2 n) = A \cos[(2\pi - \omega_0)n] = A \cos(2\pi n - \omega_0 n) = A \cos(\omega_0 n)$$

The two are the same sequence, so ω_2 is an **alias** of ω_0 . That is why we only use $0 \leq \omega \leq \pi$ (or $-\frac{1}{2} \leq f \leq \frac{1}{2}$).

IN TERMS OF ANALOG FREQUENCIES

The highest digital frequency is $f = \frac{1}{2}$, so the highest analog frequency a sampled signal can carry is $F_{\max} = F_s/2$ — the **folding frequency**. Every frequency $F_k = F_0 + k F_s$ gives the same samples as F_0 .

WORKED

10 Hz and 50 Hz at $F_s = 40$ Hz (teacher's example)

$x_1(t) = \cos(2\pi \cdot 10t)$ and $x_2(t) = \cos(2\pi \cdot 50t)$, sampled at 40 Hz.

$$\begin{aligned} x_1(n) &= \cos\left(2\pi \cdot \left(\frac{10}{40}\right)n\right) = \cos\left(\left(\frac{\pi}{2}\right)n\right) \\ x_2(n) &= \cos\left(2\pi \cdot \left(\frac{50}{40}\right)n\right) = \cos\left(\left(\frac{5\pi}{2}\right)n\right) = \cos\left(2\pi n + \left(\frac{\pi}{2}\right)n\right) = \cos\left(\left(\frac{\pi}{2}\right)n\right) \end{aligned}$$

So $x_1(n) = x_2(n)$: **50 Hz is an alias of 10 Hz at 40 Hz sampling**. Once sampled, nothing can tell them apart.

1.8

Harmonically related complex exponentials

A family of exponentials whose frequencies are whole-number multiples of one fundamental — the bricks every Fourier method is built from.

Module 01 §24-25 · Teacher module M01L003 §9-10

Continuous time — infinitely many

$$s_k(t) = e^{jk\Omega_0 t} = e^{j2\pi k F_0 t}, k = 0, \pm 1, \pm 2, \dots$$

Each is periodic with period T_p/k , and every different k is a different signal, so there are **infinitely many**. A linear combination

$$x_a(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t}$$

with period $T_p = 1/F_0$ is the **Fourier series**.

Discrete time — only N of them

With $f_0 = 1/N$, $s_k(n) = e^{j2\pi kn/N}$. But now

$$s_{k+N}(n) = e^{\frac{j2\pi(k+N)n}{N}} = e^{\frac{j2\pi kn}{N}} \cdot e^{j2\pi n} = s_k(n)$$

THE FACT THAT LEADS TO THE DFT

In discrete time there are only **N distinct harmonics**, $k = 0, 1, \dots, N-1$, all sharing the period N. So the discrete Fourier series is a *finite* sum:

$$x(n) = \sum_{k=0}^{N-1} c_k e^{\frac{j2\pi kn}{N}}$$

This is why the DFT of N samples has exactly N values (Unit 5).

CHECK YOURSELF

Teacher's examples on periodicity of DT signals

(1) Is $\cos(2\pi F n)$ periodic for $F = 0.32$? For $F = \sqrt{3}$?

$$0.32 = \frac{8}{25} \rightarrow \text{periodic, } N = 25$$

3 is irrational $\rightarrow$ not periodic

(2) Period of $e^{j0.2\pi n} + e^{j0.3\pi n}$?

$$f_1 = 0.1 = 1/10 \rightarrow N_1 = 10; \quad f_2 = 0.15 = 3/20 \rightarrow N_2 = 20;$$

$$N = \text{LCM} = 20$$

(3) $x(t) = 2\cos(40\pi t) + \sin(60\pi t)$ sampled at 75 Hz. Common period of $x(n)$?

$$f_1 = \frac{20}{75} = \frac{4}{15} \rightarrow 15; \quad f_2 = \frac{30}{75} = \frac{2}{5} \rightarrow 5; \quad N = 15 \text{ samples} = 0.2s = 4 \text{ full cycles of the } 20 \text{ Hz part}$$

1.9

Sampling and the sampling theorem

Take a reading every T seconds. Do it faster than twice the highest frequency and you lose nothing.

We use **uniform (periodic) sampling**:

$$x(n) = x_a(nT), F_s = \frac{1}{T}, t = nT = \frac{n}{F_s}$$

CONTINUOUS-TIME	DISCRETE-TIME	LINK
Ω rad/s, F Hz	ω rad/sample, f cycles/sample	$\omega = \Omega T, f = F/F_s$
$-\infty < F < \infty$	$-\frac{1}{2} \leq f \leq \frac{1}{2}$	$-F_s/2 \leq F \leq F_s/2$

THE SAMPLING THEOREM

If the highest frequency in the analog signal is $F_{\max}$, choose

$$F_s > 2F_{\max}$$

so that every component has $|f| = \frac{|F|}{F_s} < \left(\frac{1}{2}\right)$

$F_N = 2F_{\max}$ is the **Nyquist rate**. Sample slower and aliasing folds the high components down onto the low ones, permanently.

WORKED

Teacher's review examples on sampling

(A) $x_a(t) = 3\cos(100\pi t)$, so $F = 50$ Hz.

1. Minimum rate to avoid aliasing: $F_s = 2F = 100$ Hz.
2. At $F_s = 200$ Hz: $x(n) = 3\cos(\pi n/2) = 3\cos(2\pi \cdot \frac{1}{4}n)$.
3. At $F_s = 75$ Hz: $x(n) = 3\cos(4\pi n/3) = 3\cos(2\pi n - 2\pi n/3) = 3\cos(2\pi \cdot \frac{1}{3}n)$.
4. The sinusoid that gives the same samples as (3) without aliasing:
 $F = f \cdot F_s = \frac{1}{3} \times 75 = 25$ Hz, i.e. $y_a(t) = 3\cos(50\pi t)$.

(B) $x_a(t) = 3\cos(2000\pi t) + 5\sin(6000\pi t) + 10\cos(12000\pi t)$

$$F_1 = 1 \text{ kHz}, F_2 = 3 \text{ kHz}, F_3 = 6 \text{ kHz} \Rightarrow \text{Nyquist rate} = 12 \text{ kHz}$$

Sample at only 5 kHz (folding frequency 2.5 kHz). 1 kHz survives; 3 kHz folds to $3 - 5 = -2$ kHz; 6 kHz folds to $6 - 5 = 1$ kHz:

$$x(n) = 3\cos(2\pi \cdot \frac{1}{5}n) - 5\sin(2\pi \cdot \frac{1}{5}n) + 10\cos(2\pi \cdot \frac{1}{5}n) = 13\cos(2\pi \cdot \frac{1}{5}n) - 5\sin(2\pi \cdot \frac{1}{5}n)$$

Ideal interpolation then gives back $y_a(t) = 13\cos(2000\pi t) - 5\sin(4000\pi t)$ — two components instead of three, and the 6 kHz one has landed on top of the 1 kHz one. That is the cost of aliasing.

(C) **Module 01 exercise.** An analog signal has frequencies up to 10 kHz.

1. Exact reconstruction needs $F_s \geq 2 \times 10 = 20$ kHz.
2. At $F_s = 8$ kHz the folding frequency is 4 kHz, so $F_1 = 5$ kHz appears as $8 - 5 = 3$ kHz.
3. $F_2 = 9$ kHz appears as $9 - 8 = 1$ kHz.

1.10

Quantization, coding and D/A

Sampling fixed the time axis. Quantization rounds each sample to the nearest allowed level, and coding writes that level in bits.

$$\Delta = \left(\frac{x_{\max} - x_{\min}}{L - 1} \right) \quad \text{step size (resolution), } L = \text{number of levels}$$

$$e_q(n) = x_q(n) - x(n), -\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2} \quad \text{for rounding}$$

Rounding picks the nearest level; **truncation** throws the extra digits away. More levels $\rightarrow$ smaller $\Delta \rightarrow$ smaller error $\rightarrow$ a more accurate quantizer.

WORKED

Quantizing $x(n) = 0.9^n u(n)$ to one decimal place (teacher's table) -

$x_{\min} = 0, x_{\max} = 1, L = 11$ levels, so $\Delta = 1/10 = 0.1$.

N	$x(N)$	$x_q(N)$	$e_q(N)$
0	1.0000	1.0	0
2	0.8100	0.8	-0.0100
3	0.7290	0.7	-0.0290
4	0.6561	0.7	+0.0439
6	0.5314	0.5	-0.0314
10	0.3487	0.3	-0.0487

Every error sits inside $\pm 0.05 = \pm \Delta/2$ ✓

SIGNAL-TO-QUANTIZATION-NOISE RATIO

With b bits over a range $2A$, $\Delta = 2A/2^b$. Treating the error as a small ramp between levels, its power is $P_q = \Delta^2/12$, and for a full-scale sinusoid $P_x = A^2/2$:

$$\text{SQNR(dB)} = 10 \log_{10} \left(\frac{P_x}{P_q} \right) = 1.76 + 6.02b$$

Every extra bit is worth about 6 dB.

Coding gives each of the L levels a b -bit binary word, $b \geq \lceil \log_2 L \rceil$.

Digital-to-analog conversion

All D/A converters interpolate. The simplest is the **zero-order hold** — hold each sample's value until the next one arrives, giving a staircase. **Linear interpolation** joins successive samples with

straight lines, but needs a delay of T so it knows the next sample. Higher-order interpolation does better still.

1.11

Writing down a discrete-time signal

Four ways to write the same list of numbers, and a few words for describing its shape.

Teacher module M01L001 §1-3

A discrete-time signal $x(n)$ is a function of an integer n , the time index, with $x(n) = x_a(nT_s)$. It is **not defined** for non-integer n . The teacher's example $x(n) = \{0, 0, 1, 4, 1, 0, 1\}$ can be written:

1. **Graphically** — a stem plot.
2. **Functionally** — $x(n) = 1$ for $n = -1, 1, 3$; 4 for $n = 2$; 0 elsewhere.
3. **In a table** — a row of n , a row of $x(n)$.
4. **As a sequence** — $\{\dots, 0, 0, 1, \uparrow 4, 1, 0, 1, 0, \dots\}$ with an arrow at $n = 0$.

WATCH THE ARROW

With no arrow, the sequence starts at $n = 0$. Get the origin wrong and every later convolution and DFT answer shifts with it.

NAME	MEANING
Right-sided	$x(n) = 0$ for $n < N$
Causal	$x(n) = 0$ for $n < 0$
Left-sided	$x(n) = 0$ for $n > N$
Anti-causal	$x(n) = 0$ for $n \geq 0$
Periodic, period N	$x(n + kN) = x(n)$ for all n and whole k ; N is always a whole number of samples

1.12

Measuring a signal: sums, energy, power

In discrete time, adding up replaces integrating — and three different sums tell you three different things.

Teacher module M01L001 §4

discrete sum $S_D = \sum_{n=-\infty}^{\infty} x(n)$

absolute sum $S_A = \sum_{n=-\infty}^{\infty} |x(n)|$
finite $\Rightarrow$ "absolutely summable"

cumulative sum $S_C(n) = \sum_{k=-\infty}^n x(k)$ the running total

ENERGY AND POWER

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$P = \left(\frac{1}{N}\right) \sum_{n=0}^{N-1} |x(n)|^2$$

periodic, over one period

$$P = \lim_{N \rightarrow \infty} \left(\frac{1}{2N+1}\right) \sum_{n=-N}^N |x(n)|^2 \quad \text{general}$$

- $0 < E < \infty \rightarrow$ an **energy signal** (then $P = 0$).
- $0 < P < \infty \rightarrow$ a **power signal** (then $E = \infty$).
- **All periodic signals are power signals.** Their energy is infinite, so we use averages instead.

WORKED Teacher's examples: energy and power

(3) $x(n) = 6\cos(2\pi n/4)$, period 4. Samples over one period: 6, 0, -6, 0.

$$\text{average} = \frac{6+0-6+0}{4} = 0 \quad P = \frac{36+0+36+0}{4} = 18 = \frac{A^2}{2}$$

(4) $x(n) = 6e^{j2\pi n/4}$, period 4. $|x(n)| = 6$ always, so

$$P = \left(\frac{1}{4}\right) (4 \times 36) = 36$$

$= A^2$ a complex exponential has twice the power of the cosine

CHECK YOURSELF

Tutorial 1, Q6 — energy or power signal?

(a) $(-0.5)^n u(n)$: $E = \sum (0.25)^n = 1/(1 - 0.25) = 4/3$ — finite, so an **energy signal**.

(b) $u(n)$: $E = \infty$; $P = \lim_{N \rightarrow \infty} (N+1)/(2N+1) = 1/2$ — a **power signal**.

(c) $e^{j(\pi n/4 + \pi/3)}$: magnitude 1, period 8, $P = 1$ — a **power signal**.

1.13

Operations on signals

Change the index n — shift, fold, scale — or change the value $x(n)$ — scale, add, multiply.

Teacher module M01L001 §5-6

Time transformations of n

Time shift $y(n) = x(n - k)$

$k > 0$: a **delay** — the signal moves right. $k < 0$: an **advance**. If x starts at $n = N$, $x(n-k)$ starts at $N + k$. Check: a sample at old index n lands at the new index that makes the argument equal n .

Folding $y(n) = x(-n)$

A mirror image about $n = 0$.

Time scaling $y(n) = x(Dn)$

$D > 1$ whole number: **decimation** — keep every D th sample. Information is lost. $0 < D < 1$:

interpolation — insert new samples (zero, step, or linearly interpolated values) between existing ones.

TWO ORDERS THAT MATTER

Fold and delay do not commute. With TD for delay and FD for fold:

$$\text{FD}\{\text{TD}_k[x(n)]\} = x(-n - k) \text{ but } \text{TD}_k\{\text{FD}[x(n)]\} = x(-n + k)$$

Decimation is the inverse of interpolation, but not the other way round. Decimate then interpolate loses samples; interpolate then decimate returns the original. If you need both, *interpolate first*.

WORKED

Fractional delay, the teacher's way

$x(n) = \{3, 4, 5, 6\}$. Find $y(n) = x(2n/3 - 1)$ — wait, better: find $x(n - \frac{1}{2})$, a half-sample delay.

A fractional argument needs three whole-number steps, in this order: **interpolate** → **shift** → **decimate**, so that each step involves only integers.

interpolate by 2 (step) : $\{3, 3, 4, 4, 5, 5, 6, 6\} = x\left(\frac{n}{2}\right)$

delay by 1 : $\{0, 3, 3, 4, 4, 5, 5, 6, 6\}$

decimate by 2 : $\{0, 3, 4, 5, 6\} = x\left(n - \left(\frac{1}{2}\right)\right)$, using step interpolation

Operations on the value $x(n)$

scaling : $y(n) = A \cdot x(n)$

sum : $y(n) = x_1(n) + x_2(n)$

product : $y(n) = x_1(n) \cdot x_2(n)$

CHECK YOURSELF

Tutorial 1, Q2 and Q5 — sketch the transformed signals

Signal from the tutorial: $x(n) = \{1, 1, 1, 1, \frac{1}{2}\}$ at $n = -1, 0, 1, 2, 3$, zero elsewhere.

ASKED	METHOD	RESULT
$x(n - 2)$	delay by 2 — slide right	1,1,1,1,½ at $n = 1 \dots 5$
$x(4 - n)$	write as $x(-(n-4))$: fold, then delay by 4	½,1,1,1,1 at $n = 1 \dots 5$
$x(n + 2)$	advance by 2 — slide left	1,1,1,1,½ at $n = -3 \dots 1$
$x(2n)$	keep even indices only	$y(0)=x(0)=1$, $y(1)=x(2)=1$; rest 0
$x(n) u(2 - n)$	$u(2 - n) = 1$ for $n \leq 2$	keeps 1,1,1,1 at $n = -1 \dots 2$; drops the ½
$x(n-1) \delta(n-3)$	impulse picks one value: $x(2)\delta(n-3)$	a single impulse of height 1 at $n = 3$
$x(n^2)$	evaluate: $y(\pm 1)=x(1)=1$, $y(0)=x(0)=1$, $y(\pm 2)=x(4)=0$	automatically even

Check the fold-then-delay result: the original sample at $n = 3$ (the ½) should appear where $4 - n = 3$, i.e. $n = 1$ ✓

1.14

Symmetry: even and odd parts

Any signal is one mirror-image part plus one anti-mirror part.

Teacher module M01L001 §7-8

A real signal is **even (symmetric)** if $x(-n) = x(n)$ and **odd (anti-symmetric)** if $x(-n) = -x(n)$. The two kinds are mutually exclusive, and every signal splits uniquely into one of each:

$$x_e(n) = \left(\frac{1}{2}\right) [x(n) + x(-n)] \quad x_o(n) = \left(\frac{1}{2}\right) [x(n) - x(-n)] \quad x(n) = x_e(n) + x_o(n)$$

If x is already even, $x_o = 0$; if already odd, $x_e = 0$. And $x_o(0) = 0$ always.

CHECK YOURSELF

Tutorial 1, Q5(g),(h) — even and odd parts

$x(n) = \{1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}\}$ at $n = -1 \dots 4$.

N	-4	-3	-2	-1	0	1	2	3	4
x(n)	0	0	0	1	1	1	1	1/2	1/2
x(-n)	1/2	1/2	1	1	1	1	0	0	0
x_e(n)	1/4	1/4	1/2	1	1	1	1/2	1/4	1/4
x_o(n)	-1/4	-1/4	-1/2	0	0	0	1/2	1/4	1/4

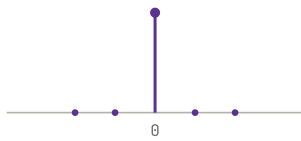
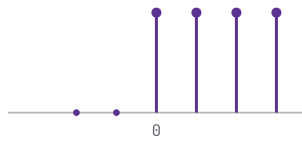
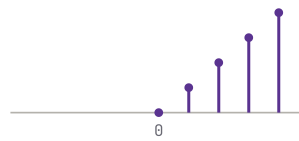
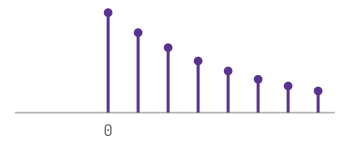
x_e reads the same backwards ✓; x_o reads the same backwards with flipped signs ✓; the two rows add back to $x(n)$ ✓

1.15

Elementary signals

The impulse, the step, the ramp, the pulse, the sinc and the exponential — everything else is built from these.

Teacher module M01L001 §9-14

IMPULSE $\delta(n)$ 1 at $n = 0$, else 0STEP $u(n)$ 1 for $n \geq 0$; $u(0) = 1$ RAMP $r(n)$  n for $n \geq 0$ EXPONENTIAL $a^n u(n)$  $0 < a < 1$ shown

The basic building blocks. Relations: $\delta(n) = u(n) - u(n-1)$, $u(n) = \sum_{k \geq 0} \delta(n-k)$,
 $r(n) = n \cdot u(n)$.

PROPERTIES OF THE DISCRETE IMPULSE

Product : $x(n) \cdot \delta(n-k) = x(k) \cdot \delta(n-k)$ an impulse of strength $x(k)$

Shifting (sifting) : $\sum_{n=-\infty}^{\infty} x(n) \cdot \delta(n-k) = x(k)$ the impulse extracts $x(k)$

So any signal is a sum of shifted, weighted impulses:

$$x(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n-k)$$

This one line is the starting point of convolution in Unit 2.

WORKED

Teacher's example: an impulse times a signal

$x(n) = 2^n$ and $y(n) = \delta(n-3)$. Find $z(n) = x(n)y(n)$ and $A = \sum z(n)$.

$$z(n) = 2^n \delta(n-3) = 2^3 \delta(n-3) = 8 \delta(n-3) \quad A = 8$$

The product is a single impulse of strength $x(3) = 8$; the sum just returns that strength.

Pulses and the sinc

$$\text{rect}\left(\frac{n}{2N}\right) = 1 \quad \text{for } |n| \leq N, \quad \text{else } 0$$

$2N+1$ samples; the $2N$ avoids half-integers

$$\text{tri}\left(\frac{n}{N}\right) = 1 - \frac{|n|}{N} \quad \text{for } |n| \leq N, \quad \text{else } 0$$

$2N+1$ samples, ends are zero

$$\text{sinc}\left(\frac{n}{N}\right) = \left(\frac{\sin\left(\frac{\pi n}{N}\right)}{\frac{\pi n}{N}}\right), \text{sinc}(0) = 1$$

zero at $n = kN$; a main lobe with decaying side lobes; $\text{sinc}(n) = \delta(n)$

The exponential

$$x(n) = a^n u(n)$$

real a : decays for $0 < a < 1$, grows for $a > 1$, alternates sign for $a < 0$

$$a = re^{j\theta}: \quad x(n) = r^n [\cos(\theta n) + j \sin(\theta n)] u(n)$$

For complex a : $0 < r < 1$ gives decaying cosine and sine parts, $r = 1$ pure ones, $r > 1$ growing ones. Remember this — it comes back as the rule for where poles may sit (Unit 3).

UNIT 2 · TEACHER MODULES M02L002, M02L003, M02L004, M02L005, M02L005B

Discrete-time systems

A system is a rule that turns one sequence into another. This unit is about the rules worth studying — linear, time-invariant ones — and the three ways to find their output: recursion, the formal solution, and convolution.

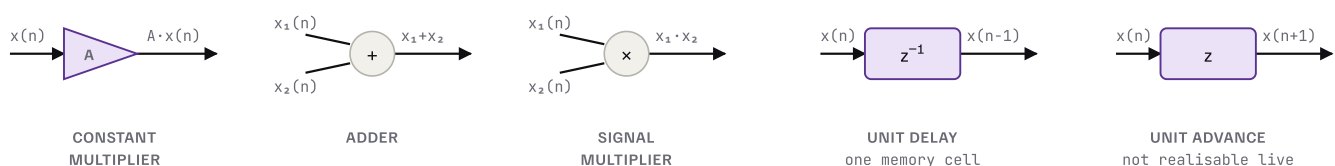
2.1

Building blocks and the input–output description

Every discrete-time system is wired from five parts, and its wiring can be written as a difference equation.

Teacher module M02L002 §3–5

In the time domain most discrete-time systems are modelled by a **difference equation** — for example $y(n) = y(n-1) + x(n)$, whose output is the previous output plus the present input. Such an equation only ever uses shifted signals, added signals, multiplied signals and scaled signals, so five blocks are enough to draw any of them.



Any difference equation is a wiring of these five. Real-time hardware can use every block except the advance.

The five blocks, as drawn in M02L002. The delay is written z^{-1} because, in Unit 3, delaying by one sample multiplies the z -transform by z^{-1} .

Input–output description

The rule that turns the input (excitation) into the output (response) is written with an operator T :

$$y(n) = T[x(n)] \quad \text{or} \quad x(n) \xrightarrow{T} y(n)$$

Teacher's reading example: $T\{\text{input}\} = 4z^{-3}\{\text{input}\} + 6$ means $y(n) = 4x(n-3) + 6$ — delay the input by 3, multiply by 4, add 6.

WORKED

One input, six systems (M02L002)

$x(n) = |n|$ for $-3 \leq n \leq 3$, zero elsewhere: $x = \{3, 2, 1, \uparrow 0, 1, 2, 3\}$.

N	-4	-3	-2	-1	0	1	2	3	4
$x(n)$	0	3	2	1	0	1	2	3	0
(a) $y = x(n)$	0	3	2	1	0	1	2	3	0
(b) $y = x(n-1)$	0	0	3	2	1	0	1	2	3
(c) $y = x(n+1)$	3	2	1	0	1	2	3	0	0
(d) $\frac{1}{3}[x(n+1)+x(n)+x(n-1)]$	1	5/3	2	1	2/3	1	2	5/3	1
(e) MAX of the same three	3	3	3	2	1	2	3	3	3
(f) $\sum_{k \leq n} x(k)$	0	3	5	6	6	7	9	12	12

(b) is a delay (the identity system with one memory cell), (c) an advance, (d) a *3-point moving average* — e.g. $y(0) = \frac{1}{3}[x(1)+x(0)+x(-1)] = \frac{1}{3}[1+0+1] = \frac{2}{3}$. (e) is a median-type non-linear system. (f) is the **accumulator**:
 $y(0) = x(0)+x(-1)+x(-2)+x(-3) = 0+1+2+3 = 6$, and it stays at 12 once the input has ended.

The accumulator, two ways

Written with inputs only it is **non-recursive**: $y(n) = \sum_{k=-\infty}^n x(k)$. Split off the last term and it becomes **recursive**:

$$y(n) = \sum_{k=-\infty}^{n-1} x(k) + x(n) = y(n-1) + x(n)$$

The recursive form needs an **initial condition** $y(-1)$ — the state of the memory when the input starts — and that value changes the output.

WORKED

**Accumulator
with**
 $x(n) = n u(n)$, two initial conditions
(M02L002)

$$y(n) = \sum_{k=-\infty}^{-1} x(k) + \sum_{k=0}^n x(k) = y(-1) + \sum_{k=0}^n k = y(-1) + \left(\frac{n(n+1)}{2} \right)$$

(a) $y(-1) = 0$: $y(n) = n(n+1)/2, n \geq 0$.

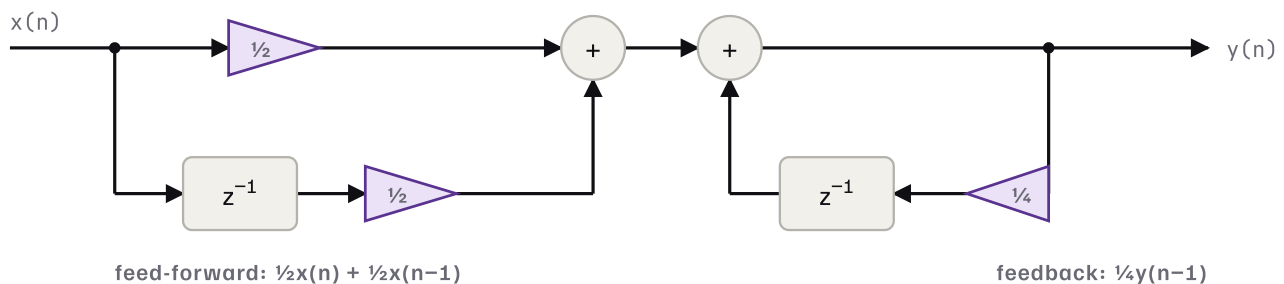
(b) $y(-1) = 1$: $y(n) = 1 + n(n+1)/2 = (n^2 + n + 2)/2, n \geq 0$.

Same system, same input, different output — the initial condition is part of the problem.

TWO REPRESENTATIONS

- **Non-recursive:** output written from *input samples only*.
- **Recursive:** output written from input samples *and previous outputs*. Needs initial conditions.

Block diagram from the equation



M02L002 block-diagram example: $y(n) = \frac{1}{4}y(n-1) + \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$. The left half builds $\frac{1}{2}x(n) + \frac{1}{2}x(n-1)$ from the input; the right loop takes the output, scales it by $\frac{1}{4}$, delays it and adds it back. That feedback loop is what makes the system recursive.

2.2

Linearity and superposition

A linear system treats a sum of inputs as a sum of outputs. That one property makes every method in this course work.

additive : $T\{x_1(n) + x_2(n)\} = T\{x_1(n)\} + T\{x_2(n)\}$

homogeneous : $T\{K \cdot x(n)\} = K \cdot T\{x(n)\}$

linear = both: $T\{A \cdot x_1(n) + B \cdot x_2(n)\} = A \cdot T\{x_1(n)\} + B \cdot T\{x_2(n)\}$

Anything that fails this is **non-linear**. Superposition of linear transformations is itself linear, so a chain of linear systems is linear.

WHY WE CARE

If $x(n) = \sum a_k x_k(n)$ then $y(n) = \sum a_k y_k(n)$. So: **break the input into simple pieces, find the response to each piece, add.** Convolution does this with impulses as the pieces (§2.11); the DTFT does it with complex exponentials as the pieces (Unit 4).

2.3

Classifying systems

Static or dynamic, time-invariant or not, linear or not, causal or not, stable or not — five yes/no questions, each with a one-line test.

Teacher module **M02L002** §7-9

QUESTION	YES WHEN...	TEACHER'S EXAMPLES
Static (memoryless)	output uses only $x(n)$ at the same n	$y = ax(n); y = nx(n) + bx^3(n)$
Dynamic (has memory)	output uses past or future samples	$y = ax(n) + 3x(n-1); \sum_{k=0}^n x(n-k)$
Time-invariant	delaying the input only delays the output: $y(n, k) = y(n-k)$ for all k	differentiator $x(n) - x(n-1)$
Linear	superposition holds (§2.2)	constant-coefficient difference equations
Causal	output depends on present and past inputs only	$y(n) - y(n-1) = x(n)$
BIBO stable	every bounded input gives a bounded output	roots of the characteristic equation inside the unit circle

The time-invariance test

Delay the input by k and compute the response $y(n, k) = T[x(n-k)]$. Separately delay the original output to get $y(n-k)$. Equal for every $k \rightarrow$ time-invariant. Not equal for even one $k \rightarrow$ time-variant.

WORKED Three tests from M02L002

Differentiator $y(n) = x(n) - x(n-1)$: $y(n, k) = x(n-k) - x(n-k-1)$ and $y(n-k) = x(n-k) - x(n-k-1)$. Equal $\rightarrow$ **time-invariant**.

Time multiplier $y(n) = n x(n)$: $y(n, k) = n x(n-k)$ but $y(n-k) = (n-k)x(n-k)$. Different $\rightarrow$ **time-variant**.

Folder $y(n) = x(-n)$: $y(n, k) = x(-n-k)$ but $y(n-k) = x(-n+k)$. Different $\rightarrow$ **time-variant**.

LTI SYSTEMS, AND HOW TO SPOT A NON-LTI EQUATION AT A GLANCE

A **linear time-invariant (LTI)** system is one described by a **linear constant-coefficient difference equation**:

$$y(n) + a_1 y(n-1) + \dots + a_N y(n-N) = b_0 x(n) + b_1 x(n-1) + \dots + b_M x(n-M)$$

Every term contains x or y , every coefficient is a constant, and the leading coefficient is 1. N (the largest output delay) is the **order**.

- **Non-linear** if any term is a constant, or a product / power / function of x or y (e.g. $x^2(n)$, $x(n)y(n-1)$, $+3$).
- **Time-varying** if any coefficient is a function of n (e.g. $n \cdot x(n)$, $\cos(\omega n)x(n)$), or if the index is scaled or folded ($x(2n)$, $x(-n)$).

Causality from the equation

It is non-causal if the least-delayed output term sits beside an input term $x(n+k)$ with $k > 0$.

Teacher's list:

EQUATION	VERDICT
$y(n) - y(n-1) = x(n)$	causal
$y(n+1) - y(n) = x(n+1)$	causal — shift everything back by 1: $y(n) - y(n-1) = x(n)$
$y(n) - y(n-2) = x(n+1)$	non-causal — needs a future input
$y(n+1) - y(n) = x(n+2)$	non-causal — shifted: $y(n) - y(n-1) = x(n+1)$
$y(n+4) + y(n+3) = x(n+2)$	causal — identical to $y(n) + y(n-1) = x(n-2)$

The same test in z-terms (Unit 3): with $H(z) = B(z)/A(z)$ written in positive powers, a numerator of higher degree than the denominator means non-causal.

Stability from the characteristic roots

BIBO STABILITY

For an LTI difference equation, a necessary and sufficient condition is that **every root of the characteristic equation has magnitude less than 1**.

- $y(n) - \frac{1}{2}y(n-1) - \frac{1}{2}y(n-2) = x(n)$: characteristic equation $z^2 - \frac{1}{2}z - \frac{1}{2} = 0$, roots $\frac{1}{2}$ and $-\frac{1}{2} \rightarrow$ **stable**.
- $y(n) - 2y(n-1) + y(n-2) = x(n)$: repeated root at 1 $\rightarrow$ **unstable**.

CHECK YOURSELF

Tutorial 2, Q1–Q4 — classify these systems

Q1 — linear? Test $T[a_1x_1 + a_2x_2]$ against $a_1T[x_1] + a_2T[x_2]$.

SYSTEM	VERDICT	WHY
$y(n) = \sin(x(n))$	Non-linear	$\sin(a_1x_1+a_2x_2) \neq a_1\sin x_1 + a_2\sin x_2$
$y(n) = x^3(n)$	Non-linear	the cube of a sum is not the sum of cubes
$y(n) = 2x(n) + 3$	Non-linear	the constant 3: zero input should give zero output, but gives 3
$y(n) = \sum_{k=n-a}^{n+a} x(k)$	Linear	a plain sum of input samples

Q2 — time-invariant?

$y = n \cdot x(n)$: $y(n, k) = n \cdot x(n - k) \neq (n - k) \cdot x(n - k) = y(n - k) \rightarrow$ time-varying

$y = x^2(n)$: $y(n, k) = x^2(n - k) = y(n - k) \rightarrow$ time-invariant
(non-linear, but that is a separate question)

$y = \sum_{n-a}^{n+a} x(k)$: the window slides with $n \rightarrow$ time-invariant

Q3 — static or dynamic? $x(n-1)$: dynamic. $x^2(n)$: static. $\sum_{k \leq n} x(k)$: dynamic, infinite memory.

Q4 — causal, memory, time-invariant?

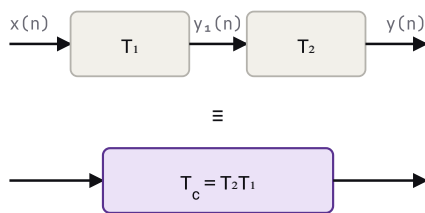
SYSTEM	CAUSAL	MEMORY	TIME-INV.
$y = x(n)\cos(\omega_c n)$	yes	memoryless	no — the coefficient depends on n
$y = \cos(x(n)) - 2\sin(x(n-1))$	yes	has memory	yes (non-linear though)
$y = x(n-2)$	yes	has memory	yes

Interconnected systems

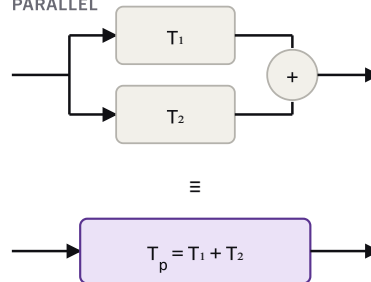
Chain two systems, or put them side by side, and you get one bigger system.

Teacher module M02L002 §8

CASCADE (series)



PARALLEL



Series and parallel, as in M02L002. In cascade $y(n) = T_2[T_1[x(n)]]$; in parallel $y(n) = T_1[x(n)] + T_2[x(n)]$. **Only if both are LTI** does the cascade order not matter ($T_1T_2 = T_2T_1$). In terms of impulse responses (§2.13): cascade $\rightarrow h_1 * h_2$, parallel $\rightarrow h_1 + h_2$.

CHECK YOURSELF

Tutorial 2, Q6 — impulse response of a cascade

$h_1(n) = (\frac{1}{2})^n u(n)$ and $h_2(n) = (\frac{1}{4})^n u(n)$ in cascade. Cascade means convolve (you will meet the formula in §2.12; the algebra is short):

$$\begin{aligned} h(n) &= \sum_{k=0}^n \left(\left(\frac{1}{2} \right) \right)^k \left(\left(\frac{1}{4} \right) \right)^{n-k} \\ &= \left(\left(\frac{1}{4} \right) \right)^n \sum_{k=0}^n 2^k \\ &= \left(\left(\frac{1}{4} \right) \right)^n (2^{n+1} - 1) \\ &= \left[2 \left(\left(\frac{1}{2} \right) \right)^n - \left(\left(\frac{1}{4} \right) \right)^n \right] u(n) \end{aligned}$$

Check $n = 0: 2 - 1 = 1 = h_1(0)h_2(0) \checkmark$. $n = 1: 1 - \frac{1}{4} = \frac{3}{4}$, and directly $\frac{1}{4} + \frac{1}{2} = \frac{3}{4} \checkmark$

2.5

FIR and IIR systems

Feed the system one impulse. If the output dies out after a finite number of samples it is FIR; if it goes on forever it is IIR.

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k) \Rightarrow y(n) = \frac{1}{a_0} \left[\sum_{k=0}^M b_k x(n-k) - \sum_{k=1}^N a_k y(n-k) \right]$$

The x terms are the **feed-forward** path (present and past inputs); the y terms are the **feedback** path (previous outputs). Feedback → recursive.

THE IMPULSE RESPONSE DECIDES

Apply $x(n) = \delta(n)$ to a relaxed LTI system; the output is $h(n)$, the **impulse response**.

- **FIR** — $h(n) = 0$ for $n > N$: a finite number of non-zero samples.
- **IIR** — $h(n) \neq 0$ for arbitrarily large n : the response never fully stops.

Teacher's first-order example $y(n) = a y(n-1) + b x(n)$ with $x = \delta$: $h(0) = b, h(1) = ab, h(2) = a^2b, \dots$ so $h(n) = b a^n u(n)$ — the feedback keeps generating new samples: IIR. Truncate any IIR response after N samples and you get an FIR system $y(n) = \sum_{k=0}^N h(k)x(n-k)$.

FIR	IIR
no feedback	feedback present
finite impulse response	infinite impulse response
usually non-recursive	recursive
always BIBO stable (finite coefficients)	stability depends on the poles

2.6

Solving by recursion

Start from the initial conditions and turn the handle: $y(0)$, then $y(1)$, then $y(2)$... Simple, but it gives numbers, not a formula.

Two methods find the output of a causal IIR system from its difference equation: **(1) recursion**, and **(2) the formal solution** (total = natural + forced, §2.7). Recursion means computing each output from the ones before it.

WORKED

Step, impulse and ramp responses by recursion (M02L003)

System $y(n) = a_1 y(n-1) + b_0 x(n)$, $y(-1) = 0$.

(1) Step input $x(n) = u(n)$:

$$y(0) = b_0, y(1) = a_1 b_0 + b_0 = b_0(1 + a_1), y(2) = b_0(1 + a_1 + a_1^2), \dots$$

$$y(n) = b_0(1 + a_1 + \dots + a_1^n) = b_0 \left(\frac{1 - a_1^{n+1}}{1 - a_1} \right) \quad \text{the step response}$$

(3) Impulse input $x(n) = \delta(n)$: $y(0) = b_0$, $y(1) = a_1 b_0$, $y(2) = a_1^2 b_0 \dots$ so
 $h(n) = b_0 a_1^n u(n)$, IIR whenever $a_1 \neq 0$.

(2) Ramp input $x(n) = n u(n)$:

$y(0) = 0$, $y(1) = b_0$, $y(2) = a_1 b_0 + 2b_0$, $y(3) = a_1^2 b_0 + 2a_1 b_0 + 3b_0$ — the pattern $y(n) = b_0 \sum_{k=1}^n k a_1^{n-k}$, which the slides sum to a closed form. The point: recursion always *works*, but the formula is only obvious for simple inputs.

(4) $y(n) = y(n-1) + x(n) - x(n-3)$ with $x = \delta$: $h = \{1, 1, 1, 0, 0, \dots\}$. Only three non-zero terms — a **recursive equation with a finite impulse response**. Recursive and IIR are not the same thing (more in §2.9).

WORKED

The cumulative-average filter (M02L003)

Non-recursive: $y(n) = \frac{1}{n+1} \sum_{k=0}^n x(k)$ — needs every past input stored. Pull out the last

term and use the previous average:

$$\begin{aligned} y(n) &= \left(\frac{1}{n+1} \right) \left[\sum_{k=0}^{n-1} x(k) + x(n) \right] \\ &= \left(\frac{1}{n+1} \right) [ny(n-1) + x(n)] \\ &= \left(\frac{n}{n+1} \right) y(n-1) + \left(\frac{1}{n+1} \right) x(n) \end{aligned}$$

Recursive form: two multiplications, one addition, **one** memory cell, instead of a growing list. $y(0) = x(0)$, $y(1) = \frac{1}{2}y(0) + \frac{1}{2}x(1)$, $y(2) = \frac{2}{3}y(1) + \frac{1}{3}x(2)$, ... Note its coefficients depend on n , so this useful system is linear but *time-varying*.

The formal solution

Total response = natural (from the roots) + forced (from the input). Or, cut another way: zero-input (from the initial conditions) + zero-state (from the input).

Teacher module M02L003 §7-11

What recursion reveals

Run $y(n) = a y(n-1) + x(n)$ from the initial condition $y(-1)$:

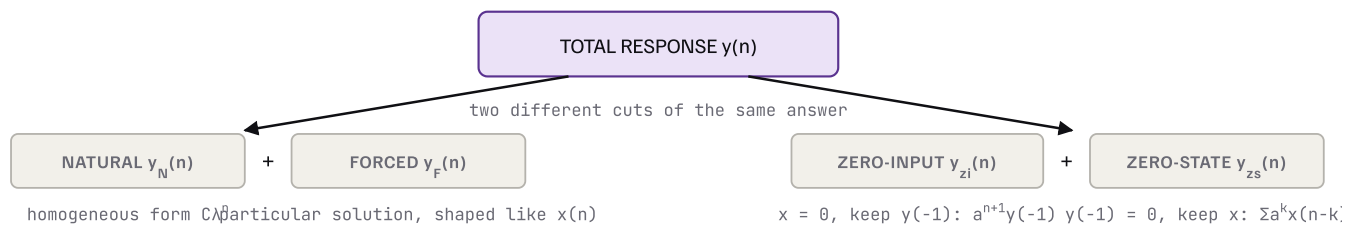
$$y(0) = ay(-1) + x(0)$$

$$y(1) = ay(0) + x(1) = a^2y(-1) + ax(0) + x(1)$$

$$y(n) = a^{n+1}y(-1) + \sum_{k=0}^n a^k x(n-k), n \geq 0$$

The output splits cleanly into a part driven by the *initial condition* and a part driven by the *input*.

Four names describe the pieces:



M02L003's "point to remember". Natural–forced = homogeneous + particular. Zero–input–zero–state = initial-condition response + input response. The zero-input response is the natural response *after* the initial condition is applied; the zero-state response contains the forced response *plus* whatever natural term is needed to start from rest. So $y_F \neq y_{ZS}$ in general.

Step 1 — the natural response from the characteristic equation

Set the input to zero and try $y_n(n) = \lambda^n$:

$$\lambda^n + a_1 \lambda^{n-1} + \dots + a_N \lambda^{n-N} = 0$$

$$\Rightarrow \lambda^N + a_1 \lambda^{N-1} + \dots + a_N = 0$$

the characteristic polynomial, N roots

ROOTS	FORM OF THE NATURAL RESPONSE
Real, distinct r	$C r^n$ (one term per root)
Complex pair $r = e^{\pm j\theta}$	$r^n [C_1 \cos(\theta n) + C_2 \sin(\theta n)]$
Real, repeated $p+1$ times	$r^n [C_0 + C_1 n + \dots + C_p n^p]$
Complex, repeated	each of the cos and sin terms gets its own polynomial in n

The constants C_i come from the initial conditions.

WORKED
Natural response of $y(n) - 3y(n-1) - 4y(n-2) = 0$ (M02L003)

$$\lambda^2 - 3\lambda - 4 = 0 \Rightarrow (\lambda + 1)(\lambda - 4) = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = 4$$

$$y_h(n) = C_1(-1)^n + C_2(4)^n$$

To fix the constants, run the equation once from $y(-1)$, $y(-2)$:

$$y(0) = 3y(-1) + 4y(-2) \text{ and } y(1) = 3y(0) + 4y(-1) = 13y(-1) + 12y(-2).$$

Match with the formula $y(0) = C_1 + C_2$, $y(1) = -C_1 + 4C_2$:

$$C_1 = -\left(\frac{1}{5}\right)y(-1) + \left(\frac{4}{5}\right)y(-2), C_2 = \left(\frac{16}{5}\right)y(-1) + \left(\frac{16}{5}\right)y(-2)$$

The root at 4 means this system is unstable — any non-zero start grows without limit.

Step 2 — the forced response from the input's shape

INPUT $x(n)$	TRY $y_p(n) =$
A (constant)	K
$A n^n$	$K n^n$
$A n^M$	$K_0 n^M + K_1 n^{M-1} + \dots + K_M$
$A^n n^M$	$A^n [K_0 n^M + \dots + K_M]$
$A \cos(\omega_0 n)$ or $A \sin(\omega_0 n)$	$K_1 \cos(\omega_0 n) + K_2 \sin(\omega_0 n)$

Substitute the guess into the difference equation at an n large enough that every step function equals 1, and solve for K .

WORKED

Particular solution for $x(n) = 2^n u(n)$ (M02L003)

System $y(n) = \frac{1}{3}y(n-1) - \frac{1}{6}y(n-2) + x(n)$. Try $y_p(n) = K \cdot 2^n$ and evaluate at $n = 2$:

$$4K = \frac{1}{3}(2K) - \frac{1}{6}(K) + 4 \Rightarrow 4K - \frac{5}{3}K + \frac{1}{6}K = 4 \Rightarrow \frac{5}{2}K = 4 \Rightarrow K = \frac{8}{5}$$

$$y_p(n) = \left(\frac{8}{5}\right) \cdot 2^n, n \geq 0$$

Step 3 — add, then fit the initial conditions

RECIPE FOR THE TOTAL RESPONSE

- 1 Roots of the characteristic equation $\rightarrow y_h(n)$ with unknown Cs.
- 2 Guess $y_p(n)$ from the table, substitute, solve for K.
- 3 $y(n) = y_h(n) + y_p(n)$.
- 4 Compute $y(0)$, $y(1)$, ... *directly from the difference equation* (using the given $y(-1)$, $y(-2)$), set them equal to the formula, and solve for the Cs.

WORKED

Second-order system with $x(n) = 4^n u(n)$ (M02L003 Example 2)

$$y(n) = 0.7y(n-1) - 0.1y(n-2) + 2x(n) - x(n-2), \text{ relaxed}$$

1

$$\lambda^2 - 0.7\lambda + 0.1 = 0 \Rightarrow \lambda = 0.5, 0.2 \Rightarrow y_h(n) = C_1(0.5)^n + C_2(0.2)^n$$

2

Try $y_p = K \cdot 4^n$; at $n = 2$ every term is live:

$$16K - 0.7(4K) + 0.1K = 2(16) - 1 \Rightarrow 13.3K = 31 \Rightarrow K = 2.331$$

3

$$y(n) = C_1(0.5)^n + C_2(0.2)^n + 2.331(4)^n$$

4

From the equation: $y(0) = 2x(0) = 2$; $y(1) = 0.7(2) + 2(4) = 9.4$. From the formula:

$$\begin{aligned} C_1 + C_2 + 2.331 &= 2 & 0.5C_1 + 0.2C_2 + 9.323 &= 9.4 \\ \Rightarrow C_2 &= -0.807, C_1 &= 0.476 \end{aligned}$$

$$y(n) = [0.476(0.5)^n - 0.807(0.2)^n + 2.331(4)^n]u(n)$$

Check: $0.476 - 0.807 + 2.331 = 2.000 = y(0) \checkmark$ (the slide prints $C_1 = 0.466$, which misses $y(0)$ by 0.01 — always run this check).

Impulse response by the same method

With $x(n) = \delta(n)$ the input is zero for $n > 0$, so the **particular solution is zero** and $h(n)$ is purely the natural form, with constants fixed by $h(0)$, $h(1)$, ... computed from the relaxed equation.

WORKED

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1) \quad (\text{M02L003})$$

$$h(n) = C_1(-1)^n + C_2(4)^n$$

from the equation, relaxed: $h(0) = 1, h(1) = 3h(0) + 2 = 5$

$$C_1 + C_2 = 1, -C_1 + 4C_2 = 5 \Rightarrow C_2 = \left(\frac{6}{5}\right), C_1 = -\left(\frac{1}{5}\right)$$

$$h(n) = \left[-\left(\frac{1}{5}\right)(-1)^n + \left(\frac{6}{5}\right)(4)^n \right] u(n)$$

WORKED

Two from the class notebook: particular solutions by the method of undetermined coefficients

(A) $y(n) = (5/6)y(n-1) - (1/6)y(n-2) + x(n)$ with $x(n) = 2^n, n \geq 0$. The input is an exponential, so try $y_p(n) = K \cdot 2^n u(n)$. Substitute for $n \geq 2$, where every u is 1:

$$\begin{aligned} K 2^n &= \frac{5}{6} K 2^{n-1} - \frac{1}{6} K 2^{n-2} + 2^n \quad \div 2^{n-2} & 4K &= \frac{5}{6} \cdot 2K - \frac{1}{6} K + 4 \\ 4K &= \frac{10}{6} K - \frac{1}{6} K + 4 \Rightarrow \frac{24-9}{6} K = 4 \Rightarrow K = \frac{24}{15} = \frac{8}{5} \end{aligned}$$

so $y_p(n) = (8/5) \cdot 2^n u(n)$. (The homogeneous part would be $C_1(1/2)^n + C_2(1/3)^n$, from $\lambda^2 - (5/6)\lambda + 1/6 = 0$.)

(B) $y(n) = 0.7y(n-1) - 0.1y(n-2) + 2x(n) - x(n-2)$ with $x(n) = 4^n u(n)$, relaxed. Find $y(n)$.

$$\begin{aligned} \text{homogeneous: } \lambda^2 - 0.7\lambda + 0.1 &= 0 \Rightarrow \lambda_1 = 0.5, \lambda_2 = 0.2, & y_h(n) &= c_1(0.5)^n + c_2(0.2)^n \\ \text{particular: } y_p(n) = K 4^n u(n): & K 4^n = 0.7K 4^{n-1} - 0.1K 4^{n-2} + 2 \cdot 4^n - 4^{n-2} \\ & \div 4^{n-2}: 16K = 2.8K - 0.1K + 32 - 1 \Rightarrow 13.3K = 31 \Rightarrow K = 2.3308 \\ \text{constants from the recursion: } y(0) = 2x(0) &= 2, & y(1) &= 0.7(2) + 2(4) = 9.4 \\ c_1 + c_2 + 2.3308 &= 2, & 0.5c_1 + 0.2c_2 + 4(2.3308) &= 9.4 \\ &\Rightarrow c_1 = 0.476, & c_2 &= -0.807 \end{aligned}$$

$$y(n) = [0.476(0.5)^n - 0.807(0.2)^n + 2.331(4)^n] u(n)$$

Check at $n = 2$: the recursion gives

$y(2) = 0.7(9.4) - 0.1(2) + 2(16) - 1 = 37.38$, and the formula gives

$0.119 - 0.032 + 37.29 = 37.38 \checkmark$. The notebook writes $c_1 = 0.466$. Solving the two equations exactly gives 0.476, and the check at $n = 2$ only passes with 0.476. The other two constants are right.

2.8

Impulse response from the equation

A two-line recipe that gives $h(n)$ for any linear constant-coefficient difference equation.

Teacher module M02L004 §1-5

Both responses below are defined only for a *relaxed* LTI system (all initial conditions zero):

- **Impulse response** $h(n)$ — output when the input is $\delta(n)$.
- **Step response** $s(n)$ — output when the input is $u(n)$.

Non-recursive (FIR): read it off

$$y(n) = b_0x(n) + b_1x(n-1) + \dots + b_Mx(n-M) \Rightarrow h(n) = \{b_0, b_1, \dots, b_M\}$$

The impulse response is the list of coefficients.

Recursive: recursion, or the natural response

Teacher's example $y(n) - a y(n-1) = x(n)$: put $x = \delta$, $h(-1) = 0$, and turn the handle: $h(0) = 1$, $h(1) = a$, $h(2) = a^2$, ... so $h(n) = a^n u(n)$. Recursion gives as many terms as you like but no formula. The formula comes from noticing that for $n > 0$ the impulse has gone, so $h(n)$ must be a natural response:

RECIPE (SINGLE INPUT TERM)

For $y(n) + a_1y(n-1) + \dots + a_Ny(n-N) = x(n)$: solve the homogeneous equation $h(n) + a_1h(n-1) + \dots + a_Nh(n-N) = 0$ with $h(0) = 1$ and every other initial condition zero.

RECIPE (GENERAL CASE)

For a right-hand side $b_0x(n) + \dots + b_Mx(n-M)$:

1. Find $h_0(n)$ for the single-input version as above.
2. Superpose: $h(n) = b_0h_0(n) + b_1h_0(n-1) + \dots + b_Mh_0(n-M)$.

WORKED

Two examples from M02L004

(1) $y(n) - 0.6y(n-1) = x(n)$. Natural form $h(n) = C(0.6)^n$; $h(0) = 1$ gives $C = 1$: $h(n) = (0.6)^n u(n)$.

(2) $y(n) - \frac{1}{6}y(n-1) - \frac{1}{3}y(n-2) = x(n)$.

$$\begin{aligned} \lambda^2 - \left(\frac{1}{6}\right)\lambda - \left(\frac{1}{3}\right) &= 0 \Rightarrow \lambda = \left(\frac{1}{2}\right), -\left(\frac{1}{3}\right) \Rightarrow h(n) = C_1 \left(\left(\frac{1}{2}\right)\right)^n + C_2 \left(-\left(\frac{1}{3}\right)\right)^n \\ h(0) = 1: \quad C_1 + C_2 &= 1 \quad h(-1) = 0: \quad 2C_1 - 3C_2 = 0 \\ \Rightarrow C_1 &= 0.6, C_2 = 0.4 \quad h(n) = \left[0.6 \left(\left(\frac{1}{2}\right)\right)^n + 0.4 \left(-\left(\frac{1}{3}\right)\right)^n\right] u(n) \end{aligned}$$

Check $h(1)$ by recursion: $\frac{1}{6}(1) = 0.1667$; formula: $0.3 - 0.1333 = 0.1667 \checkmark$

CHECK YOURSELF

Tutorial 3, Q1 — structure, impulse response, step response, initial conditions

(a) $y(n) - 0.6y(n-1) = x(n)$. Recursive structure: one delay, one multiplier (0.6), one adder. $h(n) = C(0.6)^n$, $h(0) = 1 \rightarrow h(n) = (0.6)^n u(n)$, IIR.

(b) $y(n) - \frac{1}{2}y(n-1) - \frac{1}{3}y(n-2) = 4x(n)$. **Impulse and step response.** Roots $\frac{1}{2}$, $-\frac{1}{3}$ (stable). Here $h(0) = 4$ and, from the equation, $h(1) = \frac{1}{2}(4) = \frac{2}{1}$:

$$C_1 + C_2 = 4, \left(\frac{1}{2}\right)C_1 - \left(\frac{1}{3}\right)C_2 = \left(\frac{2}{3}\right) \Rightarrow C_1 = 2.4, C_2 = 1.6$$

$$h(n) = \left[2.4 \left(\left(\frac{1}{2}\right)\right)^n + 1.6 \left(-\left(\frac{1}{3}\right)\right)^n \right] u(n)$$

(Same answer as $4 \times$ the teacher's example (2) above — superposition.) Step response: particular solution a constant K : $K - K/6 - K/6 = 4 \Rightarrow K = 6$. Then $s(0) = 4$, $s(1) = 4 + \frac{1}{2}(4) = 4.667$:

$$s(n) = 6 + D_1 \left(\left(\frac{1}{2}\right)\right)^n + D_2 \left(-\left(\frac{1}{3}\right)\right)^n, D_1 + D_2 = -2, \left(\frac{1}{2}\right)D_1 - \left(\frac{1}{3}\right)D_2 = -1.333$$

$$\Rightarrow D_1 = -2.4, D_2 = 0.4 \quad s(n) = \left[6 - 2.4 \left(\left(\frac{1}{2}\right)\right)^n + 0.4 \left(-\left(\frac{1}{3}\right)\right)^n \right] u(n)$$

It settles at 6 — the DC gain $4/(1 - \frac{1}{2} - \frac{1}{3}) = 6$.

(c) **Same system as (a)**, $x(n) = (0.4)^n$ for $n \geq 0$, $y(-1) = 10$. (This is also the teacher's Assignment 1.)

natural: $C(0.6)^n$ particular: $K(0.4)^n \rightarrow K(0.4)^n - 0.6K(0.4)^{n-1} = (0.4)^n \Rightarrow 0.4K - 0.6K = 0.4 \Rightarrow K = -2$

$y(0) = 0.6(10) + 1 = 7 = C - 2 \Rightarrow C = 9$

$y(n) = [9(0.6)^n - 2(0.4)^n]u(n)$

Check $y(1)$: formula $5.4 - 0.8 = 4.6$; equation $0.6(7) + 0.4 = 4.6 \checkmark$

(d) $y(n) - \frac{1}{2}y(n-1) - \frac{1}{3}y(n-2) = 2x(n) - 6x(n-1)$. **Impulse response.** Roots as before. $h(0) = 2$; $h(1) = \frac{1}{2}(2) - 6 = -5.667$:

$$C_1 + C_2 = 2, \left(\frac{1}{2}\right)C_1 - \left(\frac{1}{3}\right)C_2 = -5.667 \Rightarrow C_1 = -6, C_2 = 8$$

$$h(n) = \left[-6 \left(\left(\frac{1}{2}\right)\right)^n + 8 \left(-\left(\frac{1}{3}\right)\right)^n \right] u(n)$$

Or by the general recipe: $h = 2h_0(n) - 6h_0(n-1)$ with h_0 from example (2). Same result.

2.9

Recursive forms, anti-causal responses, stability

Three things the impulse response teaches: FIR can be written recursively, one equation has two impulse responses, and stability is about the roots.

Teacher module M02L004 §6-8

A non-recursive filter in recursive form

"FIR" and "non-recursive" mean the same thing; "IIR" and "recursive" usually do, but not always.

Teacher's example: $y(n) = x(n) + x(n-1) + x(n-2)$, so $h = \{1, 1, 1\}$. Write the equation one step back, $y(n-1) = x(n-1) + x(n-2) + x(n-3)$, and subtract:

$$y(n) - y(n-1) = x(n) - x(n-3) \quad \text{recursive equation, same finite } h(n)$$

Recursion makes a 3-term sum cost one add and one subtract, whatever the window length.

The anti-causal response

The same difference equation can be solved *backwards* in time. For the causal response we assume $h(n) = 0$ for $n < 0$; for the anti-causal response $h_A(n)$ we assume $h(n) = 0$ for $n \geq 0$ and solve for $n < 0$.

WORKED $y(n) - 0.4y(n-1) = x(n)$, both ways (M02L004)

Causal: $h(n) = 0.4h(n-1) + \delta(n)$ forwards from $h(0) = 1$: $h(n) = (0.4)^n u(n)$.

Anti-causal: rearrange for the earlier sample, $h(n-1) = 2.5[h(n) - \delta(n)]$, and step backwards from $h(0) = 0$:

$$h(-1) = 2.5[h(0) - 1] = -2.5 \quad h(-2) = 2.5h(-1) = -(2.5)^2 \dots$$

$$h_A(n) = -(2.5)^{-n} = -(0.4)^n u(-n-1)$$

One equation, two impulse responses. Which one you mean is decided by the region of convergence of the z-transform — Unit 3.

Stability

THREE EQUIVALENT STATEMENTS

- BIBO stable: every bounded input gives a bounded output.

- For an LTI difference equation: **every characteristic root has magnitude < 1** (complex roots: $|\lambda| < 1$). Then the natural response always decays and the forced response stays bounded.
- For an impulse response: $\sum |h(n)| < \infty$ — absolutely summable.

Roots *on* the unit circle make the system unstable: a simple unit-magnitude root gives a bounded natural response but a ramp or growing sinusoid when driven at the same frequency; a repeated one grows on its own. **FIR filters are always stable** — a finite weighted sum of bounded inputs is bounded.

TEACHER'S EXAMPLE	ROOTS	VERDICT
$y - \frac{1}{2}y(n-1) - \frac{1}{4}y(n-2) = x$	$\frac{1}{2}, -\frac{1}{2}$	stable
$y - y(n-1) = x$	1	unstable — a step input gives $C n u(n)$
$y - 2y(n-1) + y(n-2) = x$	1, 1	unstable — $C_1 + C_2 n$ grows
$y - \frac{1}{2}y(n-1) = n x(n)$	—	time-varying; step input $\rightarrow$ ramp $\rightarrow$ unstable
$y = x(n) - 2x(n-1)$	—	FIR $\rightarrow$ stable

2.10

Working backwards; the inverse system

Given $h(n)$, recover the difference equation. Then swap x and y to undo the system.

Teacher module **M02L004** §9–11

The exponentials in $h(n)$ are the characteristic roots, so they give the left-hand side. Then apply that same left-hand side to $h(n)$ itself — form $h(n) + a_1 h(n-1) + \dots$ — and see what sum of impulses is left; that is the right-hand side. The z-transform does this mechanically (Unit 3: $H(z) = Y(z)/X(z)$), which is why the teacher calls the time-domain route tedious.

WORKED

Three from M02L004

(1) $h(n) = u(n)$: $h(n) - h(n-1) = u(n) - u(n-1) = \delta(n) \Rightarrow y(n) - y(n-1) = x(n)$ — the accumulator.

(2) $h(n) = 3(0.6)^n u(n)$: root 0.6, so try $h(n) - 0.6h(n-1)$:

$$3(0.6)^n u(n) - 1.8(0.6)^{n-1} u(n-1) = 3(0.6)^n [u(n) - u(n-1)] = 3\delta(n) \\ \Rightarrow y(n) - 0.6y(n-1) = 3x(n)$$

(3) $h(n) = 2(-0.5)^n u(n) + (0.5)^n u(n)$: roots -0.5 and 0.5, so the left side is $(1 + 0.5z^{-1})(1 - 0.5z^{-1}) = 1 - 0.25z^{-2}$. Working it through (or adding the two z-transforms, Unit 3):

$$y(n) - 0.25y(n-2) = 3x(n) - 0.5x(n-1)$$

Check: $h(0) = 3\sqrt{}$; $h(1) = 2(-0.5) + 0.5 = -0.5$, and the equation gives $h(1) = -0.5\sqrt{}$. (The slide's printed answer does not pass this check — always test $h(0)$, $h(1)$.)

From input–output data: if $x = \{1, 2, 3\}$ produced $y = \{3, 3\}$, then

$$H(z) = Y(z)/X(z) = (3 + 3z^{-1})/(1 + 2z^{-1} + 3z^{-2}), \text{ i.e.}$$

$$y(n) + 2y(n-1) + 3y(n-2) = 3x(n) + 3x(n-1).$$

The inverse system



M02L004: a system and its inverse in cascade give back the input. Not every system has one — distinct inputs must lead to distinct outputs.

FOR A DIFFERENCE EQUATION, INVERTING = SWAPPING X AND Y

$$\text{system : } y(n) + 2y(n-1) = 3x(n) - 4x(n-1)$$

$$\text{inverse : } 3y(n) - 4y(n-1) = x(n) + 2x(n-1)$$

Given only $h(n)$, first recover the difference equation, then swap.

Convolution: where it comes from

Break the input into impulses, let the system answer each one, add the answers. That is all convolution is.

Teacher module M02L005 §1-3

KEY RESULT

For a relaxed LTI system, the output is the input convolved with the impulse response:

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Once you know $h(n)$, you know the zero-state response to every input.

The derivation, in three lines

Line 1 — any signal is a sum of impulses (§1.15): $x(n) = \sum_k x(k)\delta(n-k)$. Teacher's example:
 $x(n) = \{2, \uparrow 4, 0, 3\} = 2\delta(n+1) + 4\delta(n) + 3\delta(n-2)$.

Line 2 — linearity lets the system act on each impulse separately:

$$y(n) = T \left\{ \sum_k x(k)\delta(n-k) \right\} = \sum_k x(k)T\{\delta(n-k)\}$$

Line 3 — time invariance says a shifted impulse gives a shifted impulse response,
 $T\{\delta(n-k)\} = h(n-k)$. Substitute and you have the convolution sum. At one instant n_0 :
 $y(n_0) = \sum_k x(k)h(n_0-k)$ — one number, from one multiply-and-add pass over k .

2.12

Computing a convolution

Fold, shift, multiply, add — once for every output sample. Or line the numbers up in a table and let the layout do the folding for you.

Teacher module M02L005 §4-5

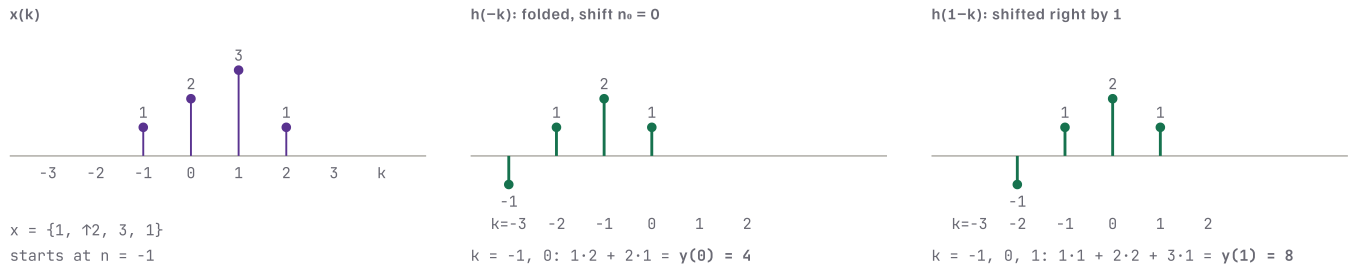
THE FIVE STEPS (M02L005)

- 1** Fold $h(k)$ about $k = 0$ to get $h(-k)$.
- 2** Shift $h(-k)$ by n_0 (right if positive, left if negative) to get $h(n_0-k)$.
- 3** Multiply $x(k)$ by $h(n_0-k)$, sample by sample.

4 Sum the products. That is $y(n_0)$.

5 Repeat 2–4 for every n_0 .

Length of the result: $n_1 + n_2 - 1$. It **starts** at (start of x) + (start of h) and **ends** at (end of x) + (end of h).



M02L005's worked example, two output samples of it. $x = \{1, 2, 3, 1\}$,
 $h = \{1, 2, 1, -1\}$. Folding h puts its samples at $k = 0, -1, -2, -3$; each shift right
 by one gives the next output. Continuing: $y(-1) = 1$, $y(2) = -1+2+6+1 = 8$,
 $y(3) = -2+3+2 = 3$, $y(4) = -3+1 = -2$, $y(5) = -1$, and zero from $n = 6$. So
 $y = \{1, 2, 8, 3, -2, -1\}$: 7 samples, from $-1+0$ to $2+3$.

The table method

Write one sequence across, the other down, fill the multiplication grid, and add along the
 diagonals. Teacher's example $x = \{2, 5, 0, 4\}$, $h = \{4, 1, 3\}$:

N	-3	-2	-1	0	1	2
×4	8	20	0	16		
×1		2	5	0	4	
×3			6	15	0	12
y(n)	8	22	11	31	4	12

Each row is x scaled by one sample of h and shifted by that sample's index. The result starts at
 $-3 + 0 = -3$.

THE FIVE-SECOND CHECK

$$\sum_n y(n) = \left(\sum_n x(n) \right) \left(\sum_n h(n) \right)$$

Above: $(11)(8) = 88$ and $8+22+11+31+4+12 = 88 \checkmark$. Do it every time — it catches
 almost every slip.

Teacher's exercise (2): $y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$, so $h = \{\frac{1}{2}, \frac{1}{2}\}$, with
 $x = \{2, 4, 6, 8, 10, 12, \dots\}$. The output is the average of each pair of neighbours:
 $y = \{1, 3, 5, 7, 9, 11, \dots\}$.

Analytical evaluation — infinite sequences

When the sequences go on forever, do the sum with algebra. The step functions set the limits:

$u(k) = 0$ for $k < 0$ gives a lower limit $k = 0$; $u(n-k) = 0$ for $k > n$ gives an upper limit
 $k = n$. Teacher's four results:

$X(N)$	$H(N)$	$Y(N)$
$u(n)$	$u(n)$	$\sum_{k=0}^n 1 = (n+1)u(n) = r(n+1)$
$a^n u(n)$	$a^n u(n)$	$\sum_{k=0}^n a^k a^{n-k} = (n+1)a^n u(n)$
$u(n)$	$a^n u(n)$	$\sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a} u(n)$
$(0.8)^n u(n)$	$(0.4)^n u(n)$	$(0.4)^n \sum_{k=0}^n 2^k = (0.4)^n (2^{n+1} - 1) u(n)$

The last one: $(0.8)^k (0.4)^{n-k} = (0.4)^n (0.8/0.4)^k = (0.4)^n 2^k$, then the geometric sum
 $\sum_0^n 2^k = (1 - 2^{n+1}) / (1 - 2)$. In general, for $a^n u(n) * b^n u(n)$ with $a \neq b$ the answer is
 $(a^{n+1} - b^{n+1}) / (a - b) u(n)$.

CHECK YOURSELF

Tutorial 2, Q5 and Tutorial 3, Q4, Q5 — convolutions by table

T2 Q5(a): $x = \{1, \uparrow 2, 1, -2\}$ ($n = -1 \dots 2$), $h = \{\uparrow 1, 2, 3, 1\}$ ($n = 0 \dots 3$). Before computing: y starts at -1 , has 7 samples, and must sum to $(2)(7) = 14$.

N	-1	0	1	2	3	4	5
x1	1	2	1	-2			
x2		2	4	2	-4		
x3			3	6	3	-6	
x1				1	2	1	-2
y(n)	1	4	8	7	1	-5	-2

$$y(n) = \{1, \uparrow 4, 8, 7, 1, -5, -2\} \quad \text{sum } 14 - \text{ok}$$

T2 Q5(b): $x = u(n)$, $h = a^n u(n)$: $y(n) = \sum_{k=0}^n a^k = (1 - a^{n+1}) / (1 - a) u(n)$, settling at $1 / (1 - a)$.

T3 Q4: $x = \{1, 1, 1, 1\}$, $h = \{1, 1, 1\}$, both from $n = 0$. Two rectangles give a trapezoid: $y = \{1, 2, 3, 3, 2, 1\}$, sum 12 = $4 \times 3 \checkmark$

T3 Q5: $x = \{1, 1\}$, $h = \{3, 2, 1\}$: rows $\{3, 2, 1\}$ and $\{-, 3, 2, 1\}$ add to $y = \{3, 5, 3, 1\}$, sum 12 = $2 \times 6 \checkmark$

CHECK YOURSELF

Tutorial 3, Q2 and Q3 — convolution sums by algebra

Q2(a) $u(n) * \delta(n-k) = u(n-k)$ — convolving with a shifted impulse just shifts.

Q2(b) $a^n u(n) * u(n) = (1 - a^{n+1}) / (1 - a) u(n)$ (table row 3).

Q2(c) $n u(n) * n u(n)$:

$$y(n) = \sum_{k=0}^n k(n-k) = n \cdot \left(\frac{n(n+1)}{2} \right) - \left(\frac{n(n+1)(2n+1)}{6} \right) = \left(\frac{n(n^2-1)}{6} \right) u(n)$$

Check $n = 2$: formula 1; directly $0 + 1 \cdot 1 + 0 = 1 \checkmark$

Q3(a)

$$(0.7)^n u(n) * (0.2)^n u(n) = \frac{(0.7)^{n+1} - (0.2)^{n+1}}{0.5} u(n) = 2[(0.7)^{n+1} - (0.2)^{n+1}] u(n)$$

Check $n = 0$: $2(0.5) = 1 = x(0)h(0) \checkmark$

2.13

Properties of convolution

Convolution behaves like multiplication — and $h(n)$ alone tells you whether the system is causal and stable.

Teacher module M02L005 §6-8

PROPERTY	STATEMENT	USE
Commutative	$x * h = h * x$	fold whichever is shorter
Associative	$(x * h_1) * h_2 = x * (h_1 * h_2)$	cascade = one system, $h = h_1 * h_2$
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	parallel = one system, $h = h_1 + h_2$
Shift	$x(n-n_0) * h(n) = x(n) * h(n-n_0) = y(n-n_0)$	shift either input, output shifts the same
Identity	$\delta * x = x, \delta * \delta = \delta$	the impulse is the "1"
Step	$x * u = \sum_{k \leq n} x(k)$	running sum; step response = running sum of h
Rectangles	$\text{rect}(n/2N) * \text{rect}(n/2N) = (2N+1) \text{tri}(n/(2N+1))$	two pulses give a triangle
Sum rule	$\sum y = (\sum x)(\sum h)$	the check

WORKED**Parallel systems (M02L005 example)**

$x = \{2, 3, 2, 1\}$ from $n = 0$; $h_1(n) = (2n+1)[u(n+1) - u(n-3)]$,
 $h_2(n) = 2(n+1)[u(n+2) - u(n-2)]$ in parallel.

$$h_1 = \{-1, 1, 3, 5\} \text{ at } n = -1 \dots 2 \quad h_2 = \{-2, 0, 2, 4\} \text{ at } n = -2 \dots 1$$

$$h = h_1 + h_2 = \{-2, -1, 3, 7, 5\} \text{ at } n = -2 \dots 2$$

distributive property

$$y = xh = \{-4, -8, -1, 19, 36, 32, 17, 5\} \text{ at } n = -2 \dots 5$$

$$\text{check : } 96 = 8 \times 12 \text{ ok}$$

CAUSALITY AND STABILITY FROM $H(N)$

$$\text{causal} \Leftrightarrow h(n) = 0 \text{ for } n < 0$$

$$\text{BIBO stable} \Leftrightarrow \sum_{k=-\infty}^{\infty} |h(k)|$$

$$< \infty$$

Why the second: if $|x(n)| < M$ then $|y(n)| \leq \sum |h(k)| |x(n-k)| < M \sum |h(k)|$, so the output is bounded exactly when h is absolutely summable. Necessary and sufficient.

For a causal system, $y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$; if the input is causal too,
 $y(n) = \sum_{k=0}^n x(k)h(n-k)$. FIR: $y(n) = \sum_{k=0}^{M-1} h(k)x(n-k)$ — a memory of M samples;

IIR: the sum runs to infinity.

TEACHER'S EXAMPLE	CAUSAL?	STABLE?
$y = x(n+1) - x(n), h = \{1, -1\}$ at $n = -1, 0$	no — $h(-1) \neq 0$	yes, $\sum h = 2$; FIR is always stable
$h = (-0.5)^n u(n)$, i.e. $y = x(n) - 0.5y(n-1)$	yes	yes, $\sum h = 1/(1-0.5) = 2$
$y - 0.5y(n-1) = n x(n)$	yes	no — step input gives a growing $n u(n)$ term (time-varying)

CHECK YOURSELF

Tutorial 1, Q1 — which impulse responses are causal (and stable)?

H(N)	CAUSAL?	STABLE?
$(\frac{1}{2})^n n u(n)$	yes — $u(n)$ kills $n < 0$	yes, $\sum n(\frac{1}{2})^n = 2$
$(\frac{1}{2})^n u(n)$	yes	yes, sum 2
$(\frac{1}{2})^{ n }$	no — $h(-1) = \frac{1}{2}$	yes, $1 + 2(1) = 3$
$u(n+2) - u(n-2)$	no — a pulse from -2 to 1	yes, finite length
$(\frac{1}{2})^n u(n) + 3^n u(-n-1)$	no — second term lives at $n < 0$	yes: for $n < 0$, 3^n shrinks, both sides sum finitely

2.14

Periodic inputs and circular convolution

A periodic input gives a periodic output. To handle repeating signals, let the index wrap around instead of running off the end.

Teacher module **M02L005b** §2-3

Teacher's example: $x = \{1, 2, -3, 1, 2, -3, \dots\}$ (period 3) into $h = \{1, 1\}$:

$$y = \{1, 3, -1, -2, 3, -1, -2, 3, -1, \dots\}$$

After the first sample the output is periodic with the same period, $y_p = \{-2, 3, -1\}$. The direct way to that steady-state period: convolve *one period* of the input with h , then **wrap** the overflow back to the front:

$$\{1, 1\} * \{1, 2, -3\} = \{1, 3, -1, -3\} \rightarrow \{1, 3, -1\} + \{-3, 0, 0\} = \{-2, 3, -1\} \quad \text{ok}$$

Wrapping N-sample sections of a signal and adding them builds its **periodic extension**

$x_{pe}(n) = \sum_k x(n+kN)$; a signal shorter than N is just zero-padded.

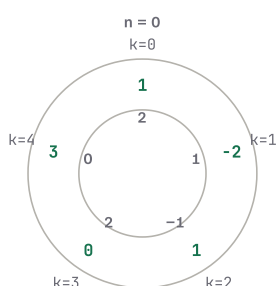
PERIODIC (CIRCULAR) CONVOLUTION

$$y_p(n) = x_p(n) \circledast h_p(n) = \sum_{k=0}^{N-1} x_p(k) h_p[(n-k) \bmod N], \quad n = 0, \dots, N-1$$

Ordinary linear convolution of two periodic signals does not exist (the sum never converges), so this is what we use. Exactly N output samples, and the result is periodic with period N.

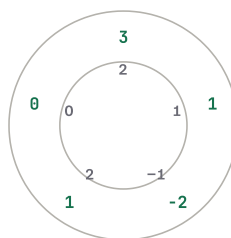
Three ways to compute it

METHOD	HOW
Wraparound	linear convolution (2N-1 samples) → append a zero → split into two halves of N → add the halves
Cyclic (circle)	$x(k)$ clockwise on an inner circle; $h(-k)$ anticlockwise on an outer circle; multiply the pairs and add → one output; rotate the outer circle one step → next output
Matrix	$y = Hx$ where each row of H is a cyclic shift of h



$$2 \cdot 1 + 1 \cdot (-2) + (-1) \cdot 1 + 2 \cdot 0 + 0 \cdot 3 = y(0) = -1$$

$n = 1$ (outer ring turned one step clockwise)



$$2 \cdot 3 + 1 \cdot 1 + (-1) \cdot (-2) + 2 \cdot 1 + 0 \cdot 0 = y(1) = 11$$

Inner ring: $x(k) = \{2, 1, -1, 2, 0\}$, clockwise
Outer ring: $h(-k) = h$ folded, anticlockwise

$h = \{1, 3, 0, 1, -2\}$

Each turn of the outer ring = one output
Five turns give all five samples:

$y_p = \{-1, 11, -2, 1, 3\}$

check: sum 12 = (4)(3) ✓

The sum rule still holds for circular convolution – wrapping moves samples, never loses them.

M02L005b's cyclic method, for $x = \{2, 1, -1, 2, 0\}$ and $h = \{1, 3, 0, 1, -2\}$.

Multiply the inner and outer values that sit together, add, rotate, repeat.

WORKED

Wraparound and matrix examples (M02L005b)

Wraparound: $x_p = \{1, 0, 1, 1\}, h_p = \{1, 2, 3, 1\}, N = 4$.

linear: $\{1, 2, 4, 4, 5, 4, 1\}$ 7 samples; append a zero $\rightarrow 8$

first half 1 2 4 4

second half 5 4 1 0

$y_p = \{6, 6, 5, 4\}$ check: $21 = 3 \times 7 \checkmark$

Matrix: $x = [1, 2, 1, 2]^T, h = [2, 1, 3, 1]^T$. Row n of H is $h[(n-k) \bmod N]$ for $k = 0 \dots 3$:

$$H = \begin{bmatrix} 2 & 1 & 3 & 1 \\ 1 & 2 & 1 & 3 \\ 3 & 1 & 2 & 1 \\ 1 & 3 & 1 & 2 \end{bmatrix} \quad y = Hx = [9, 12, 9, 12]^T \quad \text{check : } 42 = 6 \times 7 \text{ ok}$$

CHECK YOURSELF

Tutorial 3, Q6 and Q7 — circular convolution

Q6: $x_1 = \{1, 2, 3, 1\}, x_2 = \{4, 3, 2, 2\}, N = 4$.

linear: $\{4, 11, 20, 19, 13, 8, 2\}$ sum $77 = 7 \times 11 \checkmark$

first four: 4 11 20 19

wrapped: 13 8 2 0

$y = \{17, 19, 22, 19\}$ sum $77 \checkmark$

Q7(a): $x_1 = \{1, 1, 1, 1, 0, 0, 0, 0\}, x_2(n) = \sin(3\pi n/8)$ for $0 \leq n \leq 7, N = 8$. With a rectangle of four 1s, circular convolution is a *circular running sum*:

$y(n) = x_2(n) + x_2(n-1) + x_2(n-2) + x_2(n-3)$ with the indices taken mod 8. Using $x_2 = \{0, 0.924, 0.707, -0.383, -1, -0.383, 0.707, 0.924\}$:

$$y = \{1.248, 2.555, 2.555, 1.248, 0.248, -1.059, -1.059, 0.248\} \quad \text{check : } \sum 5.98 = 4 \times 1.496 \text{ ok}$$

In Unit 5 you will do this in one line: $y = \text{IDFT}[X_1(k) X_2(k)]$.

Linear from circular; link to transforms

Pad with enough zeros and circular convolution gives the linear answer. And every transform turns its kind of convolution into a plain multiplication.

Teacher module M02L005b §4-5

NO WRAPAROUND DAMAGE IF

$$L \geq N + M - 1$$

$N, M = \text{lengths of } x \text{ and } h$; zero-pad both to L , then L -point circular convolution = linear convolution

The zeros stop the wrapped tail from landing on real samples. This is how the FFT is used for filtering.

Teacher's example: $x = \{1, 2, 1, 2\}$, $h = \{2, 1, 3\}$, $L = 4 + 3 - 1 = 6$. Pad to $x_L = \{1, 2, 1, 2, 0, 0\}$, $h_L = \{2, 1, 3, 0, 0, 0\}$; the 6-point circular convolution is

$$y_L = \{2, 5, 7, 11, 5, 6\} = xh$$

check : $36 = 6 \times 6$ ok (the slide's printed row does not pass this check)

Convolution becomes multiplication

TRANSFORM	TIME DOMAIN	TRANSFORM DOMAIN
DTFS / DFT	$x_p \circledast h_p$ (periodic, discrete)	$Y_p(k) = X_p(k) H_p(k)$
DTFT	$x * h$ (regular, discrete)	$Y(F) = X(F) H(F)$
z-transform	$x * h$ (regular, discrete)	$Y(z) = X(z) H(z)$

Why (z-transform version, M02L005b): put the convolution sum inside $Y(z) = \sum_n y(n) z^{-n}$, substitute $r = n - k$, and the double sum splits into $[\sum_k x(k) z^{-k}] [\sum_r h(r) z^{-r}] = X(z) H(z)$. The DTFS proof is identical with $e^{-j2\pi kn/N}$ in place of z^{-n} . **The kind of convolution matches the transform: periodic transforms (DTFS/DFT) go with periodic convolution.**

2.16

Deconvolution

Know the input and the output, want the system? Convolution is polynomial multiplication, so deconvolution is polynomial division.

Teacher module M02L005b §6

Forward problem: $y = x * h$. Inverse problem (**system identification**): given x and y , find h .
 Since $Y(z) = X(z)H(z)$, $H(z) = Y(z)/X(z)$ — a long division of the coefficient lists.

WORKED Recovering $h = \{4, 1, 3\}$ (M02L005b)

$x = \{2, 5, 0, 4\}$, $y = \{8, 22, 11, 31, 4, 12\}$. Treat them as polynomials
 $x(w) = 2w^3 + 5w^2 + 0w + 4$ and $y(w) = 8w^5 + 22w^4 + 11w^3 + 31w^2 + 4w + 12$:

$$\frac{8w^5}{2w^3} = 4w^2: \text{ subtract } 8w^5 + 20w^4 + 0w^3 + 16w^2 \rightarrow 2w^4 + 11w^3 + 15w^2 + 4w + 12$$

$$\frac{2w^4}{2w^3} = w: \text{ subtract } 2w^4 + 5w^3 + 0w^2 + 4w \rightarrow 6w^3 + 15w^2 + 0w + 12$$

$$\frac{6w^3}{2w^3} = 3: \text{ subtract } 6w^3 + 15w^2 + 0w + 12 \rightarrow 0$$

$$h(w) = 4w^2 + w + 3 \Rightarrow h = \{4, 1, 3\}$$

A zero remainder confirms y really was $x * h$. (Compare §2.12, where the same three sequences went the other way.)

2.17

Correlation and matched filtering

Convolution asks "what does the system output?"; correlation asks "how alike are these two signals, and at what shift?"

Teacher module M02L005b §7-8

CORRELATION IS CONVOLUTION WITH A FOLDED SIGNAL

$$r_{xh}(n) = x(n) \star h(n) = x(n) * h(-n) = \sum_k x(k) h(k-n)$$

$$r_{hx}(n) = h(n) * x(-n) \quad r_{xh}(n) = r_{hx}(-n) \quad N_r = N_x + N_h - 1$$

So: **flip one signal, then convolve as usual**. At each shift, multiply the overlapping samples and add. Unlike convolution it is *not* commutative — $r_{xh} \neq r_{hx}$; one is the other reversed.

Autocorrelation

$$r_{xx}(n) = x(n) * x(-n) = \sum_k x(k)x(k-n)$$

$$r_{xx}(n) = r_{xx}(-n) \quad \text{even}$$

$$r_{xx}(0) = \sum_k x^2(k) = E_x \quad \text{the energy}$$

$$|r_{xx}(n)| \leq r_{xx}(0) \quad \text{maximum at the origin}$$

WORKED A small correlation and autocorrelation

$x = \{1, 2, 3\}$, $h = \{1, 1\}$, both from $n = 0$.

$$h(-n) = \{1, 1\} \quad \text{at } n = -1, 0 \Rightarrow r_{xh} = \{1, 2, 3\} * \{1, 1\} \text{ starting at } -1 = \{1, 3, 5, 3\} \quad \text{at } n = -1 \dots 2$$

$$r_{xx} = \{1, 2, 3\} * \{3, 2, 1\} \text{ starting at } -2 = \{3, 8, 14, 8, 3\} \quad \text{at } n = -2 \dots 2$$

r_{xx} is even, and its peak $14 = 1 + 4 + 9$ is the energy $\checkmark$. Both have length $N_x + N_h - 1$.

WHY IT MATTERS: DETECTION IN NOISE, MATCHED FILTERING, RANGING

Noise is essentially uncorrelated with a signal. So correlating a noisy recording with the clean template produces a **sharp peak** where the template is present — the signal shows up even when it is buried. A filter whose impulse response is the folded template, $h(n) = s(-n)$, does exactly this correlation by convolution: the **matched filter**. In radar, correlate the echo with the transmitted pulse; the lag of the peak is the round-trip delay, hence the target's range.

UNIT 3 · TUTORIALS 4 AND 5 (THE Z-TRANSFORM APPEARS INSIDE M02L002, M02L005B AND M04L001 TOO)

The z-transform

A change of variable that turns convolution into multiplication and a difference equation into algebra. No slide module was issued for it — this unit follows your two tutorials, and uses every place the lecture modules lean on it.

3.1

The z-transform and its ROC

Each sample $x(k)$ becomes a term $x(k)z^{-k}$. The formula only means something together with the set of z where the sum converges.

Tutorial 04 Q1-Q4 · Tutorial 05

DEFINITION

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}, z \text{ complex}$$

The set of z for which the sum is finite is the **region of convergence (ROC)**. A z-transform without its ROC is incomplete.

For a finite-length signal it is just a list: $x = \{1, \uparrow 2, 5, 7, 0, 1\}$ gives $X(z) = z + 2 + 5z^{-1} + 7z^{-2} + z^{-4}$. Positive $n \rightarrow$ negative powers of z ; negative $n \rightarrow$ positive powers.

CHECK YOURSELF

Tutorial 4, Q1–Q4 — direct z-transforms

Q1 — finite signals. Only the arrow moves between (a)–(d), and the answer changes completely.

SIGNAL	$X(z)$	ROC
$\{\uparrow 1, 2, 5, 7, 0, 1\}$	$1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$	all z except 0
$\{1, 2, \uparrow 5, 7, 0, 1\}$	$z^2 + 2z + 5 + 7z^{-1} + z^{-3}$	all z except 0 and ∞
$\{\uparrow 0, 0, 1, 2, 5, 7, 0, 1\}$	$z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-7}$	all z except 0
$\{2, 4, \uparrow 5, 7, 0, 1\}$	$2z^2 + 4z + 5 + 7z^{-1} + z^{-3}$	all z except 0 and ∞
$\delta(n)$	1	the entire plane
$\delta(n-k), k>0$	z^{-k}	all except 0
$\delta(n+k), k>0$	z^k	all except ∞

Rule for finite signals: whole plane, minus $z = 0$ if any sample sits at $n > 0$, minus $z = \infty$ if any sits at $n < 0$.

Q2 $x = a^n u(n)$: a geometric series in az^{-1} , converging when $|az^{-1}| < 1$:

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \left(\frac{1}{1 - az^{-1}} \right), \text{ROC } |z| > |a|$$

Q3 $x = (-a)^n u(-n-1)$, non-zero only for $n \leq -1$. Put $m = -n$:

$$X(z) = \sum_{m=1}^{\infty} \left(\left(\frac{z}{-a} \right) \right)^m = - \left(\frac{1}{1 + az^{-1}} \right), \text{ROC } |z| < |a|$$

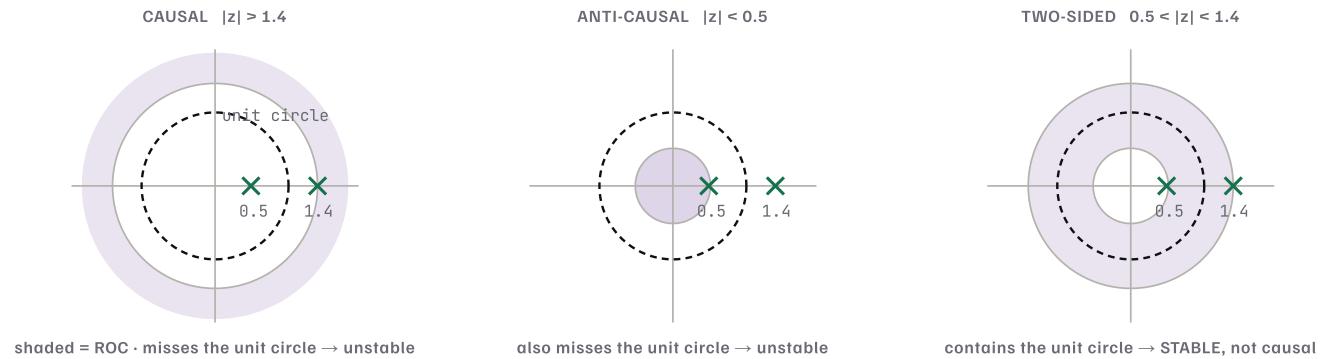
Q4 $x = a^n u(n) + b^n u(-n-1)$: transform each piece and intersect the ROCs:

$$X(z) = \left(\frac{1}{1 - az^{-1}} \right) - \left(\frac{1}{1 - bz^{-1}} \right), \text{ROC } |a| < |z| < |b|$$

If $|b| \leq |a|$ the regions do not overlap and **the transform does not exist** — say so explicitly.

Rules for the ROC

1. A ring or disc centred at the origin, and it **never contains a pole**.
2. Finite-length signal $\rightarrow$ the whole plane (minus 0 and/or ∞).
3. Right-sided (causal) $\rightarrow$ *outside* the outermost pole, $|z| > r_{\max}$.
4. Left-sided (anti-causal) $\rightarrow$ *inside* the innermost pole, $|z| < r_{\min}$.
5. Two-sided $\rightarrow$ a ring between two poles.
6. The DTFT exists (Unit 4) exactly when the **ROC contains the unit circle**.



One formula, three systems. Poles at 0.5 and 1.4 cut the plane into three possible ROCs, and each gives a different $h(n)$ (§2.9 showed this in the time domain: one equation, two impulse responses). Only the ring containing the dashed unit circle is stable; only the outside region is causal. **With a pole outside the unit circle you can be causal or stable, not both.**

THE THREE CONDITIONS

YOU WANT	ROC MUST BE
Causal	$ z > r$, the outside region
Stable	a region containing $ z = 1$
Causal and stable	$ z > r$ with $r < 1$: every pole strictly inside the unit circle — the same rule as "every characteristic root has magnitude < 1 " (§2.9)

3.2

Standard pairs and properties

A short table and a short list of rules — between them they do almost every problem.

Tutorials 04, 05 · convolution property: M02L005b §5

$X(N)$	$X(Z)$	ROC
$\delta(n)$	1	all z
$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$a^n u(n)$	$\frac{1}{1 - az^{-1}} = \frac{z}{z - a}$	$ z > a $
$-a^n u(-n-1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$n a^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2} = \frac{az}{(z - a)^2}$	$ z > a $
$(n+1)a^n u(n)$	$\frac{1}{(1 - az^{-1})^2}$	$ z > a $
$\cos(\omega_0 n) u(n)$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
$\sin(\omega_0 n) u(n)$	$\frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
$a^n \cos(\omega_0 n) u(n)$	$\frac{1 - az^{-1} \cos \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

PROPERTY	TIME DOMAIN	Z-DOMAIN
Linearity	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(z) + a_2 X_2(z)$
Time shift	$x(n - k)$	$z^{-k} X(z)$ — the delay block z^{-1} of §2.1
Scaling in z	$a^n x(n)$	$X(z/a)$
Time reversal	$x(-n)$	$X(z^{-1})$
Differentiation	$n x(n)$	$-z dX(z)/dz$
Convolution	$x_1(n) * x_2(n)$	$X_1(z) X_2(z)$
Correlation	$r_{xy}(l)$	$X(z) Y(z^{-1})$
Initial value	$x(0)$	$\lim_{z \rightarrow \infty} X(z)$ (causal)
Final value	$\lim_{n \rightarrow \infty} x(n)$	$\lim_{z \rightarrow 1} (z-1)X(z)$

THE PROPERTY EVERYTHING RESTS ON

$$y(n) = x(n) * h(n) \quad \rightarrow \quad Y(z) = X(z) H(z)$$

(Proved in §2.15.) $H(z) = Y(z)/X(z)$ is the **system function**: the z-transform of the impulse response. For a difference equation it is read off directly — transform both sides using the shift rule.

3.3

The inverse z-transform

Going back from $X(z)$ to $x(n)$. Three methods — and before any of them, one question: where is the ROC?

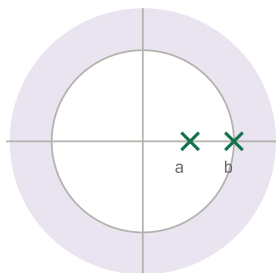
Tutorial 04 Q5-Q7 · the whole sheet is about this

THE ROC PICKS THE ANSWER — APPLY IT TO EACH POLE SEPARATELY

- ROC **outside** that pole's circle $\rightarrow$ the term is **causal**: $a^n u(n)$.
- ROC **inside** that pole's circle $\rightarrow$ the term is **anti-causal**: $-a^n u(-n-1)$.

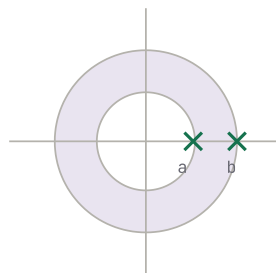
Note the minus sign on the anti-causal one — forgetting it is the most common error in the topic.

ROC OUTSIDE BOTH



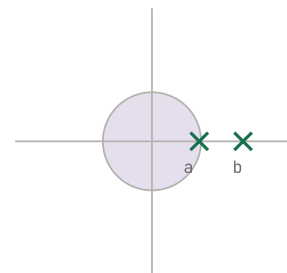
both terms CAUSAL

ROC IN BETWEEN



a causal, b ANTI-causal $\rightarrow$ two-sided

ROC INSIDE BOTH



both terms ANTI-causal

Two poles, three possible answers. Same algebra every time — only the causal/anti-causal choice changes. So the first thing to do in any inverse-z question is mark the poles and shade the ROC.

METHOD	USE IT WHEN	GIVES
1. Partial fractions	$X(z)$ is a ratio of polynomials — almost always	a closed formula
2. Power series (long division)	you want the first few samples, or $X(z)$ is not rational	numbers, one at a time
3. Residues (inversion integral)	the question asks for it (Tutorial 4 Q5 does)	a closed formula, more work

Method 1 — partial fractions

RECIPE

- 0** If the numerator degree is not smaller than the denominator's, divide once first; each constant $c \cdot z^{-k}$ peeled off is an impulse $c\delta(n-k)$.
- 1** Factor the denominator $\rightarrow$ poles. Note their radii against the ROC.
- 2** Split into one simple fraction per pole (cover-up rule for the constants).
- 3** Invert each piece from the table, choosing the causal or anti-causal row *for that pole*.
- 4** Check $x(0)$ against $\lim_{z \rightarrow \infty} X(z)$.

THE $X(z)/z$ TRICK

The table pair is $a^n u(n) \leftrightarrow z/(z-a)$ — there is a z on top. Plain partial fractions of $X(z)$ in positive powers would give $1/(z-a)$, which is not in the table. So expand $X(z)/z$, then multiply every term by z at the end. If $X(z)$ is already written in z^{-1} , treat z^{-1} as the variable and expand directly.

WORKED

Case A — distinct real poles, and what the ROC changes

$X(z) = \frac{z^2}{(z - 0.5)(z - 0.25)}$, ROC $|z| > 0.5$ — outside both poles, so both terms causal.

$$\begin{aligned} \frac{X(z)}{z} &= \frac{z}{(z - 0.5)(z - 0.25)} = \frac{A}{z - 0.5} + \frac{B}{z - 0.25} \\ A &= \frac{0.5}{0.5 - 0.25} = 2, \quad B = \frac{0.25}{0.25 - 0.5} = -1 \\ X(z) &= \frac{2z}{z - 0.5} - \frac{z}{z - 0.25} \Rightarrow x(n) = [2(0.5)^n - (0.25)^n]u(n) \end{aligned}$$

Check: $x(0) = 1 = \lim_{z \rightarrow \infty} X(z) \checkmark$

Now change only the ROC. With $0.25 < |z| < 0.5$ the ROC is inside the pole at 0.5 but outside the one at 0.25: $x(n) = -2(0.5)^n u(-n-1) - (0.25)^n u(n)$. With $|z| < 0.25$ both flip: $x(n) = [-2(0.5)^n + (0.25)^n]u(-n-1)$.

WORKED

Case B — a repeated pole

$X(z) = \frac{z^2}{(z - 0.5)^2}$, ROC $|z| > 0.5$. A double pole needs two terms:

$$\begin{aligned} \left(\frac{X(z)}{z} \right) &= \left(\frac{z}{(z - 0.5)^2} \right) = \left(\frac{A}{z - 0.5} \right) + \left(\frac{B}{(z - 0.5)^2} \right) \\ \Rightarrow z &= A(z - 0.5) + B \Rightarrow B = 0.5 (\text{put } z = 0.5), A = 1 (\text{match the } z \text{ terms}) \\ X(z) &= \left(\frac{z}{z - 0.5} \right) + \left(\frac{0.5z}{(z - 0.5)^2} \right) \\ \Rightarrow x(n) &= (0.5)^n u(n) + n(0.5)^n u(n) = (n + 1)(0.5)^n u(n) \end{aligned}$$

Check with the table pair $(n+1)a^n u(n) \leftrightarrow 1/(1 - az^{-1})^2 \checkmark$. A pole repeated m times brings factors $1, n, \dots, n^{m-1}$ — the same n -multiplying as repeated characteristic roots in §2.7.

WORKED

Case C — complex conjugate poles

Complex poles come in pairs and always give a decaying (or growing) cosine; no complex numbers survive in the answer. $X(z) = 1/(1 - z^{-1} + 0.5z^{-2})$, causal.

$$z^2 - z + 0.5 = 0 \Rightarrow z = 0.5 \pm j0.5 = 0.707 e^{\pm j\pi/4} \quad \text{radius } 0.707, \text{ angle } \pi/4$$

shape: $x(n) = A(0.707)^n \cos(\pi n/4 + \varphi) u(n)$ radius sets the decay, angle sets the frequency

from the equation $x(n) = x(n-1) - 0.5x(n-2) + \delta(n)$: $x(0) = 1, x(1) = 1$

$$A \cos \varphi = 1, \quad A(0.707) \cos(\pi/4 + \varphi) = 1 \Rightarrow 0.5(A \cos \varphi - A \sin \varphi) = 1 \Rightarrow A \sin \varphi = -1$$

$$A = \sqrt{2}, \quad \varphi = -\pi/4 \Rightarrow x(n) = \sqrt{2} (0.707)^n \cos(\pi n/4 - \pi/4) u(n)$$

Check: $x(0) = \sqrt{2} \cos(-\pi/4) = 1 \checkmark, x(1) = \sqrt{2}(0.707) \cos(0) = 1 \checkmark$

CHECK YOURSELF

Tutorial 4, Q6 and Q7 — inverse z-transform with different ROCs

Q6: $X(z) = 1/(1 - 1.5z^{-1} + 0.5z^{-2}) = 1/[(1 - z^{-1})(1 - 0.5z^{-1})]$, poles at 1 and 0.5.

$$X(z) = \left(\frac{A}{1 - z^{-1}} \right) + \left(\frac{B}{1 - 0.5z^{-1}} \right)$$

put $z^{-1} = 1 \Rightarrow A(0.5) = 1 \Rightarrow A = 2$
 put $z^{-1} = 2 \Rightarrow B(-1) = 1 \Rightarrow B = -1$

(a) ROC $|z| > 1$ — both causal: $x(n) = [2 - (0.5)^n]u(n)$. Check $x(0) = 1 = \lim_{z \rightarrow \infty} X(z) \checkmark$

(b) ROC $|z| < 0.5$ — both anti-causal: $x(n) = [-2 + (0.5)^n]u(-n-1)$.

(The third region, $0.5 < |z| < 1$, gives $-2u(-n-1) - (0.5)^n u(n)$ — the only stable one of the three.)

Q7: causal $x(n)$ for $X(z) = (1 + z^{-1})/(1 - z^{-1} + 0.5z^{-2})$. Same poles as Case C: $0.707e^{\pm j\pi/4}$, so the shape is $A(0.707)^n \cos(\pi n/4 + \phi)u(n)$. From $x(n) = x(n-1) - 0.5x(n-2) + \delta(n) + \delta(n-1)$: $x(0) = 1, x(1) = 2$.

$$A \cos \phi = 1, 0.5(A \cos \phi - A \sin \phi) = 2 \Rightarrow A \sin \phi = -3$$

$$A = 10 = 3.162, \phi = \tan^{-1}(-3) = -71.57^\circ$$

$$x(n) = 3.162(0.707)^n \cos\left(\frac{\pi n}{4} - 71.57^\circ\right) u(n)$$

Check $x(0) = 3.162 \times 0.3162 = 1 \checkmark$

Method 2 — power series (long division)

The z-transform is a power series, $X(z) = \dots + x(-1)z + x(0) + x(1)z^{-1} + \dots$, so once it is expanded the coefficients are the signal. The ROC decides the direction of division: **causal** → arrange in ascending powers of z^{-1} and get $x(0), x(1), \dots$; **anti-causal** → ascending powers of z and get $x(-1), x(-2), \dots$

WORKED

The same $X(z)$ divided both ways

$$X(z) = 1/(1 - 0.5z^{-1}).$$

(a) ROC $|z| > 0.5$: $1 \div (1 - 0.5z^{-1}) = 1 + 0.5z^{-1} + 0.25z^{-2} + \dots \rightarrow$

$$x = \{1, 0.5, 0.25, \dots\} = (0.5)^n u(n).$$

(b) ROC $|z| < 0.5$: rewrite

$$X(z) = z/(z - 0.5) = -2z/(1 - 2z) = -2z(1 + 2z + 4z^2 + \dots) = -2z - 4z^2 - 8z^3 - \dots$$

A term cz^k means $x(-k) = c$: $x(-1) = -2$, $x(-2) = -4$, $\dots \rightarrow$

$$x(n) = -(0.5)^n u(-n-1). \text{ Exactly the table's anti-causal row, minus sign included.}$$

LONG DIVISION GIVES NUMBERS, NOT A FORMULA

Use it for "find the first four samples", to check another method, or when $X(z)$ is not a ratio of polynomials (e.g. $\log(1 + az^{-1})$). For "find $x(n)$ ", use partial fractions.

Method 3 — the inversion integral (residues)

FORMULA

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz = \sum_{\text{poles inside } C} \text{Res}[X(z) z^{n-1}]$$

C is an anticlockwise loop inside the ROC, once round the origin. Simple pole at p :

$\text{Res} = [(z-p)X(z)z^{n-1}]_{z=p}$. Pole of order m : differentiate $(z-p)^m X(z) z^{n-1}$ $m-1$ times, divide by $(m-1)!$, substitute. For $n < 0$ the factor z^{n-1} adds a pole at the origin — treat $n \geq 0$ and $n < 0$ separately.

CHECK YOURSELF

Tutorial 4, Q5 — inverse by the inversion integral

$$X(z) = 1/(1 - az^{-1}) = z/(z - a), \text{ROC } |z| > |a|.$$

$$x(n) = \frac{1}{2\pi j} \oint_C \frac{z^n}{z - a} dz, \quad C \text{ inside the ROC, so it}$$

encloses $z = a$

$$n \geq 0: \quad \text{Res}_{z=a} = \left[(z - a) \cdot \frac{z^n}{z - a} \right]_{z=a} = a^n$$

$n < 0$: z^n adds a pole at the origin whose residue cancels the one at $a \Rightarrow 0$

$$x(n) = a^n u(n)$$

3.4

System function, poles, zeros, stability

Everything about an LTI system — its response, its stability, its transient — can be read off the positions of its poles and zeros.

Tutorial 05 Q1-Q4, Q6, Q7 · M02L002 §9 (causality via $H(z)$) · M02L004 §8

Transform a difference equation with the shift rule and solve for $Y(z)/X(z)$:

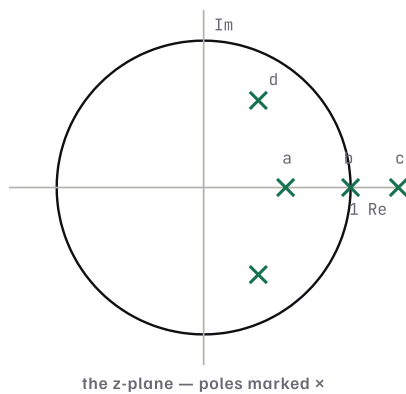
$$H(z) = \left(\frac{B(z)}{A(z)} \right) = \left(\frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}} \right)$$

Zeros ○

roots of the numerator — where $H(z) = 0$

Poles ×

roots of the denominator — where $H(z)$ blows up. These are exactly the characteristic roots of §2.7, seen again.



a inside, real ($0 < r < 1$) → decays



b ON the circle ($r = 1$) → steady



c outside ($r > 1$) → blows up



d complex pair, inside → decaying wobble



Distance from the origin sets the DECAY.

Angle from the real axis sets the FREQUENCY.

The pole map. A real pole at radius r gives r^n ; a complex pair at $re^{\pm j\omega_0}$ gives $r^n \cos(\omega_0 n + \psi)$. A repeated pole multiplies by n , so even a repeated pole *on* the circle is unstable — the same table as §2.7, drawn.

READING $H(z)$

- **Causal** $\Leftrightarrow$ ROC is the outside region; for a rational $H(z)$ in positive powers, the numerator degree must not exceed the denominator degree (M02L002: $M > N$ means non-causal).
- **Stable** $\Leftrightarrow$ ROC contains the unit circle.
- **Both** $\Leftrightarrow$ every pole strictly inside the unit circle. Zeros can be anywhere and never affect stability.
- **Second-order shortcut:** for $1 + a_1 z^{-1} + a_2 z^{-2}$, both poles are inside iff $|a_2| < 1$ and $|a_1| < 1 + a_2$ (the stability triangle).
- **Response:** $Y(z) = H(z)X(z)$. After partial fractions, terms from the poles of H are the *natural* (transient) response; terms from the poles of X are the *forced* (steady-state) response.

CHECK YOURSELF

Tutorial 5, Q1 and Q6 — ROC, causal, stable, anti-causal

Q1: $H(z) = (3 - 4z^{-1})/(1 - 3.5z^{-1} + 1.5z^{-2})$.

$$z^2 - 3.5z + 1.5 = 0 \Rightarrow z = 3, 0.5$$

$$H(z) = \left(\frac{3 - 4z^{-1}}{(1 - 3z^{-1})(1 - 0.5z^{-1})} \right) = \left(\frac{2}{1 - 3z^{-1}} \right) + \left(\frac{1}{1 - 0.5z^{-1}} \right)$$

CASE	ROC	H(N)
(ii) causal	$ z > 3$	$[2(3)^n + (0.5)^n]u(n)$ — unstable
(i) stable	$0.5 < z < 3$	$-2(3)^n u(-n-1) + (0.5)^n u(n)$ — two-sided
(iii) anti-causal	$ z < 0.5$	$[-2(3)^n - (0.5)^n]u(-n-1)$ — unstable

One pole outside the unit circle $\Rightarrow$ causal or stable, never both. Check $h(0) = 3 = b_0$ for the causal case $\checkmark$

Q6: $H(z) = (2 - \frac{1}{2}z^{-1})/(1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2})$. Denominator $(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})$, poles $\frac{1}{2}$ and $\frac{1}{4}$, both inside.

$$H(z) = \left(\frac{1}{1 - (\frac{1}{2})z^{-1}} \right) + \left(\frac{1}{1 - (\frac{1}{4})z^{-1}} \right)$$

(i) stable and causal, $|z| > \frac{1}{2}$: $h(n) = [(\frac{1}{2})^n + (\frac{1}{4})^n]u(n)$; (ii) anti-causal, $|z| < \frac{1}{4}$: $h(n) = -[(\frac{1}{2})^n + (\frac{1}{4})^n]u(-n-1)$. Check $h(0) = 2 = b_0 \checkmark$

CHECK YOURSELF

Tutorial 5, Q2, Q3, Q4, Q7 — system functions

Q2: $y(n) = y(n-1) + y(n-2) + x(n-1)$.

$$H(z) = \left(\frac{z^{-1}}{1 - z^{-1} - z^{-2}} \right), \text{poles } z = \frac{1 \pm \sqrt{5}}{2} = 1.618, -0.618$$

(b) causal, $|z| > 1.618$: $h(n) = (1/\sqrt{5})[(1.618)^n - (-0.618)^n]u(n)$ — the Fibonacci numbers; unstable. (c) stable, $0.618 < |z| < 1.618$:
 $h(n) = -(1/\sqrt{5})(1.618)^n u(-n-1) - (1/\sqrt{5})(-0.618)^n u(n)$ — two-sided, not causal.

Q3: $H(z) = (z^3 - 2z^2 + z)/(z^2 + \frac{1}{3}z + \frac{1}{3})$. Numerator degree 3 > denominator degree 2 $\Rightarrow$ dividing out gives a term in z , i.e. $x(n+1)$: **non-causal**.

Q4: the block diagram with gains $-k/3$ (feedback) and $-k/4$ (feed-forward):

$w(n) = x(n) - (k/3)w(n-1)$, $y(n) = w(n) - (k/4)w(n-1)$.

$$H(z) = \left(\frac{1 - (\frac{k}{4})z^{-1}}{1 + (\frac{k}{3})z^{-1}} \right): \text{pole at } -\frac{k}{3}, \text{zero at } \frac{k}{4}; (b) \text{ stable iff } \left| \frac{k}{3} \right| < 1 \Rightarrow |k| < 3$$

(c) $k = 1$, $x(n) = (\frac{1}{3})^n$ for all n (two-sided, so steady state): the output is the input times H at $z = \frac{1}{3}$: $H(\frac{1}{3}) = (1 - 0.375)/(1 + 0.5) = 0.4167 \Rightarrow y(n) = 0.4167(\frac{1}{3})^n$.

Q7: $2y(n) = ay(n-2) - 2x(n) + \beta x(n-1)$. Characteristic $\lambda^2 - a/2 = 0$,
 $\lambda = \pm\sqrt{a/2}$; stability needs $|a/2| < 1$: $|a| < 2$, β any finite value — it only moves a zero.

3.5

One-sided z-transform and initial conditions

A version that remembers $y(-1)$, so it solves a difference equation with initial conditions in one pass.

Tutorial 03 Q1(c) redone · complements M02L003 §7

$$X^+(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

THE SHIFT RULE THAT CARRIES THE INITIAL CONDITIONS

$$Z^+\{y(n-1)\} = z^{-1}Y^+(z) + y(-1)$$

$$Z^+\{y(n-2)\} = z^{-2}Y^+(z) + z^{-1}y(-1) + y(-2)$$

Transform the whole equation, solve for $Y^+(z)$, invert: the complete response (zero-input + zero-state) falls out with no guessing.

WORKED
Tutorial 3 Q1(c) the fast way

$$y(n) - 0.6y(n-1) = x(n), x = (0.4)^n u(n), y(-1) = 10.$$

$$\begin{aligned} Y(z) - 0.6[z^{-1}Y(z) + 10] &= X(z) \Rightarrow Y(z)(1 - 0.6z^{-1}) = \left(\frac{1}{1 - 0.4z^{-1}}\right) + 6 \\ Y(z) &= \left(\frac{1}{(1 - 0.6z^{-1})(1 - 0.4z^{-1})}\right) + \left(\frac{6}{1 - 0.6z^{-1}}\right) \\ &= \left(\frac{3}{1 - 0.6z^{-1}}\right) - \left(\frac{2}{1 - 0.4z^{-1}}\right) + \left(\frac{6}{1 - 0.6z^{-1}}\right) \\ y(n) &= [9(0.6)^n - 2(0.4)^n]u(n) \end{aligned}$$

Same as §2.8, and the two parts are visible: $y_{zi} = 6(0.6)^n$ from the initial condition, $y_{zs} = 3(0.6)^n - 2(0.4)^n$ from the input.

3.6

Transient and steady-state response

Switch a sinusoid on and the output has a part that fades (the system's poles) and a part that stays (the input's poles).

Tutorial 05 Q8, Q9 · M04L001 §7 (steady state via H at $z = e^{j2\pi F}$)

WHERE EACH PART COMES FROM

- **Transient:** from the poles of $H(z)$. Each pole with $|p| < 1$ gives a term p^n that dies out; the closer to the circle, the slower.
- **Steady state:** from the poles of $X(z)$, which sit on the unit circle for a step ($z = 1$) or a sinusoid ($e^{\pm j\omega_0}$). For a sinusoid it is the input scaled by $|H(\omega_0)|$ and shifted by $\angle H(\omega_0)$ — Unit 4's recipe.

CHECK YOURSELF

Tutorial 5, Q8 and Q9

Q8: step response of $y(n) = y(n-1) + x(n)$ (the accumulator), relaxed:

$y = 1, 2, 3, \dots \Rightarrow y(n) = (n+1)u(n)$. The pole is at $z = 1$, on the circle; $h = u(n)$,
 $\sum |h| = \infty$: **unstable** — a bounded input gives an unbounded output.

Q9: $y(n) = 0.5y(n-1) + x(n)$, $x = 10\cos(\pi n/4)u(n)$, relaxed.

$$H(\omega) = \left(\frac{1}{1 - 0.5e^{-j\omega}} \right)$$

$$\text{at } \omega = \frac{\pi}{4}: 1 - 0.5e^{-j\frac{\pi}{4}} = 0.6464 + j0.3536 = 0.7367 \angle 28.68^\circ$$

$$|H| = 1.357, \angle H = -28.68^\circ \Rightarrow y_{ss}(n) = 13.57 \cos\left(\frac{\pi n}{4} - 28.68^\circ\right)$$

transient from the pole at 0.5: $C(0.5)^n$

$$y(0) = x(0) = 10 = C + 13.57 \cos(-28.68^\circ) = C + 11.91 \Rightarrow C = -1.91$$

$$y(n) = \left[-1.91(0.5)^n + 13.57 \cos\left(\frac{\pi n}{4} - 28.68^\circ\right) \right] u(n)$$

The first term fades — transient. The second lasts — steady state, at the input's frequency.

UNIT 4 · TEACHER MODULE M04L001, FREQUENCY ANALYSIS OF SIGNALS AND SYSTEMS

Frequency analysis

Which sinusoids are inside a signal, and how strongly. Then the same question for a system: what does it do to each frequency? The DTFS, the DTFT and the frequency response answer them.

4.1

The four Fourier transforms

There are four, not one. They differ in only two ways: is time continuous or sampled, and does the signal repeat?

Teacher module [M04L001](#) §1 · general summary in [M04L002](#)

		IS THE SIGNAL PERIODIC?		
		YES — it repeats	NO — it happens once	
TIME	CONTINUOUS	FOURIER SERIES (FS) continuous + repeating → spectrum is DISCRETE, goes on forever c_k — separate lines	FOURIER TRANSFORM (FT) continuous + one-off → spectrum is CONTINUOUS, goes on forever $X(F)$ — a smooth curve	ONE RULE MAKES THE TABLE If a signal REPEATS in one domain, its spectrum is made of LINES in the other. If a signal is SAMPLED in one domain, its spectrum REPEATS in the other. Only the bottom-left box is discrete and finite in BOTH domains — that is the one a computer can do (Unit 5).
	DISCRETE	DTFS (→ the DFT) discrete + repeating → spectrum is DISCRETE and REPEATS N numbers. THE ONLY COMPUTABLE ONE.	DTFT discrete + one-off → spectrum is CONTINUOUS and REPEATS $X(\omega)$ — period 2π	

Everything follows from one sentence. Sampling in time makes the spectrum repeat — that is aliasing (§1.7). Sampling the *spectrum* makes the time signal repeat — that is where the DFT comes from (§5.2).

Why bother (M04L001): a frequency-domain view is another way to see a signal or a system; it turns convolution into multiplication; and it reveals a system's frequency response.

4.2

DTFS — Fourier series for DT periodic signals

A repeating sampled signal is a sum of N harmonics. N numbers, $c_0 \dots c_{N-1}$, say how much of each.

Teacher module **M04L001** §2

THE DTFS PAIR

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j\frac{2\pi kn}{N}} \quad \text{synthesis}$$

$$c_k = \left(\frac{1}{N}\right) \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} \quad \text{analysis}$$

Each c_k belongs to the harmonic at digital frequency $F = k/N$, i.e. $\omega_k = 2\pi k/N$. Because the basis functions repeat after N , so do the coefficients: $c_{k+N} = c_k$. One period is enough — $k = 0 \dots N-1$, or $-N/2 \leq k < N/2$ if you prefer it centred. The teacher notes: apart from the $1/N$, **this pair is the DFT** (Unit 5).

WORKED Spectrum of $x(n) = \cos(\pi n/3)$ (M04L001)

$\omega_0 = \pi/3$, $f_0 = 1/6$, so $N = 6$. Do not grind the sum — use Euler:

$$x(n) = \cos\left(\left(\frac{2\pi n}{6}\right)\right) = \left(\frac{1}{2}\right) e^{j\frac{2\pi n}{6}} + \left(\frac{1}{2}\right) e^{-j\frac{2\pi n}{6}}, \quad \text{and} \quad e^{-j\frac{2\pi n}{6}} = e^{j\frac{2\pi(5)n}{6}} \quad \text{so} \quad c_{-1} = c_5$$

$$c_0 = 0, c_1 = \left(\frac{1}{2}\right), c_2 = c_3 = c_4 = 0, c_5 = \left(\frac{1}{2}\right)$$

A pure cosine is two lines, at $\pm\omega_0$, nothing else. (The slides confirm the same numbers with a six-line MATLAB loop.)

WORKED

Spectrum of $x(n) = \{1, 1, 0, 0\}$, $N = 4$ (M04L001)

$$c_k = \left(\frac{1}{4}\right) \sum_{n=0}^3 x(n) e^{-j2\pi kn/4} = \left(\frac{1}{4}\right) (1 + e^{-j\pi k/2})$$

$$c_0 = \left(\frac{1}{2}\right), c_1 = \left(\frac{1}{4}\right) (1 - j), c_2 = 0, c_3 = \left(\frac{1}{4}\right) (1 + j)$$

K	0	1	2	3
$ c_k $	0.5	$\sqrt{2}/4 = 0.354$	0	0.354
$\angle c_k$	0	$-\pi/4$	—	$+\pi/4$

x is real, so $c_3 = c_1^*$: magnitude even, phase odd $\checkmark$. Power check:

$$\sum |c_k|^2 = 0.25 + 0.125 + 0.125 = 0.5 = \frac{1}{4}(1+1) \checkmark$$

4.3

DTFT — Fourier transform for DT signals

The spectrum of a sampled, non-repeating signal: a continuous function of ω that repeats every 2π .

Teacher module M04L001 §3

DEFINITION

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$x(n) = \left(\frac{1}{2\pi}\right) \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

For a finite-energy (absolutely summable) signal. $X(\omega)$ is periodic with period 2π even though $x(n)$ is not periodic:

$$X(\omega + 2\pi k) = \sum_n x(n) e^{-j(\omega + 2\pi k)n} = X(\omega) \quad \text{because } e^{-j2\pi kn} = 1 \text{ for every whole } n$$

Link back to Unit 3: $X(\omega) = X(z)$ at $z = e^{j\omega}$, the z -transform on the unit circle — which is why the DTFT exists only when the ROC contains that circle.

4.4

Properties of the DTFT

The teacher's table, plus the symmetry rule that halves the work for real signals.

Teacher module M04L001 §4

PROPERTY	TIME DOMAIN	FREQUENCY DOMAIN
Linearity	$ax_1(n) + bx_2(n)$	$aX_1(\omega) + bX_2(\omega)$
Time shifting	$x(n - k)$	$e^{-j\omega k}X(\omega)$
Time reversal	$x(-n)$	$X(-\omega)$
Convolution	$x_1(n) * x_2(n)$	$X_1(\omega)X_2(\omega)$
Correlation	$r_{x_1x_2}(\ell) = x_1(\ell) * x_2(-\ell)$	$S_{x_1x_2}(\omega) = X_1(\omega)X_2(-\omega) = X_1(\omega)X_2^*(\omega)$
Frequency shifting	$e^{j\omega_0 n}x(n)$	$X(\omega - \omega_0)$
Modulation	$x(n) \cos(\omega_0 n)$	$\frac{1}{2}X(\omega + \omega_0) + \frac{1}{2}X(\omega - \omega_0)$
Multiplication	$x_1(n)x_2(n)$	$(1/2\pi) \int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega - \lambda) d\lambda$
Differentiation in frequency	$n x(n)$	$j dX(\omega)/d\omega$
Conjugation	$x^*(n)$	$X^*(-\omega)$
Parseval	$\sum_n x_1(n)x_2^*(n) = (1/2\pi) \int_{-\pi}^{\pi} X_1(\omega)X_2^*(\omega) d\omega$	

SYMMETRY FOR A REAL SIGNAL

$$X(-\omega) = X^*(\omega) \Rightarrow |X(-\omega)| = |X(\omega)|$$

magnitude even, $\angle X(-\omega) = -\angle X(\omega)$ phase odd

So for real data the spectrum from 0 to π tells you everything. A real *even* signal has a real spectrum (phase only 0 or π).

4.5

Worked spectra

The teacher's five examples — a two-sided exponential, two one-sided ones, and a pulse.

WORKED Example 1 — $x(n) = 0.8^{|n|}$

Split by linearity into $x_1 = 0.8^n$ for $n \geq 0$ and $x_2 = 0.8^{-n}$ for $n < 0$:

$$X_1(\omega) = \sum_{n=0}^{\infty} (0.8e^{-j\omega})^n = \left(\frac{1}{1 - 0.8e^{-j\omega}} \right)$$

$$X_2(\omega) = \sum_{m=1}^{\infty} (0.8e^{j\omega})^m = \left(\frac{0.8e^{j\omega}}{1 - 0.8e^{j\omega}} \right)$$

$$X(\omega) = X_1 + X_2 = \left(\frac{1 - 0.8^2}{1 - 2(0.8)\cos\omega + 0.8^2} \right) = \left(\frac{0.36}{1.64 - 1.6\cos\omega} \right)$$

Real and positive for every ω — as it must be for an even signal. Value 9 at $\omega = 0$, 0.111 at $\omega = \pi$: a low-pass shape. Example 2 is the same with 0.5: $X(\omega) = 0.75/(1.25 - \cos\omega)$. In general $a^{|n|} \leftrightarrow (1 - a^2)/(1 - 2a\cos\omega + a^2)$.

WORKED Examples 3, 4 — $0.5^n u(n)$ and $(-0.5)^n u(n)$

$$0.5^n u(n): X(\omega) = \sum_{n=0}^{\infty} (0.5e^{-j\omega})^n = \left(\frac{1}{1 - 0.5e^{-j\omega}} \right) = \left(\frac{e^{j\omega}}{e^{j\omega} - 0.5} \right)$$

$$|X| = 2 \quad \text{at } \omega = 0, 0.667 \quad \text{at } \pi: \text{ low-pass}$$

$$(-0.5)^n u(n): X(\omega) = \left(\frac{1}{1 + 0.5e^{-j\omega}} \right) = \left(\frac{e^{j\omega}}{e^{j\omega} + 0.5} \right) \quad \text{largest at } \omega = \pi: \text{ high-pass}$$

Flipping the sign of a flips the spectrum end for end — an alternating sequence is the highest digital frequency (§1.6).

WORKED Example 5 — a pulse, $x(n) = 2$ for $0 \leq n \leq 5$

$$X(\omega) = 2 \sum_{n=0}^5 e^{-j\omega n} = 2 \cdot \left(\frac{1 - e^{-j6\omega}}{1 - e^{-j\omega}} \right)$$

$$= 2e^{-j\frac{5\omega}{2}} \cdot \left(\frac{\sin(3\omega)}{\sin(\frac{\omega}{2})} \right)$$

(Pull $e^{-j3\omega}$ out of the top, $e^{-j\omega/2}$ out of the bottom, use $e^{j\theta} - e^{-j\theta} = 2j \sin\theta$.) The $\sin(3\omega/2)/\sin(\omega/2)$ shape — a main lobe with side lobes — returns in Unit 5 as the cause of leakage. The $e^{-j5\omega/2}$ is linear phase: it only says the pulse is centred at $n = 2.5$.

4.6

Frequency response of a system

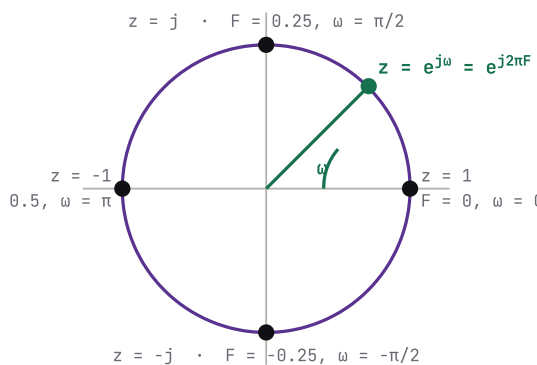
Evaluate $H(z)$ on the unit circle and you get the system's gain and phase shift at every frequency.

Teacher module M04L001 §6

DEFINITION

$$H_p(F) = H(z)|_{z=e^{j2\pi F}} = e^{j\omega} = |H_p(F)| e^{j\varphi(F)}$$

It is the DTFT of $h(n)$. $|H_p|$ is the **magnitude response** (gain), φ the **phase response**. It is periodic with period 1 in F (2π in ω), and for real $h(n)$ it has conjugate symmetry: magnitude even, phase odd over $-0.5 \leq F \leq 0.5$. Most useful for stable systems, whose natural response dies away.



WALKING ROUND THE CIRCLE IS SWEEPING FREQUENCY

Start at $z = 1$: DC, $\omega = 0$, $F = 0$.

A quarter turn, $z = j$: $\omega = \pi/2$, $F = 0.25$.

Half a turn, $z = -1$: the highest frequency, $\omega = \pi$, $F = 0.5$.

Keep going and you come back to the start – which is exactly why the spectrum repeats every 2π .

M04L001's unit-circle picture. The point $z = e^{j\omega} = e^{j2\pi F}$ is the frequency you are evaluating at. It ties together the z -plane (Unit 3), the DTFT (§4.3) and the frequency response.

TWO SHORTCUTS

$$H_p(0) = H(z)|_{z=1} \quad \text{DC gain}$$

at the highest frequency

$$H_p(0.5) = H(z)|_{z=-1} \quad \text{gain}$$

WORKED

Frequency response of the recursive filter

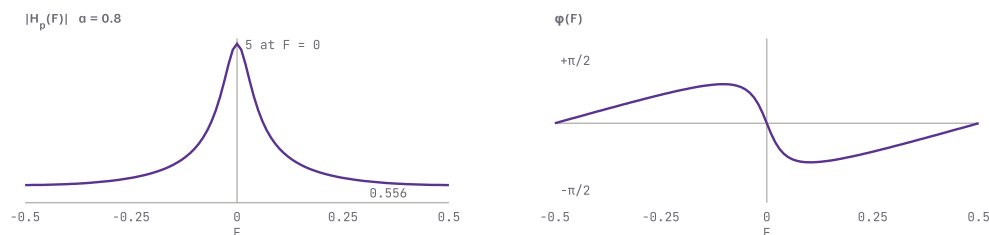
$$y(n) = a y(n-1) + x(n) \quad (\text{M04L001})$$

$$H(z) = \left(\frac{1}{1 - az^{-1}} \right) \Rightarrow H_p(F) = \left(\frac{1}{1 - ae^{-j2\pi F}} \right) = \left(\frac{1}{1 - a \cos(2\pi F) + ja \sin(2\pi F)} \right)$$

$$|H_p(F)| = \left(\frac{1}{1 - 2a \cos(2\pi F) + a^2} \right) \quad \text{even in } F$$

$$\phi(F) = -\tan^{-1} \left(\frac{a \sin(2\pi F)}{1 - a \cos(2\pi F)} \right) \quad \text{odd in } F$$

With $a = 0.8$: gain 5 at DC, 0.556 at $F = 0.5$; the pole at $z = 0.8$ is near $z = 1$, so low frequencies are boosted — a low-pass filter.



M04L001's plots, redrawn from the formula. Magnitude is even about $F = 0$, phase is odd; both repeat with period 1 in F . The phase is negative for positive F — the filter delays.

4.7

Steady-state response to a sinusoid

Put a sinusoid in, the same sinusoid comes out — scaled by $|H|$ and shifted by $\angle H$ at that frequency.

Teacher module M04L001 §7

Why: feed $e^{j\omega n}$ into the convolution sum and it comes straight back out:

$$y(n) = \sum_k h(k) e^{j\omega(n-k)} = e^{j\omega n} \cdot \sum_k h(k) e^{-j\omega k} = e^{j\omega n} \cdot H(\omega)$$

An LTI system cannot change a complex exponential's shape, only multiply it by the complex number $H(\omega)$. A real cosine is two exponentials, so:

THE RECIPE (M04L001)

input $x(n) = A \cos(2\pi F_0 n + \theta)$

evaluate $H(z)$ at $z = e^{j2\pi F_0} \rightarrow H_0 \angle \psi_0$

output $y_{ss}(n) = A H_0 \cos(2\pi F_0 n + \theta + \psi_0)$

A constant input is $F_0 = 0$: multiply by $H(1)$. A periodic input: do this harmonic by harmonic with the DTFS — the output is periodic with the same N ; an LTI system never creates a new frequency.

WORKED The two examples from M04L001

(1) $y(n) = 0.5y(n-1) + x(n)$, $x(n) = 10\cos(0.5\pi n + 60^\circ)$.

$$H(z) = \left(\frac{1}{1 - 0.5z^{-1}} \right); F_0 = 0.25 \Rightarrow z = e^{j\frac{\pi}{2}} = j, z^{-1} = -j$$

$$H_p(0.25) = \left(\frac{1}{1 + j0.5} \right) = 0.8 - j0.4 = 0.8944 \angle -26.6^\circ$$

$$y_{ss}(n) = 8.944 \cos(0.5\pi n + 60^\circ - 26.6^\circ) = 8.944 \cos(0.5\pi n + 33.4^\circ)$$

(2) $H(z) = (2z - 1)/(z^2 + 0.5z + 0.5)$, $x(n) = 6u(n)$. A constant $\rightarrow F = 0 \rightarrow z = 1$:

$$H(1) = \left(\frac{2 - 1}{1 + 0.5 + 0.5} \right) = 0.5 \Rightarrow y_{ss} = 6 \times 0.5 = 3$$

Before it settles there is also a transient from the system's poles — Tutorial 5 Q9 in §3.6 shows the full answer with both parts.

4.8

Between the z-transform and the DTFT

Two substitution rules, and one warning about signals whose poles sit on the circle.

Teacher module M04L001, last slide

THE RULES

- **From $X(z)$ to the DTFT:** if $x(n)$ is absolutely summable, replace z by $e^{j2\pi F}$ (or $e^{j\omega}$).
- **From the DTFT to $X(z)$:** delete any impulse terms, then replace $e^{j2\pi F}$ (or $e^{j\omega}$) by z .

Why impulses appear: $u(n)$ and $\cos(\omega_0 n)$ are not absolutely summable — their z-transforms have poles *on* the unit circle, exactly where the DTFT is evaluated. Their DTFTs exist only if we allow impulses in frequency: a cosine is two impulses at $\pm\omega_0$ (repeating every 2π), which is the "two lines" of §4.2 seen from the DTFT side. Those impulse terms have no z-transform counterpart, hence the "delete" in the second rule.

UNIT 5 · TEACHER MODULES M04L002 (INTRODUCTION TO DFT VIA DTFT) AND M04L004 (FAST FOURIER TRANSFORM)

The DFT and the FFT

The DTFT is a continuous curve — a computer cannot hold it. Sample it at N points and you have the DFT: N numbers, storable, computable. Then find the arithmetic that computes those N numbers hundreds of times faster.

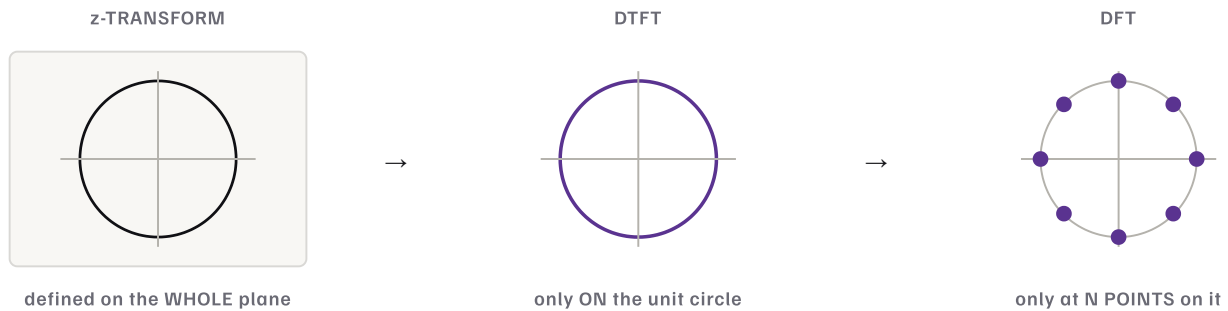
5.1 Why the DFT

Of the four Fourier transforms, only one is discrete and finite in both domains — the one a digital processor can actually compute.

Teacher module M04L002 §1

Many applications need a signal in the frequency domain — to read its frequencies, its periodicity, its energy or power spectrum — and that analysis is most conveniently done on a digital signal processor. Applications the module lists: **spectral analysis, convolution of signals, partial differential equations, multiplication of large integers, data compression.**

TIME DOMAIN	FREQUENCY DOMAIN	TRANSFORM
Continuous, periodic	Discrete, non-periodic	Fourier series
Continuous, aperiodic	Continuous, non-periodic	Fourier transform
Discrete, non-periodic	Continuous, periodic	DTFT
Discrete, periodic	Discrete, periodic	DFT



Three views of one function. Restrict the z-transform to the unit circle $\rightarrow$ the DTFT (§4.3); sample that at N equally spaced points $\rightarrow$ the DFT. In symbols:

$$X(k) = X(\omega) \big|_{\omega=2\pi k/N} = X(z) \big|_{z=e^{j2\pi k/N}}, \text{ and compared with the DTFS, } X(k) = N c_k.$$

5.2

Frequency-domain sampling; choosing N

Two things stop a computer computing the DTFT. Fix both by sampling — and use a small trick with the index k .

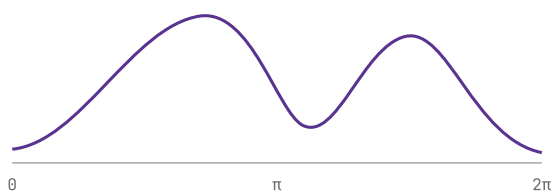
Teacher module M04L002 §2-3

WHY WE CAN'T JUST COMPUTE THE DTFT (M04L002)

1. The sum $X(\omega) = \sum x(n) e^{-j\omega n}$ runs over **infinitely many** n . Solved if the sequence is finite — the realistic case.
2. $X(\omega)$ must be evaluated at **infinitely many** ω in $(-\pi, \pi)$. Still a problem.

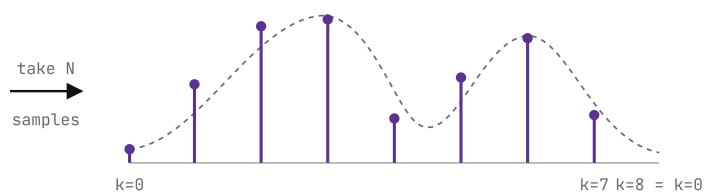
So we sample the spectrum: since $X(\omega)$ repeats every 2π , take N equidistant samples in $0 \leq \omega < 2\pi$ with spacing $\Delta\omega = 2\pi/N$ — the **frequency resolution**.

THE DTFT $X(\omega)$ — a continuous curve



infinitely many values — cannot be stored

THE DFT $X(k)$, $k = 0 \dots N-1$



N numbers — storable; the FFT makes them fast

The DFT is the sampled DTFT (M04L002, $N=8$ case: spacing $2\pi/8 = \pi/4$). The point at $k = N$ is not computed — it equals $k = 0$.

Selecting N , and the mirror trick

Take a finite signal, $x(n) = 0$ for $n < 0$ and $n \geq N$. We are really interested in $-\pi \leq \omega < \pi$; everywhere else the spectrum just repeats. Computing it at 8 points would be far too few — in practice thousands. Then the trick: **do not compute the negative- ω points; compute their**

mirror images between π and 2π instead. Same numbers, but now the index runs $0, 1, 2, \dots$ instead of straddling zero, which makes every later formula tidy. So with M frequency points:

$$\omega_k = k \cdot \left(\frac{2\pi}{M} \right), k = 0, 1, \dots, M-1 \Rightarrow X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi kn}{M}}$$

Padding zeros so that $M = N$ gives the DFT proper, and forcing $M = N$ is what makes the inverse work. The "upper" DFT points are really the negative frequencies — move them down when you plot.

WHAT SAMPLING THE SPECTRUM DOES TO THE TIME SIGNAL

The IDFT returns $x_p(n)$, which is **periodic with period N** . Sampling in frequency makes the time signal repeat (§4.1's rule), so if the signal is longer than N its tail wraps onto its head — time-domain aliasing, the mirror image of §1.7. Recovery is exact only when $N \geq$ the signal length.

5.3

The DFT and the IDFT

N numbers in, N numbers out, and the twiddle factor W_N does all the work.

Teacher module M04L002 §3-4

THE PAIR

$$\text{DFT: } X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi kn}{N}} = \sum_{n=0}^{N-1} x(n) W_N^{kn}, k = 0 \dots N-1$$

$$\text{IDFT: } x(n) = \left(\frac{1}{N} \right) \sum_{k=0}^{N-1} X(k) W_N^{-kn}, n = 0 \dots N-1$$

$$W_N = e^{-j \frac{2\pi}{N}}$$

the twiddle factor

The inverse differs in only two ways: a $1/N$ and a sign in the exponent. $X(0) = \sum x(n)$ always (every twiddle is 1 at $k = 0$) — a free check on any answer. Bin k stands for the analog frequency $F_k = kF_s/N$.

THREE PROPERTIES OF W_N (M04L004 ASKS YOU TO PROVE THEM)

$$\text{periodicity } W_N^{k+N} = W_N^k$$

$$\text{symmetry } W_N^{k+\frac{N}{2}} = -W_N^k$$

$$\text{half-point } W_N^2 = W_{\frac{N}{2}}$$

Direct computation ignores all three. The FFT is nothing but these three facts used over and over.

Summary from the module: given N data points we can compute the DFT (efficiently, by the FFT); given N DFT points we get the data back (by the IFFT). And from §2.15: circular convolution in time is $X_1(k)X_2(k)$ in the DFT domain — the shortcut for Tutorial 3 Q7.

WORKED

A 4-point DFT by hand: $x = \{1, 2, 3, 4\}$ (M04L002 Example 2)

$$W_4 = e^{-\frac{j\pi}{2}} = -j; \text{ powers: } 1, -j, -1, j$$

$$X(0) = 1 + 2 + 3 + 4 = 10$$

$$X(1) = 1 + 2(-j) + 3(-1) + 4(j) = -2 + 2j$$

$$X(2) = 1 - 2 + 3 - 4 = -2$$

$$X(3) = 1 + 2(j) + 3(-1) + 4(-j) = -2 - 2j$$

Checks: real input $\Rightarrow X(3) = X(1)^*$. Parseval: $\sum |x|^2 = 30$ and $\frac{1}{4}\sum |X|^2 = \frac{1}{4}(100 + 8 + 4 + 8) = 30 \checkmark$

5.4

The DFT as a matrix; twiddle factors

The DFT is a matrix multiplication, and the matrix entries are just points going round the unit circle.

Teacher module M04L002 §5

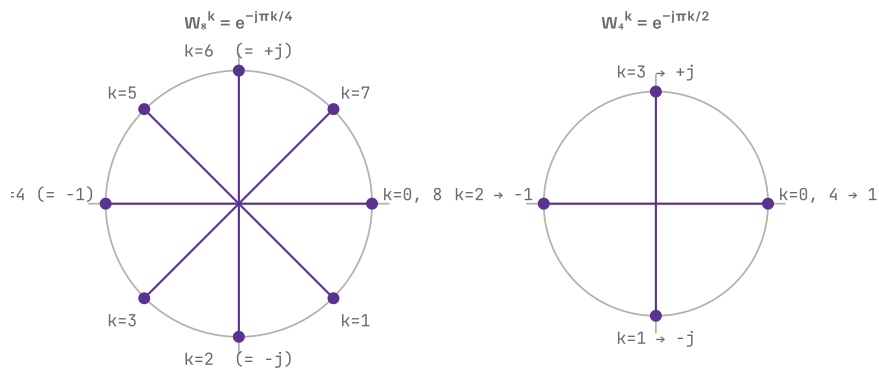
$$X_N = W_N x_N \quad x_N = \left(\frac{1}{N}\right) W_N^{-1} X_N = \left(\frac{1}{N}\right) W_N^* X_N$$

entry (k, n) is W_N^{kn}

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

symmetric; first row and column all 1s; inverse =

conjugate / N

**WHY THIS SAVES WORK**

Only N different twiddle values exist, however large k gets ($W_8^9 = W_8^1$).

And half of them are the negatives of the other half.

Magnitude is always 1.

M04L002's phasor diagrams. W_8^k steps round by $-\pi/4$ each time and repeats every 8 — $W_8^{k+8} = W_8^k$. The four values of W_4 are $1, -j, -1, +j$, free to multiply by — which is why radix-4 (§5.17) is cheap.

5.5**Worked DFTs**

The six examples from the module, done by the matrix or the formula.

Teacher module **M04L002** §5, Examples 1–6

WORKED

Examples 1–4: two DFTs, a zero-padded one, and an IDFT

(1) $x = \{2, 3, 4, 4\}$, by the matrix (rows 1, $-j$, -1 , j):

$$X(0) = 13, X(1) = 2 - 3j - 4 + 4j = -2 + j, X(2) = 2 - 3 + 4 - 4 = -1, X(3) = -2 - j$$

(2) $x = \{1, 2, 3, 4\}$: $\{10, -2 + 2j, -2, -2 - 2j\}$ (§5.3).

(3) 8-point DFT of $x = \{1, 1, 1, 1\}$: pad four zeros, then

$$X(k) = \sum_{n=0}^3 W_8^{kn} = (1 - (-1)^k) / (1 - e^{-j\pi k/4}):$$

K	0	1	2	3	4	5	6	7
$X(k)$	4	$1 - 2.414j$	0	$1 - 0.414j$	0	$1 + 0.414j$	0	$1 + 2.414j$
$ X(k) $	4	2.613	0	1.082	0	1.082	0	2.613
$\angle X(k)$	0	-67.5°	—	-22.5°	—	22.5°	—	67.5°

Even k (other than 0) vanish because $1 - (-1)^k = 0$; the odd ones are $2/(1 - W_8^k)$.

Conjugate pairs $X(7) = X(1)^*$ etc. ✓ (The slide's printed answer is the answer to Example 1, pasted by mistake.)

(4) IDFT of $X = \{10, 2 + 2j, 2, 2 - 2j\}$. Same matrix conjugated (rows 1, j , -1 , $-j$) and divided by 4:

$$x(0) = \left(\frac{1}{4}\right) [10 + 2 + 2j + 2 + 2 - 2j] = 4$$

$$x(1) = \left(\frac{1}{4}\right) [10 + j(2 + 2j) - 2 - j(2 - 2j)] = \left(\frac{1}{4}\right) [10 - 2 - 2 - 2] = 1$$

$$x(2) = \left(\frac{1}{4}\right) [10 - (2 + 2j) + 2 - (2 - 2j)] = 2$$

$$x(3) = \left(\frac{1}{4}\right) [10 - j(2 + 2j) - 2 + j(2 - 2j)] = \left(\frac{1}{4}\right) [10 + 2 - 2 + 2] = 3$$

$$x = \{4, 1, 2, 3\}$$

TRANSFORM BACK TO CHECK

The DFT of $\{4, 1, 2, 3\}$ is $X(1) = 4 - j - 2 + 3j = 2 + 2j$ ✓ — it returns the given $X(k)$. The slide prints $\{1, 2, 3, 4\}$, but that sequence transforms to $\{10, -2 + 2j, -2, -2 - 2j\}$ (Example 2). One sign in $X(k)$ rotates the whole answer, so always transform back.

WORKED

Examples 5, 6: a^n and a short cosine

(5) $x(n) = a^n, 0 \leq n \leq N-1$: a finite geometric series,

$$X(k) = \sum_{n=0}^{N-1} \left(a e^{-j \frac{2\pi k}{N}} \right)^n = \left(\frac{1 - a^N}{1 - a e^{-j \frac{2\pi k}{N}}} \right)$$

$$a = 0.5, N = 4: \quad \frac{0.9375}{1 - 0.5 e^{-j \frac{\pi k}{2}}}$$

(6) 4-point DFT of $\cos(\pi n/4)$: samples $\{1, 0.707, 0, -0.707\}$,

$$\begin{aligned} X(0) &= 1 \\ X(1) &= 1 + 0.707(-j) - 0.707(j) = 1 - 1.414j \\ X(2) &= 1 - 0.707 + 0.707 = 1 \\ X(3) &= 1 + 1.414j \end{aligned}$$

Not two clean lines: $\cos(\pi n/4)$ has period 8 and we took only 4 samples — half a cycle. The DFT treats those 4 samples as repeating, which is not a pure cosine, so the energy spreads. That is leakage (§5.9), met early.

5.6

Properties of the DFT

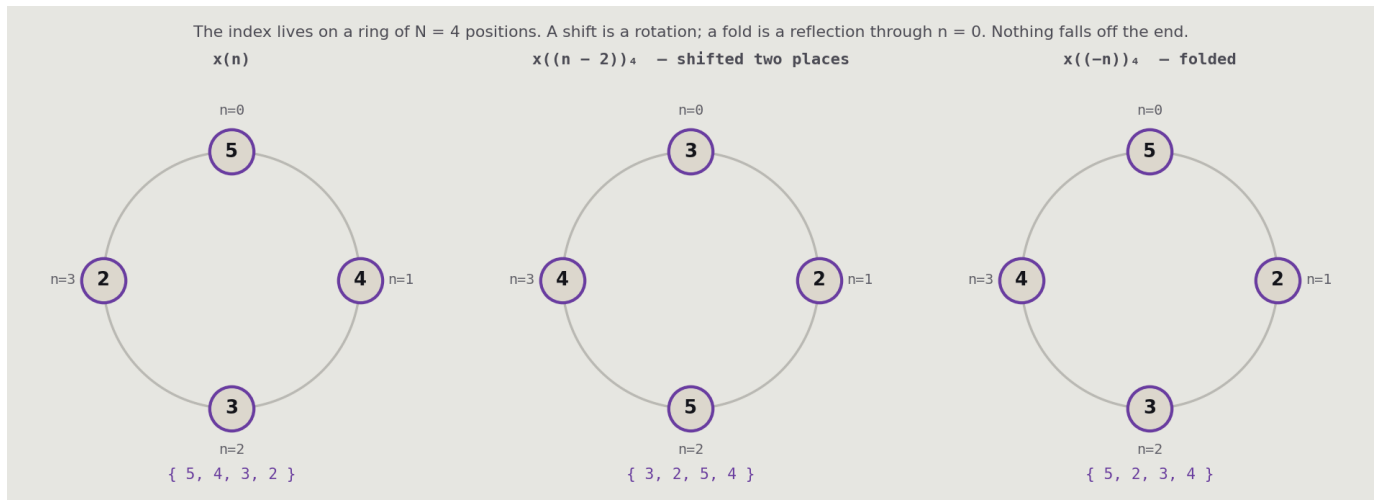
What the DFT does to a shifted, folded, conjugated or multiplied sequence — with the index always counted round a ring of N .

Slides [propertis-of-discrete-fourier-transform.pdf](#) · Proakis §7.2 · class notebook

THE ONE RULE BEHIND ALL OF THEM

A DFT of length N only ever sees $x(0), \dots, x(N-1)$, and it treats them as **one period of a repeating sequence**. So every index is read **modulo** N : $x((n))_N = x(n \bmod N)$. Shift a sequence and the samples that fall off one end come back at the other. Fold it and $x((-n))_N = x(N-n)$. Every property below is the ordinary DTFT property with this wrap-around built in.

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The index lives on a ring. $x(n) = \{5, 4, 3, 2\}$ at positions 0–3, clockwise from the top. A circular shift by two, $x((n-2))_4$, turns the ring: $\{3, 2, 5, 4\}$. A circular fold, $x((-n))_4$, reflects it through position 0: $\{5, 2, 3, 4\}$ — the sample at 0 stays where it is, the others reverse.

PROPERTY	SEQUENCE	DFT
Periodicity	$x(n + N) = x(n)$	$X(k + N) = X(k)$
Linearity	$a x_1(n) + b x_2(n)$	$a X_1(k) + b X_2(k)$
Circular time shift	$x((n - m))_N$	$W_N^{km} X(k) = X(k) e^{-j2\pi km/N}$
Circular frequency shift	$x(n) e^{j2\pi \ell n/N} = x(n) W_N^{-\ell n}$	$X((k - \ell))_N$
Time reversal (circular fold)	$x((-n))_N = x(N - n)$	$X((-k))_N = X(N - k)$
Complex conjugate	$x^*(n)$	$X^*((-k))_N = X^*(N - k)$
Conjugate and fold	$x^*((-n))_N$	$X^*(k)$
Real sequence	$x(n)$ real	$X(N - k) = X^*(k)$: $ X $ even, $\angle X$ odd about $N/2$
Real and even	$x(n) = x(N - n)$	$X(k) = \sum x(n) \cos(2\pi kn/N)$, real and even
Real and odd	$x(n) = -x(N - n)$	$X(k) = -j \sum x(n) \sin(2\pi kn/N)$, imaginary and odd
Circular convolution	$x_1(n) \otimes x_2(n)$	$X_1(k) X_2(k)$
Multiplication	$x_1(n) x_2(n)$	$(1/N) X_1(k) \otimes X_2(k)$
Circular correlation	$r_{xy}(\ell) = \sum_n x(n) y^*((n - \ell))_N$	$X(k) Y^*(k)$
Parseval	$\sum_n x(n) y^*(n)$	$(1/N) \sum_k X(k) Y^*(k)$; energy: $\sum x ^2 = (1/N) \sum X ^2$

The three proofs the exam asks for

Periodicity. Put $k + N$ into the definition and use $e^{-j2\pi n} = 1$ for every whole number n :

$$X(k + N) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}(k+N)n} = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn} e^{-j2\pi n} = X(k) \underset{=1}{=} X(k)$$

The DFT repeats in k with period N . This is why §5.5 insisted that a 4-point DFT has exactly four different values.

Circular shift. Shifting by m multiplies the DFT by a linear phase, W_N^{km} . Substitute $r = n - m$; the modulo- N index carries the sum round the ring, so the range is still one full period:

$$\sum_{n=0}^{N-1} x((n-m))_N W_N^{kn} = \sum_r x(r) W_N^{k(r+m)} = W_N^{km} \sum_{r=0}^{N-1} x(r) W_N^{kr} = W_N^{km} X(k)$$

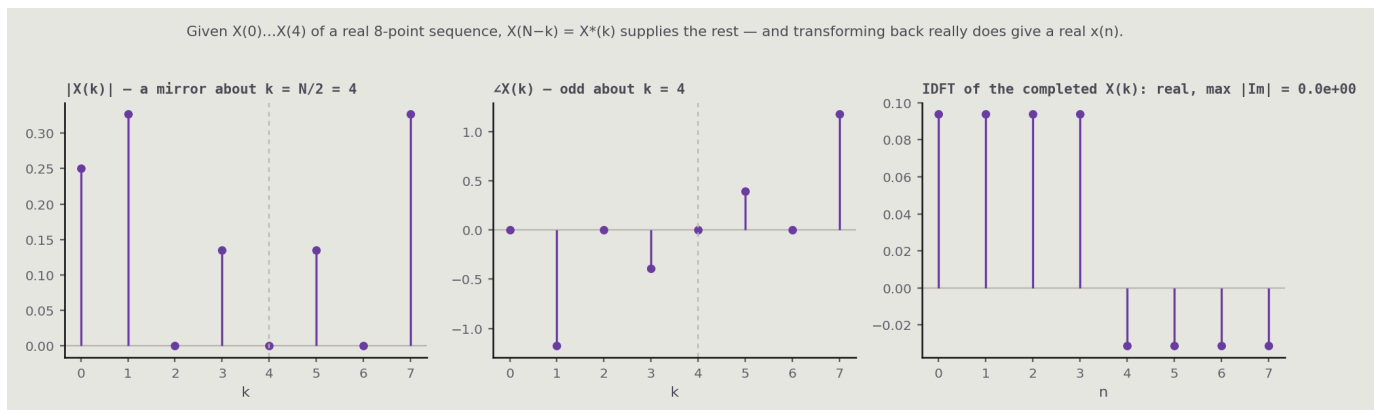
The magnitude is untouched ($|W_N^{km}| = 1$); only the phase tilts, by $-2\pi km/N$. By the same substitution, multiplying $x(n)$ by $e^{j2\pi l n/N}$ turns the spectrum round: $X((k-l))_N$.

Symmetry of a real sequence. Replace k by $N-k$:

$$X(N-k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}(N-k)n} = \sum_{n=0}^{N-1} x(n) e^{+j\frac{2\pi}{N}kn} e^{-j2\pi n} = \left[\sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn} \right]^* = X^*(k)$$

The last step works because $x(n)$ is real, so conjugating the sum only conjugates the exponentials. What follows: $|X(N-k)| = |X(k)|$ (the magnitude is a mirror image about $N/2$), $\angle X(N-k) = -\angle X(k)$ (the phase flips sign), $X(0)$ is real, and for even N so is $X(N/2)$, because it must equal its own conjugate.

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The symmetry, and the example below, computed. Left and middle: $|X(k)|$ mirrors about $k = N/2 = 4$ and $\angle X(k)$ changes sign about it. Right: the inverse DFT of the completed eight values is real to machine precision — the check that the three missing points were filled in correctly.

WORKED

The first five points of an 8-point DFT of a real sequence are given; find the other three (slides p. 15)

$X(0) = 0.25, X(1) = 0.125 - j0.3018, X(2) = 0, X(3) = 0.125 - j0.0518,$
 $X(4) = 0.$ For real $x(n), X(k) = X^*(N - k) = X^*(8 - k):$

$$X(5) = X^*(3) = 0.125 + j0.0518$$

$$X(6) = X^*(2) = 0$$

$$X(7) = X^*(1) = 0.125 + j0.3018$$

Check: $X(0)$ and $X(4)$ are real, as they must be. The slide's second version gives $\{0.25, -j0.3018, 0, 0, 0.125 - j0.0518\}$ and asks the same. Then $X(5) = X^*(3) = 0, X(6) = 0, X(7) = +j0.3018.$ Notice the trap in that version: $X(4) = 0.125 - j0.0518$ is *not* real, so the sequence it claims to come from cannot be real. A good answer says so.

Correlation, and multiplication

Circular cross-correlation is circular convolution with one sequence conjugated and folded: $r_{xy}(l) = x(l) \circledast y^*((-l))_N.$ Apply the convolution row and the conjugate-and-fold row of the table, and the DFT of the correlation is $X(k)Y^*(k).$ With $y = x$ this gives the autocorrelation $\leftrightarrow |X(k)|^2,$ which is how a power spectrum is computed. Multiplying two sequences sample by sample convolves their DFTs *circularly* and divides by N — the property behind windowing (§5.9 and §6.6).

CIRCULAR, EVERY TIME

Every shift, fold and convolution in this table is circular. If a question gives you $x(n) = \{1, 2, 3, 4\}$ and asks for the DFT of $x(n - 1)$ “using the shift property”, the sequence that property applies to is $\{4, 1, 2, 3\}$, not $\{0, 1, 2, 3\}.$ And if it gives you $X(k) = \{10, -2+2j, -2, -2-2j\}$ and asks for the sequence whose DFT is $X^*(k),$ the answer is $x((-n))_4 = \{1, 4, 3, 2\}$ — fold, and keep $x(0)$ where it is.

5.7

Filtering a long signal with the DFT: overlap-add and overlap-save

A DFT can only convolve blocks of fixed size. Cut the long input into blocks, filter each one with a DFT, and join the outputs so nothing is lost at the joins.

Class notebook (18 September) · Proakis §7.3.2

Unit 2 showed that a circular convolution of length N equals the linear one if $N \geq L + M - 1.$ In practice the input $x(n)$ can be very long (audio, a whole recording) and the filter $h(n)$ is short (length M). You cannot take one giant DFT, and you do not want to wait for the whole signal

anyway. So you work in blocks. There are two ways to handle the joins, and the exam asks for both.

OVERLAP - ADD

Cut $x(n)$ into blocks of L samples that do *not* overlap. Convolve each block with $h(n)$ by an N -point DFT, with $N = L + M - 1$ (pad both to N with zeros). Each output block is N long, so it sticks out $M - 1$ samples past the next block's start. **Add** the parts that overlap.

OVERLAP - SAVE

Cut $x(n)$ into blocks of N samples that *do* overlap: each block starts with the last $M - 1$ samples of the previous one (the very first block starts with $M - 1$ zeros). Do an N -point circular convolution with $h(n)$. The first $M - 1$ outputs of each block are wrong (that is where the circular wrap-around lands), so **throw them away** and **save** the other $N - M + 1$. Join the saved parts end to end.

overlap-add: $N = L + M - 1$, blocks of L inputs $\rightarrow N$ outputs, add the $M - 1$ overlapping samples

overlap-save: blocks of N inputs overlapping by $M - 1 \rightarrow$ keep the last $N - M + 1$ outputs of each

The

WORKED

class $h = \{2, 2, 1\}$, $x = \{3, 0, -2, 0, 2, 1, 0, -2, -1, 0\}$
example:

$M = 3, N = 8$. The direct answer, to check against, is the linear convolution:

$$y(n) = x(n) * h(n) = \{6, 6, -1, -4, 2, 6, 4, -3, -6, -4, -1, 0\} \quad (10 + 3 - 1 = 12 \text{ samples})$$

Overlap-add. $N = L + M - 1$ gives $L = 8 - 3 + 1 = 6$: blocks of six inputs. Block 1 is $\{3, 0, -2, 0, 2, 1\}$, block 2 is $\{0, -2, -1, 0\}$ padded to six. Each block, padded to 8, is circularly convolved with h padded to 8 — and because $8 \geq 6 + 3 - 1$, each of those is a true linear convolution of the block:

$$\begin{array}{ll} y_1 = \{6, 6, -1, -4, 2, 6, 4, 1\} & \text{sits at } n = 0 \dots 7 \\ y_2 = \{0, -4, -6, -4, -1, 0, 0, 0\} & \text{sits at } n = 6 \dots 13 \\ y(6) = 4 + 0 = 4, & y(7) = 1 + (-4) = -3 \quad \text{the two overlapping samples, added} \end{array}$$

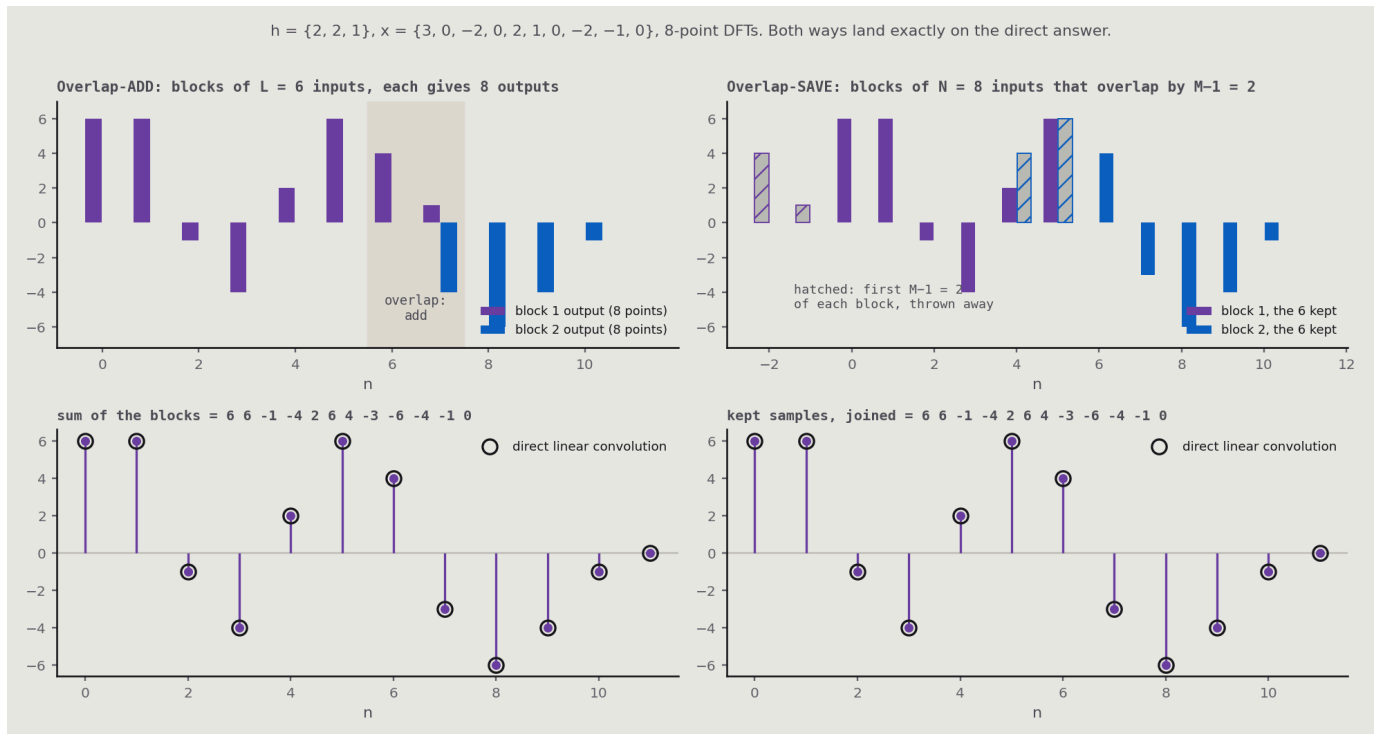
Joined: $\{6, 6, -1, -4, 2, 6, 4, -3, -6, -4, -1, 0\}$. Matches.

Overlap-save. Blocks of $N = 8$ inputs, overlapping by $M - 1 = 2$. Block 1 starts with two zeros: $\{0, 0, 3, 0, -2, 0, 2, 1\}$. Block 2 starts with the last two of block 1: $\{2, 1, 0, -2, -1, 0, 0, 0\}$.

$$\begin{array}{ll} \text{block 1 circular: } \{4, 1, 6, 6, -1, -4, 2, 6\} & \rightarrow y(0 \dots 5) = 6, 6, -1, -4, 2, 6 \\ & \text{discard} \\ \text{block 2 circular: } \{4, 6, 4, -3, -6, -4, -1, 0\} & \rightarrow y(6 \dots 11) = 4, -3, -6, -4, -1, 0 \\ & \text{discard} \end{array}$$

Joined: the same twelve numbers. The two thrown-away values in each block (4, 1 and 4, 6) are the samples the circular wrap-around spoiled — you can see they are not in the direct answer.

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Both methods, computed on the class example. Left: overlap-add — two output blocks of eight, the two samples where they overlap shaded, and below them the sum sitting exactly on the direct linear convolution (rings). Right: overlap-save — the first two outputs of each block (hatched) are thrown away and the rest are joined; again the rings show the direct answer.

WHICH ONE IS WHICH

Add: input blocks do not overlap, output blocks do, and you add. **Save:** input blocks overlap, you throw away the first $M - 1$ outputs of each, and you keep the rest. In both, the DFT size N must be at least $L + M - 1$, and the cost per output sample is about $\log_2 N$ multiplications instead of M — which is why this is worth doing when M is more than about 30.

5.8

DFT and DTFT: the finite case

If the signal really is finite, the DFT points sit exactly on its DTFT — and zero padding just puts more dots on the same curve.

Teacher module M04L002 §6-7

The practical chain (M04L002): a sensor $\rightarrow$ ADC at $F_s \rightarrow N$ samples in memory $\rightarrow$ FFT $\rightarrow N$ DFT values in memory. So we need to know what those values say about the true spectrum. Two cases: the signal is **finite** (all its non-zero samples fit in our N), or **infinite** (it goes on longer than we can capture).

FINITE CASE

If $x(n) = 0$ outside $0 \dots N-1$, the DTFT sum has nothing to leave out, and comparing it with the DFT formula:

$$X(k) = X(\omega) \Big|_{\omega = 2\pi k/N} \quad \text{exactly — the DFT points lie ON the DTFT curve}$$

N only sets how closely spaced the samples are. **Zero-padding trick:** tack zeros on the end so the total is N_z ; the sums are unchanged (zeros add nothing) but the spacing becomes $2\pi/N_z$, smaller than $2\pi/N$: a denser set of samples on the *same* curve.

WORKED The pulse example (M04L002)

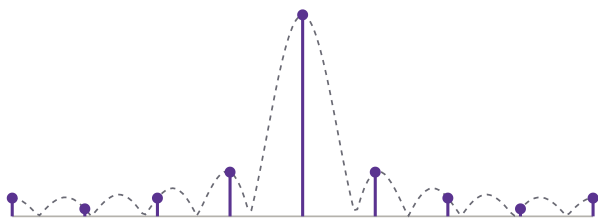
$x(n) = 1$ for $n = 0 \dots 2q$, a pulse of $N = 2q + 1$ ones — the centred pulse $x_p(n)$ delayed by q . By the delay property,

$$X_p(\omega) = \left(\frac{\sin\left(\frac{N\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} \right) = \left(\frac{\sin\left[\left(q + \frac{1}{2}\right)\omega\right]}{\sin\left(\frac{\omega}{2}\right)} \right), X(\omega) = X_p(\omega)e^{-jq\omega}$$

$$\Rightarrow X(k) = \left(\frac{\sin\left[\frac{\left(q + \frac{1}{2}\right) \cdot 2\pi k}{N}\right]}{\sin\left(\frac{\pi k}{N}\right)} \right) e^{-\frac{jq \cdot 2\pi k}{N}}$$

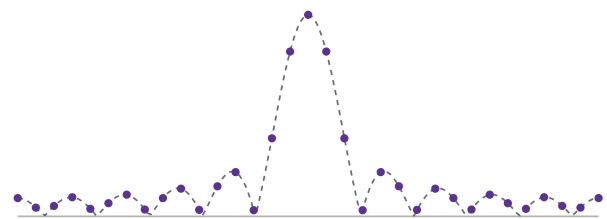
For $q = 5$ the slides compute this with 11 zeros padded ($N = 22$) and with 77 padded ($N = 88$): at 22 points the dots still do not show what the DTFT looks like; at 88 they do.

FEW POINTS — too few dots to see the shape



The side lobes are missed — from these dots you would never guess the true shape.

ZERO-PADDED, FOUR TIMES AS MANY — the shape appears



Same curve, same information — just sampled finely enough to actually see the lobes.

Zero padding is interpolation, not new information. The dashed curve is the true DTFT of an 11-sample pulse ($q = 5$), identical in both panels — it is the module's pair of plots redrawn. Padding only decides how many dots you put on it.

SUMMARY — FINITE-LENGTH SIGNALS (M04L002)

- The DFT samples the DTFT at $\omega_k = 2\pi k/N$; the points lie exactly on the curve.
- More points → a denser sampling → a better *picture* of the spectral shape.

- Zero-padding does not change the DTFT or the spectral shape; it reduces the spacing $2\pi/N$ and gives a denser set of samples of the same curve.

5.9

DFT and DTFT: leakage and the three errors

If the signal goes on longer than you can capture, the DFT shows you a smeared spectrum — and no amount of zero padding un-smears it.

Teacher module M04L002 §8

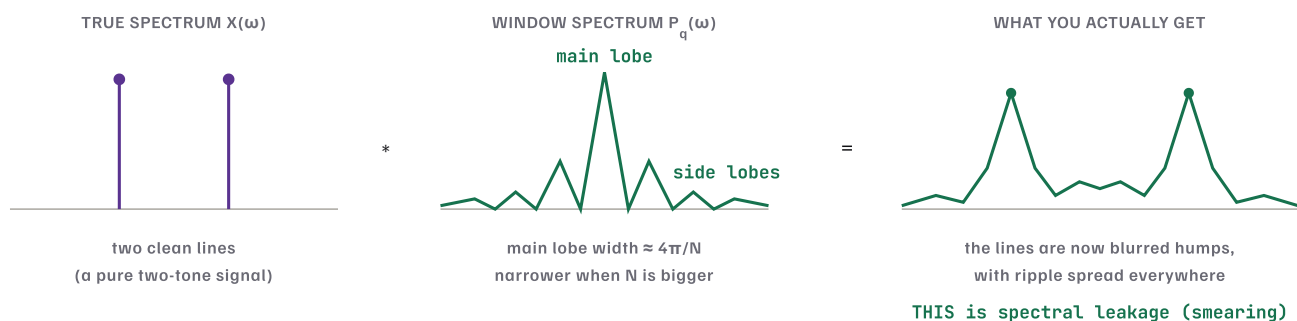
The computer cannot hold infinitely many samples, so from a signal that goes on forever we grab N : $x_N(n) = x(n)$ for $n = 0 \dots N-1$, zero otherwise. We have lost information. The DFT we compute samples the DTFT of the *truncated* signal, $X_N(\omega)$, not the true $X(\omega)$. How do they differ? Truncation is multiplication by a window pulse:

$$x_N(n) = x(n) \cdot P_q(n - q) \quad N = 2q+1 \text{ samples kept} \quad \Rightarrow \quad X_N(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\lambda) P_q(\omega - \lambda) d\lambda$$

multiplication in time = convolution in frequency (§4.4)

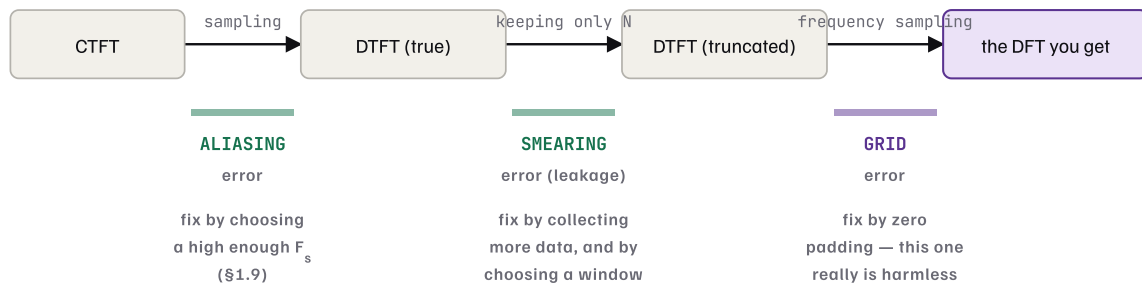
$$P_q(\omega) = \left(\frac{\sin \left[\left(q + \left(\frac{1}{2} \right) \right) \omega \right]}{\sin \left(\frac{\omega}{2} \right)} \right) e^{-jq\omega}$$

a main lobe with side lobes – the pulse spectrum of §4.5



Your DFT samples this smeared curve, not the true one. More data $\rightarrow$ narrower $P_q(\omega) \rightarrow$ less smearing.

Truncation in time is smearing in frequency. Two tones that were sharp come out as humps with ripple between them; if they are close, the humps merge and you cannot tell there were two. The only real cure is more data (a longer record). A better-shaped window reduces the side lobes at the cost of a wider main lobe.



Three steps, three errors, three different cures. Zero padding fixes only the third. Aliasing is fixed before the ADC; smearing is fixed by recording for longer.

ZERO PADDING VERSUS MORE DATA

	ZERO PADDING	COLLECTING MORE DATA
What changes	only the number of dots	the underlying curve
Leakage	unchanged	reduced
Resolution	unchanged	improved
Good for	seeing the shape clearly	actually separating two close tones

ONE LINE TO REMEMBER

Finite signal: the DFT points sit exactly on the DTFT. Infinite signal: they sit exactly on the DTFT of the truncated signal — a smeared version of the one you wanted.

5.10

Cost of the direct DFT

Straight from the definition, an N -point DFT costs N^2 complex multiplications — and N^2 grows horribly.

Teacher module M04L004 §2-3

The DFT is used for linear filtering, correlation and spectrum analysis, so computing it efficiently matters. Two families of methods exist: **divide and conquer**, and **DFT as linear filtering** (§5.11). First, what the direct method costs. Write out one output:

$$X(k) = x(0)W_N^0 + x(1)W_N^k + x(2)W_N^{2k} + \cdots + x(N-1)W_N^{(N-1)k}$$

DIRECT COMPUTATION

per k : N complex multiplications, $N-1$ complex additions
 total: N^2 complex multiplications, $N(N-1)$ complex additions

In *real* arithmetic — one complex multiply is 4 real multiplies and 2 real adds (write $x = x_R + jx_I$ and $W = \cos - j \sin$):

$4N^2$ real multiplications
 $2N(N-1)$ real additions
 $2N^2$ trigonometric evaluations

The trig functions hurt most; every real implementation stores the twiddles in a table.

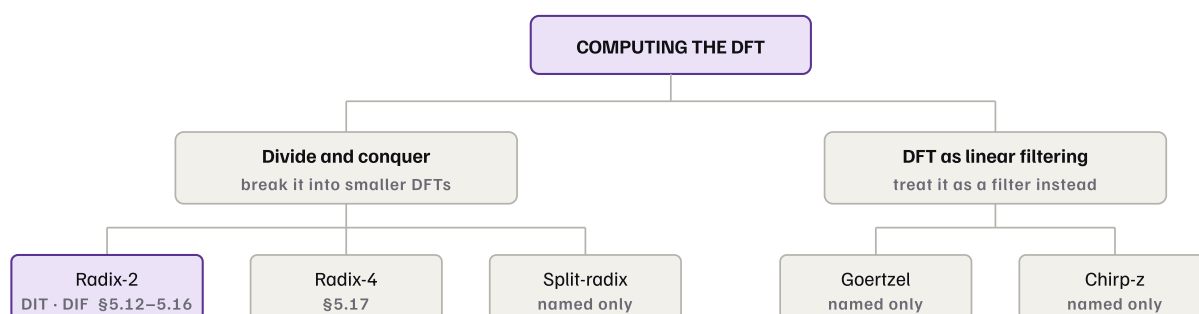
The module's "little trick": to add 1 to 100, pair them off — $1 + 100, 2 + 99, \dots$ — fifty pairs of 101, so $50 \times 101 = 5050$ in one multiplication. The FFT is the same move: the direct DFT ignores the structure in W_N (periodicity, symmetry, half-point — §5.3) and recomputes the same products over and over; the FFT does each piece of work once. **It is not a different transform** — it gives exactly the same $X(k)$.

5.11

Ways to compute the DFT

The family tree from the module, and how much each branch saves.

Teacher module **M04L004** §2, §6



Left branch: you want the **WHOLE** spectrum. Right branch: you want only **PART** of it.

M04L004's tree. The module works through radix-2 (DIT and DIF), composite radix (3 and 4) and the general divide-and-conquer method; the right-hand branch is only named.

N	DIRECT: MULTS N^2	DIRECT: ADDS $N(N-1)$	FFT: MULTS $(N/2)\log_2 N$	FFT: ADDS $N \log_2 N$	SPEED-UP (MULTS)
4	16	12	4	8	4.0
8	64	56	12	24	5.3
16	256	240	32	64	8.0
32	1 024	992	80	160	12.8
64	4 096	4 032	192	384	21.3
128	16 384	16 256	448	896	36.6
256	65 536	65 280	1 024	2 048	64.0
512	262 144	261 632	2 304	4 608	113.8
1 024	1 048 576	1 047 552	5 120	10 240	204.8

(The module's table prints 56.88 for $N = 512$; $262144/2304 = 113.8$.)

Goertzel: one bin, as a filter

The right-hand branch of the tree gets one paragraph, because the syllabus names it and the class has reached it. Write the DFT sum for a single k and multiply it by $W_N^{-kN} = 1$:

$$X(k) = \sum_{m=0}^{N-1} x(m) W_N^{km} = \sum_{m=0}^{N-1} x(m) W_N^{-k(N-m)} = y_k(N) \quad \text{where} \quad y_k(n) = \sum_{m=0}^n x(m) W_N^{-k(n-m)}$$

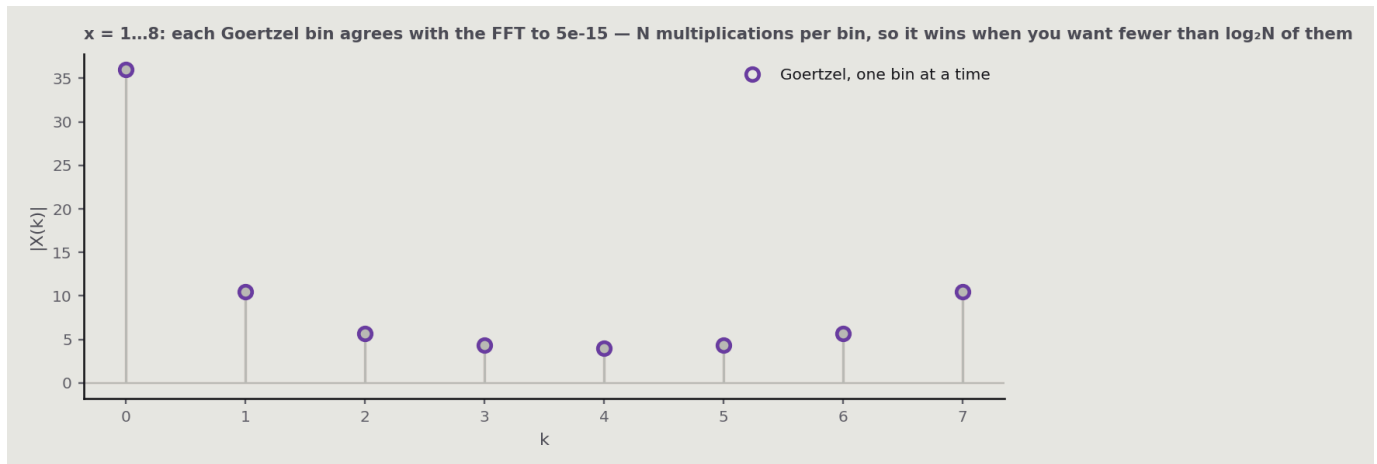
That is a convolution: $y_k(n)$ is the output of a filter with impulse response $h_k(n) = W_N^{-kn} u(n)$ — one complex pole on the unit circle at the bin's frequency — and $X(k)$ is that output **at the instant** $n = N$. Put the pole together with its conjugate to keep the arithmetic real:

$$v_k(n) = x(n) + 2 \cos\left(\frac{2\pi k}{N}\right) v_k(n-1) - v_k(n-2), \quad v_k(-1) = v_k(-2) = 0, \quad n = 0, \dots, N$$

$$X(k) = v_k(N) - W_N^k v_k(N-1)$$

One real multiplication per input sample per bin, and no twiddle table. So Goertzel costs about N multiplications for one bin, against $(N/2)\log_2 N$ for the whole spectrum by FFT. It wins when you want **fewer than about $\log_2 N$ bins** — tone detection, DTMF telephone dialling, one carrier. The **chirp-z transform** is the same idea taken further: it works out the transform along any spiral of points in the z -plane as one convolution, which the FFT can then do quickly.

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Goertzel, checked against the FFT. The same $x = 1 \dots 8$ as the DIT example below, one bin at a time through the two-line recursion. Every ring lands on its stem to machine precision. For $k = 1$ the recursion gives $-4 + 9.657j$, the value the butterfly produces in §5.13.

5.12

Radix-2 DIT

Split the input into even and odd samples, over and over, until only 2-point DFTs are left. Decimation in time.

Teacher module M04L004 §4

The building block: a 2-point DFT

$$X(0) = x(0)W_2^0 + x(1)W_2^0 = x(0) + x(1)$$

$$X(1) = x(0) + x(1)W_2^1 = x(0) - x(1)$$

$$W_2 = e^{-j\pi} = -1: \text{ no multiplications at all}$$

Decomposition

With $N = 2^v$, an N -point DFT is a weighted sum of two $N/2$ -point DFTs — one of the even-index samples, one of the odd-index samples:

$$\begin{aligned}
 X(k) &= \sum_{m=0}^{\frac{N}{2}-1} x(2m) W_N^{2mk} + \sum_{m=0}^{\frac{N}{2}-1} x(2m+1) W_N^{(2m+1)k} \\
 &= \sum_{m=0}^{\frac{N}{2}-1} x(2m) W_{\frac{N}{2}}^{mk} + W_N^k \sum_{m=0}^{\frac{N}{2}-1} x(2m+1) W_{\frac{N}{2}}^{mk}
 \end{aligned}$$

half-point : $W_N^2 = W_{\frac{N}{2}}$

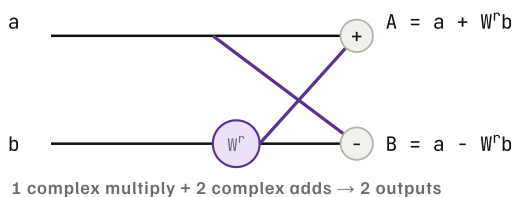
$$X(k) = E(k) + W_N^k O(k)$$

THE BUTTERFLY

$$X(k) = E(k) + W_N^k O(k) \quad X\left(k + \frac{N}{2}\right) = E(k) - W_N^k O(k), k = 0 \dots \frac{N}{2} - 1$$

Only $k = 0 \dots N/2 - 1$ needs evaluating: $E(k)$ and $O(k)$ repeat with period $N/2$, and the symmetry property $W_N^{k+N/2} = -W_N^k$ gives the upper half for free. **One complex multiplication produces two outputs.** Because $N = 2^v$, each $N/2$ -point DFT splits the same way, v times in all.

THE BUTTERFLY



IN-PLACE



A and B overwrite a and b — no extra memory

M04L004's butterfly. Each butterfly reads two values and writes two back to the same addresses — in-place computation (§5.14).

THE COUNTS (M04L004)

$$\text{stages} = \log_2 N$$

$$\text{butterflies per stage} = \frac{N}{2}$$

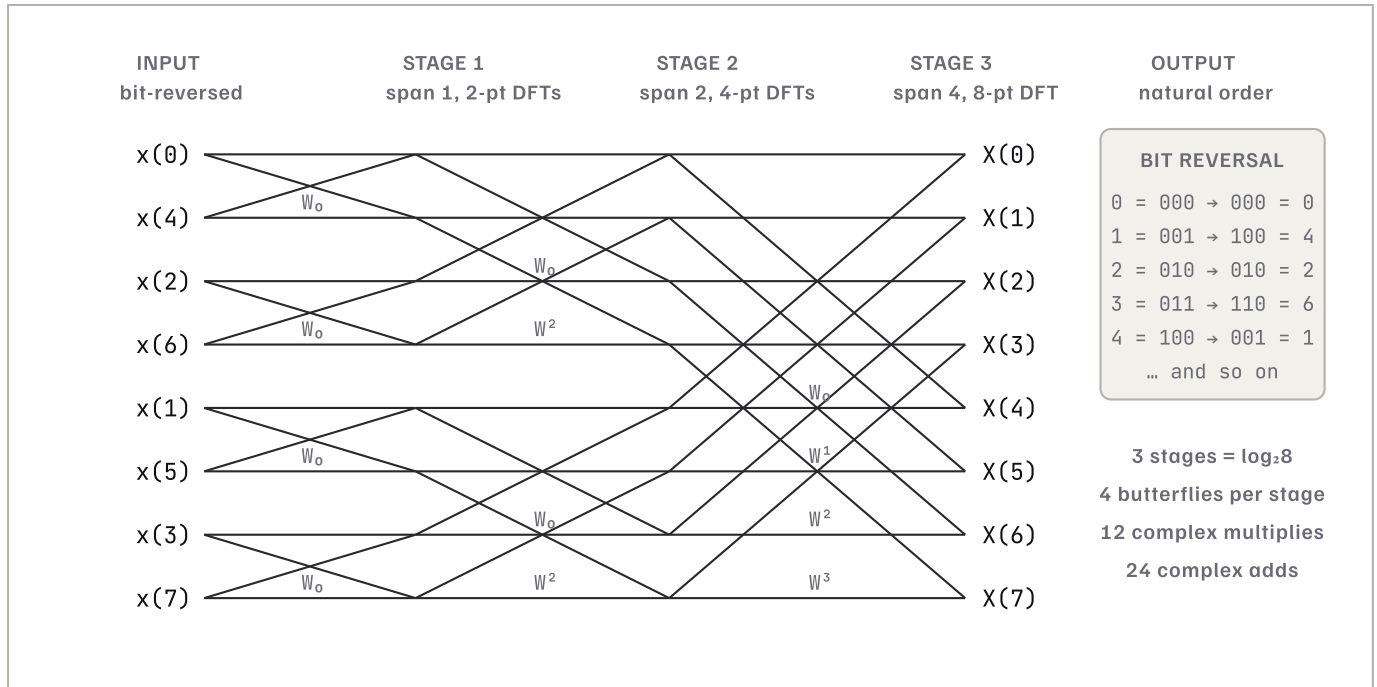
1 complex multiply and 2 complex adds per butterfly

$$\text{complex multiplications} = \log_2 N \times \left(\frac{N}{2}\right) \times 1 = \left(\frac{N}{2}\right) \log_2 N$$

$$\text{complex additions} = \log_2 N \times \left(\frac{N}{2}\right) \times 2 = N \log_2 N$$

Against N^2 and $N(N-1)$ for the direct method.

The 8-point flow graph



Radix-2 DIT, N = 8 (M04L004). Input in bit-reversed order, output in natural order. Each stage applies the twiddle factors and then 2-point DFTs. Read the twiddles across: all W^0 ; then W^0, W^2 ; then $W^0 \dots W^3$ ($W = W_8$) — butterfly span 1, 2, 4.

5.13

DIT worked example

The module's example, stage by stage — and its two self-assignment answers.

Teacher module **M04L004** §5

WORKED

8-point DIT FFT of $x = \{1, 2, 3, 4, 5, 6, 7, 8\}$ (M04L004)

1 Bit-reverse into the order 0, 4, 2, 6, 1, 5, 3, 7: $\{1, 5, 3, 7, 2, 6, 4, 8\}$.

2 Stage 1 (span 1, all twiddles $W^0 = 1$): add and subtract adjacent pairs.

$$(1, 5) \rightarrow 6, -4 \quad (3, 7) \rightarrow 10, -4 \quad (2, 6) \rightarrow 8, -4 \quad (4, 8) \rightarrow 12, -4$$

after stage 1: $\{6, -4, 10, -4, 8, -4, 12, -4\}$

3 Stage 2 (span 2, twiddles $W^0 = 1, W^2 = -j$):

$$\begin{aligned} \text{rows 0, 2: } 6 \pm 10 &\rightarrow 16, -4 & \text{rows 1, 3: } -4 \pm (-j)(-4) \\ &\rightarrow -4 + 4j, -4 - 4j \\ \text{rows 4, 6: } 8 \pm 12 &\rightarrow 20, -4 & \text{rows 5, 7: } -4 \pm (-j)(-4) \\ &\rightarrow -4 + 4j, -4 - 4j \end{aligned}$$

4 Stage 3 (span 4, twiddles $W^0 = 1, W^1 = 0.707 - 0.707j, W^2 = -j, W^3 = -0.707 - 0.707j$):

$$\begin{aligned} X(0) &= 16 + 20 = 36 \\ X(4) &= 16 - 20 = -4 \\ X(1) &= (-4 + 4j) + W^1(-4 + 4j) = -4 + 9.657j \\ X(5) &= (-4 + 4j) - W^1(-4 + 4j) = -4 - 1.657j \\ X(2) &= -4 + (-j)(-4) = -4 + 4j \\ X(6) &= -4 - 4j \\ X(3) &= (-4 - 4j) + W^3(-4 - 4j) = -4 + 1.657j \\ X(7) &= -4 - 9.657j \end{aligned}$$

Checks. $X(0) = \sum x = 36 \checkmark$; $X(4) = 1-2+3-4+5-6+7-8 = -4 \checkmark$; every $X(k)$ pairs with $X(8-k)$ as a conjugate $\checkmark$. ($9.657 = 4 + 4\sqrt{2}$, $1.657 = 4\sqrt{2} - 4$.)

CHECK YOURSELF

M04L004 self-assignments

(I) $x = \{0.707, 1, 0.707, 0, -0.707, -1, -0.707, 0\}$. Notice first that this is $\sin(2\pi n/8 + \pi/4)$ — exactly one cycle in 8 samples — so only bins 1 and 7 can be non-zero:

$$X = \{0, 2.828 - 2.828j, 0, 0, 0, 0, 2.828 + 2.828j\}$$

$$|X(1)| = 4 = N/2 \text{ for a unit sinusoid; check}$$

$$X(0) = \sum x = 0 \text{ — ok}$$

(II) $x = \{1, 1, 1, 1, 0, 0, 0, 0\}$ — the same as §5.5 Example 3:

$$X = \{4, 1 - 2.414j, 0, 1 - 0.414j, 0, 1 + 0.414j, 0, 1 + 2.414j\}$$

(III) 8 samples of $\sin(2\pi ft)$ at $F_s = 50$ Hz with one full cycle in the record ($f = F_s/8 = 6.25$ Hz): $X = \{0, -4j, 0, 0, 0, 0, 0, 4j\}$ — a sine puts $-jN/2$ in bin 1 and $+jN/2$ in bin $N-1$.

5.14

Bit reversal and memory

Write the index in binary, read it backwards — and the whole FFT fits in $5N/2$ memory locations.

Teacher module M04L004 §6

INPUT POSITION	BINARY	BITS REVERSED	SAMPLE PLACED THERE
0	000	000	x(0) stays
1	001	100	x(4) swap 1 ↔ 4
2	010	010	x(2) stays
3	011	110	x(6) swap 3 ↔ 6
4	100	001	x(1) (swapped)
5	101	101	x(5) stays
6	110	011	x(3) (swapped)
7	111	111	x(7) stays

Bit reversal rearranges the input into the DIT order, so that every butterfly in every stage finds its correct pair; the shuffle is a set of swaps, so it too is done in place.

MEMORY REQUIREMENT (M04L004)

- One butterfly: inputs (a, b), outputs (A, B). Once A, B are known, a, b are not needed — store A where a was, B where b was (**in-place**). Four locations: two for a/A, two for b/B.
- One stage: $N/2$ butterflies $\rightarrow 4 \times N/2 = 2N$ locations.
- Stages run one after another and share the same locations, so $2N$ is the total for data.
- Plus $N/2$ twiddle factors per stage.

$$\text{maximum memory} = 2N + \left(\frac{N}{2}\right) = \left(\frac{5N}{2}\right) \text{ locations, twiddles included}$$

5.15

Radix-2 DIF

The same saving, but you split the output into even and odd frequencies instead of the input into even and odd samples.

Teacher module M04L004 §7

Split the sum into first and second halves and use $W_N^{kN/2} = (-1)^k$:

$$\begin{aligned}
 X(k) &= \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{kn} + \sum_{n=\frac{N}{2}}^{N-1} x(n) W_N^{kn} \\
 &= \sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + (-1)^k x\left(n + \frac{N}{2}\right) \right] W_N^{kn}
 \end{aligned}$$

$$\begin{aligned}
 \text{even : } X(2k) &= \sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + x\left(n + \frac{N}{2}\right) \right] W_{\frac{N}{2}}^{kn} \\
 \text{odd : } X(2k+1) &= \sum_{n=0}^{\frac{N}{2}-1} \left\{ \left[x(n) - x\left(n + \frac{N}{2}\right) \right] W_N^n \right\} W_{\frac{N}{2}}^{kn}
 \end{aligned}$$

THE DIF BUTTERFLY

$$A(k) = x(k) + x\left(k + \frac{N}{2}\right) \quad B(k) = \left[x(k) - x\left(k + \frac{N}{2}\right) \right] W_N^k$$

The 2-point DFT comes first and the twiddle multiplies the *difference* branch afterwards.
 Each stage: 2-point DFTs, then twiddles. Same count as DIT — $(N/2) \log_2 N$ multiplications, $N \log_2 N$ additions.

	DIT	DIF
What is split	the input (time samples)	the output (frequency samples)
Input order	bit-reversed	natural
Output order	natural	bit-reversed
Butterfly	twiddle <i>before</i> the add/subtract	twiddle <i>after</i> , on the difference
Butterfly span per stage	1, 2, 4, ...	$N/2$, ..., 2, 1
Cost	identical	

WORKED

8-point DIF FFT of $x = \{2, 1, 2, 1\}$ zero-padded (M04L004)

Input in natural order: $\{2, 1, 2, 1, 0, 0, 0, 0\}$.

1 Stage 1 (span 4): sums on top, differences times W_8^n below.

$$\begin{aligned}\text{top: } x(n) + x(n+4) &= \{2, 1, 2, 1\} \\ \text{bottom: } [x(n) - x(n+4)]W_8^n &= \{2, 0.707 - 0.707j, -2j, -0.707 - 0.707j\}\end{aligned}$$

2 Stage 2 (span 2, twiddles $W_4^n = 1, -j$) inside each half:

$$\begin{aligned}\{2, 1, 2, 1\} &\rightarrow \text{sums } \{4, 2\}, \text{ differences } \{0, 0\} \\ \text{bottom half} &\rightarrow \text{sums } \{2 - 2j, -1.414j\} \\ \text{differences } \{2 + 2j, (0.707 - 0.707j + 0.707 + 0.707j)(-j) = -1.414j\}\end{aligned}$$

3 Stage 3 (span 1): plain add/subtract.

$$\begin{aligned}\{4, 2\} &\rightarrow 6, 2 \\ \{0, 0\} &\rightarrow 0, 0 \\ \{2 - 2j, -1.414j\} &\rightarrow 2 - 3.414j, 2 - 0.586j \\ \{2 + 2j, -1.414j\} &\rightarrow 2 + 0.586j, 2 + 3.414j\end{aligned}$$

4 Un-shuffle — DIF output is bit-reversed, so positions 0...7 hold $X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7)$:

K	0	1	2	3	4	5	6	7
X(k)	6	2 - $j(2+\sqrt{2})$	0	2 + $j(2-\sqrt{2})$	2	2 - $j(2-\sqrt{2})$	0	2 + $j(2+\sqrt{2})$

Checks: $X(0) = 6\checkmark, X(4) = 2-1+2-1 = 2\checkmark$, conjugate pairs $\checkmark$ — matching the module's table.

5.16

The inverse FFT

Same flow graph, conjugate twiddles, divide by N.

Teacher module **M04L004** §8-9

IFFT FROM EITHER GRAPH

Replace every W_N^k by W_N^{-k} and divide the result by N . Structure and cost are unchanged: $(N/2)\log_2 N$ multiplications, $N \log_2 N$ additions.

- **IFFT by DIF:** $X(k)$ in natural order, $x(n)$ out bit-reversed; each stage = 2-point DFTs, then twiddles on the difference branch.
- **IFFT by DIT:** $X(k)$ in bit-reversed order, $x(n)$ out natural; each stage = twiddles, then 2-point DFTs.

Equivalent trick with an unmodified forward routine: $x(n) = (1/N) [\text{FFT}(X^*(k))]^*$.

CHECK YOURSELF

The module's two 8-point IFFT examples

(1)

$X = \{20, 5.828 + 2.414j, 0, -0.172 - 0.414j, 0, -0.172 + 0.414j, 0, 5.828\}$
 $\rightarrow x = \{1, 2, 3, 4, 4, 3, 2, 1\}$. Before you start: $X(0) = 20 = \sum x \checkmark$ and
 $X(4) = 1 - 2 + 3 - 4 + 4 - 3 + 2 - 1 = 0 \checkmark$.

(2)

$X = \{255, 48.63 + 166.05j, -51 + 102j, -78.63 + 46.05j, -85, -78.63 - 46.05j, 51 - 102j, 78.63 - 46.05j\}$. Checks: $\sum x = 255 \checkmark$,
 $X(4) = 1 - 2 + 4 - 8 + 16 - 32 + 64 - 128 = -85 \checkmark$.

5.17

Composite radix and divide-and-conquer

Radix-2, 3 and 4 are one idea with a different number; divide-and-conquer is the general method that works for any N that factorises.

Teacher module **M04L004** §10-11

Composite radix

If $N = m_1 N_1$, split $x(n)$ into m_1 interleaved sub-sequences of N_1 samples by taking every m_1 th sample:

$$X(k) = \sum_{r=0}^{m_1-1} W_N^{rk} \sum_{n=0}^{N_1-1} x(m_1 n + r) W_{N_1}^{nk}$$

$$\text{Radix-3 } (N = 3^v): X(k) = \sum_n x(3n) W_{\frac{N}{3}}^{nk} + W_N^k \sum_n x(3n+1) W_{\frac{N}{3}}^{nk} + W_N^{2k} \sum_n x(3n+2) W_{\frac{N}{3}}^{nk}$$

$$\text{Radix-4 } (N = 4^v): X(k) = \sum_n x(4n) W_{\frac{N}{4}}^{nk} + W_N^k \sum_n x(4n+1) W_{\frac{N}{4}}^{nk} + W_N^{2k} \sum_n x(4n+2) W_{\frac{N}{4}}^{nk} + W_N^{3k} \sum_n x(4n+3) W_{\frac{N}{4}}^{nk}$$

RADIX-3 AND RADIX-4 IN PRACTICE

- **Radix-3:** $\log_3 N$ stages of 3-point butterflies; the 3-point twiddles $1, -\frac{1}{2} - j0.866, -\frac{1}{2} + j0.866$ are not free. Reordering is digit reversal in base 3.
- **Radix-4:** $\log_4 N$ stages of 4-point butterflies; the 4-point twiddles are $1, -j, -1, j$ — sign changes and real/imaginary swaps, *free*. That is why radix-4 needs about $(3N/8)\log_2 N$ multiplications, a quarter fewer than radix-2. Reordering is digit reversal in base 4.
- If N is prime there is nothing to split; pad with zeros to a convenient length (harmless — §5.8).

Divide and conquer — the general method

Take $N = LM$ (L rows, M columns) and store the N samples as an $L \times M$ table. Two natural mappings for the input index, and likewise for the output index:

$$\begin{aligned} \text{row-wise : } n &= Ml + m & \text{column-wise: } n &= l + mL & l &= 0 \dots L-1, m = 0 \dots M-1 \\ \text{row-wise : } k &= Mp + q & \text{column-wise: } k &= p + qL & p &= 0 \dots L-1, q = 0 \dots M-1 \end{aligned}$$

ROW-WISE $n = Ml + m$

$l \backslash m$	0	1	2	$M-1$
0	$x(0)$	$x(1)$	$x(2)$	$x(M-1)$
1	$x(M)$	$x(M+1)$	$x(M+2)$	$x(2M-1)$
	...			

consecutive samples run ACROSS a row

COLUMN-WISE $n = l + mL$

$l \backslash m$	0	1	2	$M-1$
0	$x(0)$	$x(L)$	$x(2L)$	...
1	$x(1)$	$x(L+1)$	$x(2L+1)$	...
	...			

consecutive samples run DOWN a column

Feed the input one way and read the output the other way (M04L004 shows both choices).

Two ways to fold a list into a table. Using one mapping for the input and the other for the output is what makes the twiddle exponent split into separate pieces.

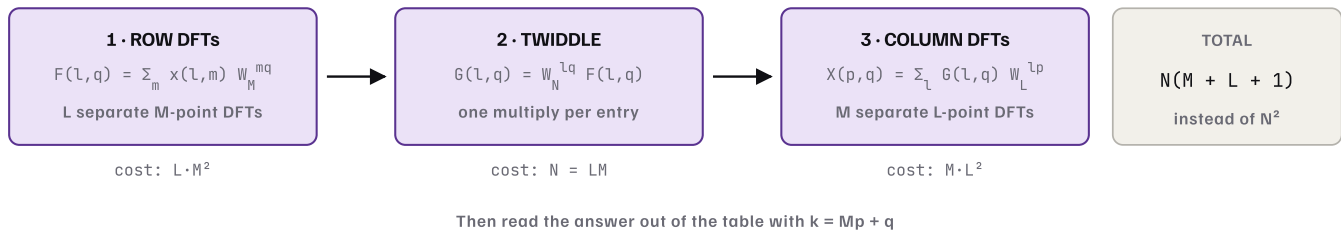
Take the input column-wise ($n = l + mL$) and the output row-wise ($k = Mp + q$) and substitute into the DFT:

$$\begin{aligned} X(p, q) &= \sum_{m=0}^{M-1} \sum_{l=0}^{L-1} x(l, m) W_N^{(Mp+q)(mL+l)} \\ W_N^{(Mp+q)(mL+l)} &= W_N^{MLmp} W_N^{mLq} W_N^{Mpl} W_N^{lq} \\ W_N^{MLmp} &= W_N^{Nmp} = 1 \cdot W_N^{mLq} = W_M^{mq} \cdot W_N^{Mpl} = W_L^{pl} \cdot W_N^{lq} \text{ stays the extra twiddle} \end{aligned}$$

RESULT

$$X(p, q) = \sum_{l=0}^{L-1} \left\{ W_N^{lq} \left[\sum_{m=0}^{M-1} x(l, m) W_M^{mq} \right] \right\} W_L^{lp}$$

One N-point DFT has become DFTs of length M and L, with a twiddle multiplication between.



Row DFTs, twiddle, column DFTs (M04L004). Cost falls from N^2 to $N(M + L + 1)$ complex multiplications and $N(M + L - 2)$ additions.

WORKED How much does one factorisation save?

$N = 1000$: direct cost $N^2 = 1\,000\,000$ multiplications.

$L \times M$	$N(M + L + 1)$	SAVING
2×500	503 000	50 %
10×100	111 000	89 %
20×50	71 000	93 %
25×40	66 000	93.4 %

Best when $L \approx M \approx \sqrt{N}$. Then factorise the M-point DFTs again, and again: for $N = r^v$ this gives $N \log_r N$ behaviour — radix-2 ($L = 2$), radix-3, radix-4 are this method with a fixed L. Radix-2 DIT is $L = 2$, $M = N/2$; radix-2 DIF is $L = N/2$, $M = 2$.

UNIT 6

SYLLABUS UNIT 3 · FIR FILTER DESIGN · CLASS NOTEBOOK · PROAKIS CH. 10

FIR filter design

Until now you were given a system and asked to study it. This unit turns the question around: you are told what the filter must do, and you have to find its coefficients. You will see why FIR filters are the ones that can have a perfectly straight phase, what that symmetry gives you and what it costs, how to cut an ideal filter down to a real one with a window, how to design by sampling the frequency response instead, and what the differentiator, the Hilbert transformer and the "optimum" method are.

6.1

Why FIR, and what “design” means

An FIR filter is a short list of numbers. Design means choosing those numbers so the filter does what you want.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** §4-7

DEFINITION

A **finite impulse response** (FIR) filter has no feedback. The output is a weighted sum of the present input and the last $M-1$ inputs. The weights *are* the impulse response:

$$y(n) = \sum_{k=0}^{M-1} b_k x(n-k), \quad h(n) = \begin{cases} b_n & 0 \leq n \leq M-1 \\ 0 & \text{otherwise} \end{cases}$$

M is the **length** (the number of taps). The **order** is $M-1$. To design the filter is to choose the M numbers $h(0), \dots, h(M-1)$.

Your notebook lists the facts that make FIR filters the first choice when the phase matters. Here they are, with the cost next to each one:

WHAT FIR GIVES YOU	WHAT IT COSTS
Linear phase — if $h(n)$ is symmetric (§6.2). Every frequency is delayed by the same number of samples, so a pulse keeps its shape. No IIR filter can do this exactly.	Length. A sharp cutoff needs many taps — often five to ten times more than an IIR filter with the same transition band.
Always stable. The only poles are at $z = 0$ (there are $M-1$ of them, all at the origin), so a finite input always gives a finite output. Nothing to check.	More multiplications per output sample, and more memory, for the same job.
Easy to design. Cut an ideal response with a window, or sample it. No analog prototype, no bilinear transform.	Some responses (a very narrow band at a very low frequency) need an FIR filter too long to be practical.
Cheap when symmetric. With $h(n) = h(M-1-n)$, mirror taps share a multiplier, so the work is of the order of $M/2$ multiplications.	Round-off errors add up over many taps, though there are no limit cycles (no feedback).

CAUSAL — THE THREE WAYS TO SAY IT (NOTEBOOK)

A real-time filter must be causal. Your notebook gives the three equivalent tests.

Input/output: $y(n)$ depends only on the present and past inputs. **Impulse response:** $h(n) = 0$ for $n < 0$. **System function:** the number of finite zeros is not more than the number of finite poles.

The specification a filter is designed to

No real filter is a brick wall. So the specification allows a small ripple in the passband, a small leftover in the stopband, and a band in between where nothing is promised:

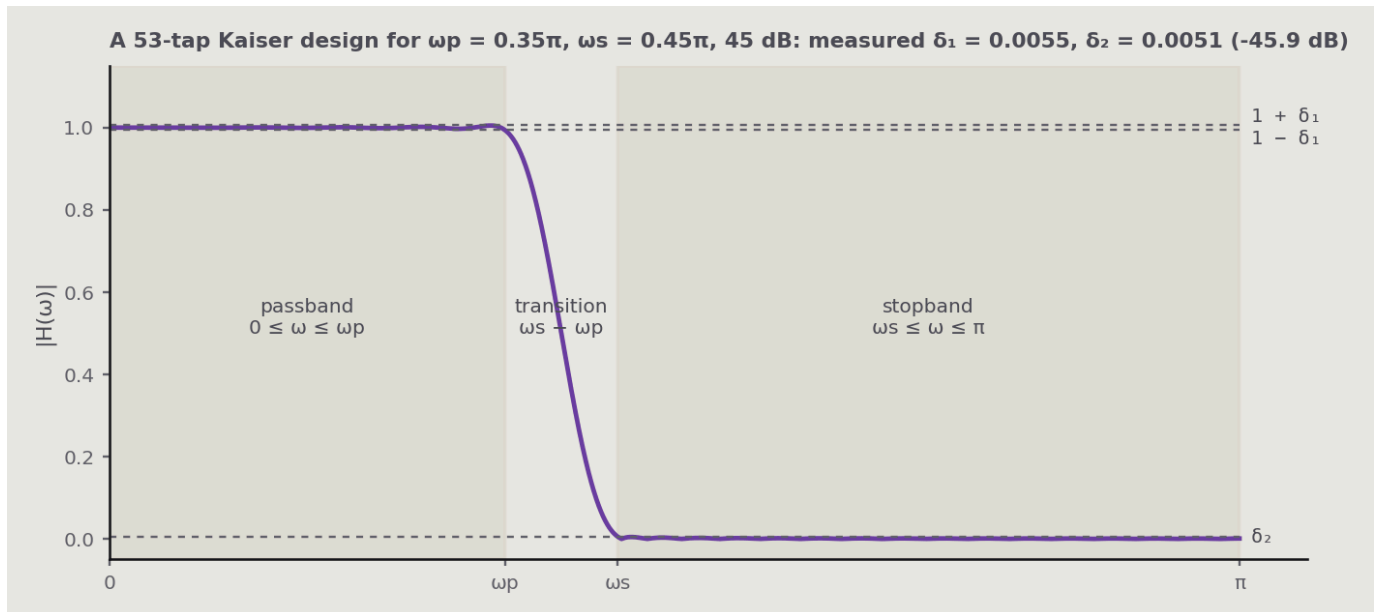
$$\begin{aligned}
 1 - \delta_1 &\leq |H(\omega)| \leq 1 + \delta_1 & 0 \leq \omega \leq \omega_p & \text{ (passband)} \\
 0 &\leq |H(\omega)| \leq \delta_2 & \omega_s \leq \omega \leq \pi & \text{ (stopband)} \\
 \Delta\omega &= \omega_s - \omega_p & & \text{ (the transition band)}
 \end{aligned}$$

δ_1 is the **passband ripple**, δ_2 the **stopband ripple**, ω_p the passband edge, ω_s the stopband edge.

In dB: passband ripple $A_p = 20 \log_{10} [(1+\delta_1)/(1-\delta_1)]$, stopband attenuation

$A_s = -20 \log_{10} \delta_2$.

COMPUTED · TOOLS/MAKE-FIGURES.PY



The specification, read off a real filter. This is a Kaiser-window low-pass designed for $\omega_p = 0.35\pi$, $\omega_s = 0.45\pi$ and 45 dB. The passband ripple δ_1 , the stopband level δ_2 and the transition band were measured from the computed response — this is the drawing in your notebook, with real numbers on it.

6.2

Linear phase and the four symmetric families

Symmetry about the middle tap is what makes the phase a straight line. There are four ways to be symmetric, and they are not interchangeable.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** §10-14

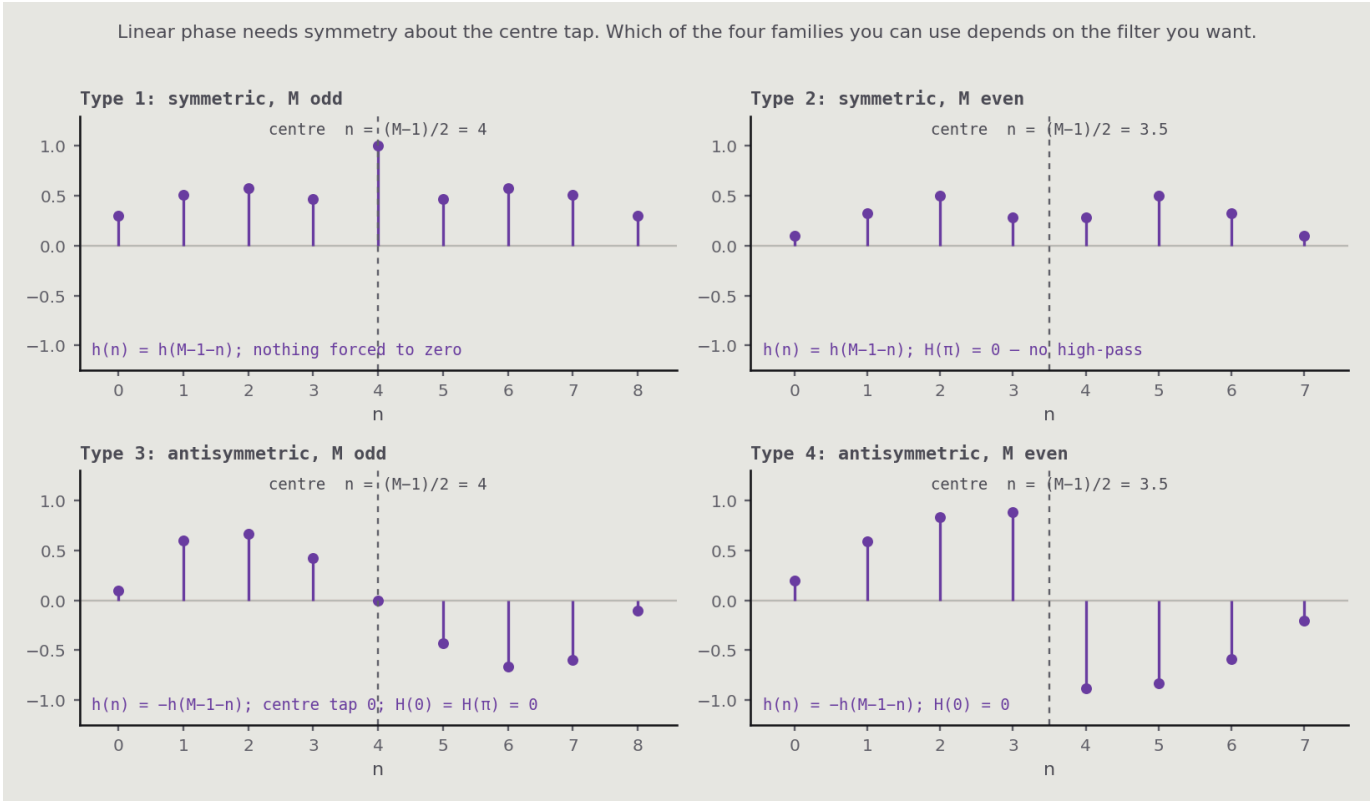
LINEAR PHASE

A filter has **linear phase** when $\angle H(\omega) = -a\omega$: a straight line through the origin (jumps of π are allowed where the magnitude changes sign). Then every sinusoid is delayed by the same a samples — the **group delay** $-d\angle H/d\omega = a$ is constant — so a waveform passes through without changing shape. An FIR filter of length M has linear phase exactly when

$$h(n) = \pm h(M-1-n), \quad n = 0, 1, \dots, M-1, \quad \text{delay } \alpha = \frac{M-1}{2} \text{ samples}$$

Plus sign: **symmetric**. Minus sign: **antisymmetric**. The centre of symmetry is $n = (M-1)/2$. It is a tap when M is odd and falls between two taps when M is even.

COMPUTED · TOOLS/MAKE-FIGURES.PY



The four linear-phase families. Symmetric or antisymmetric, length odd or even. The dashed line is the centre $(M-1)/2$. In the antisymmetric odd case the centre tap has to be zero. The lines written on each panel — which of $H(0)$ and $H(\pi)$ is forced to zero — follow from the symmetry alone, and they decide which family can build which filter.

TYPE	SYMMETRY	M	FORCED ZEROS	CAN BUILD	CANNOT BUILD
1	symmetric	odd	none	anything: LP, HP, BP, BS	—
2	symmetric	even	$H(\pi) = 0$	LP, BP	HP, BS (they need gain at π)
3	antisymmetric	odd	$H(0) = H(\pi) = 0$	BP, differentiator, Hilbert transformer	LP, HP, BS
4	antisymmetric	even	$H(0) = 0$	HP, BP, differentiator, Hilbert	LP, BS

THE QUESTION THAT CATCHES PEOPLE

“Design a linear-phase high-pass filter of even length.” A symmetric filter of even length has $H(\pi) = 0$ exactly — it is a low-pass no matter what you do. Say so, and then use odd M (Type 1) or an antisymmetric even M (Type 4). The forced zeros are the whole point of the table.

Why the forced zeros happen: in a symmetric even-length filter the taps pair up as $h(n) [z^{-n} + z^{-(M-1-n)}]$. At $z = -1$ the two powers have opposite signs (the exponents differ by the odd number $M-1-2n$), so each pair cancels and $H(\pi) = 0$. In an antisymmetric filter $H(1) = \sum h(n) = 0$, because the taps cancel in pairs at $z = 1$.

6.3

Frequency response of a linear-phase FIR filter

Pair up the taps that mirror each other. Each pair becomes a cosine (or a sine). The whole response is a real sum of cosines times a pure delay.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** §12-14, 29

Take the symmetric case with M odd. Write $\tau = (M-1)/2$. Take the middle tap out on its own, and pair every other $h(n)$ with its mirror $h(M-1-n)$:

$$\begin{aligned}
 H(z) &= \sum_{n=0}^{M-1} h(n) z^{-n} = h(\tau) z^{-\tau} + \sum_{n=0}^{\tau-1} h(n) [z^{-n} + z^{-(M-1-n)}] \\
 z^{-n} + z^{-(M-1-n)} &= z^{-\tau} [z^{\tau-n} + z^{-(\tau-n)}] \quad \begin{matrix} z=e^{j\omega} \\ e^{-j\omega\tau} \cdot 2 \cos(\omega(\tau-n)) \end{matrix} \\
 H(e^{j\omega}) &= e^{-j\omega\tau} \left[h(\tau) + 2 \sum_{n=0}^{\tau-1} h(n) \cos(\omega(\tau-n)) \right]
 \end{aligned}$$

READ IT IN TWO PARTS

The part in brackets is **real**. Call it $H_r(\omega)$, the amplitude. The factor in front is a **pure delay** of τ samples. So

$$|H(\omega)| = |H_r(\omega)|, \quad \angle H(\omega) = \begin{cases} -\omega\tau & H_r(\omega) > 0 \\ -\omega\tau + \pi & H_r(\omega) < 0 \end{cases}$$

The phase is the straight line $-\omega\tau$, with a jump of π wherever the amplitude crosses zero. That is what “linear phase” means in practice: a slope of $-\tau$ and nothing else. Your notebook puts it as: in the stopband $H(\omega)$ can be 0, and the phase there is 0 or 180°.

FAMILY	$H(e^{j\Omega})$	AMPLITUDE $H_r(\Omega)$
Type 1, symmetric, M odd	$e^{-j\omega\tau} H_r(\omega)$	$h(\tau) + 2 \sum_{n=0}^{\tau-1} h(n) \cos \omega(\tau-n)$
Type 2, symmetric, M even	$e^{-j\omega\tau} H_r(\omega)$	$2 \sum_{n=0}^{M/2-1} h(n) \cos \omega(\tau-n)$ — zero at $\omega = \pi$
Type 3, antisymmetric, M odd	$j e^{-j\omega\tau} H_r(\omega)$	$2 \sum_{n=0}^{\tau-1} h(n) \sin \omega(\tau-n)$ — zero at 0 and π
Type 4, antisymmetric, M even	$j e^{-j\omega\tau} H_r(\omega)$	$2 \sum_{n=0}^{M/2-1} h(n) \sin \omega(\tau-n)$ — zero at 0

The antisymmetric families carry an extra j . That is a fixed 90° phase shift on top of the delay — exactly what a differentiator and a Hilbert transformer need (§6.11).

WORKED**The 11-tap low-pass written as a sum of cosines (slides p. 29)**

Take

$h(n) = \{0.0637, 0, -0.1061, 0, 0.3183, 0.5, 0.3183, 0, -0.1061, 0, 0.0637\}$
the filter designed in §6.8(a). $M = 11$, $\tau = 5$, symmetric, so

$$\begin{aligned} H_r(\omega) &= h(5) + 2 \sum_{n=0}^4 h(n) \cos \omega(5 - n) \\ &= 0.5 + 2(0.0637) \cos 5\omega + 2(0) \cos 4\omega + 2(-0.1061) \cos 3\omega + 2(0) \cos 2\omega + 2(0.3183) \cos \omega \\ &= 0.5 + 0.636 \cos \omega - 0.212 \cos 3\omega + 0.127 \cos 5\omega \end{aligned}$$

At $\omega = 0$ this is $0.5 + 0.636 - 0.212 + 0.127 = 1.052$. At $\omega = \pi/2$ every cosine of an odd multiple is zero, so $H_r = 0.5$ exactly — -6.02 dB. A windowed low-pass always passes half the amplitude at its cutoff. (The slide prints 1.0151 at $\omega = 0$; the digits are swapped, the sum is 1.052.) The phase is -5ω : five samples of delay at every frequency.

6.4

Where the zeros of a linear-phase filter sit

Symmetry forces the zeros into mirror-image sets: inside and outside the unit circle in reciprocal pairs, and above and below the real axis in conjugate pairs.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf**

Symmetry $h(n) = h(M-1-n)$ means the polynomial reads the same backwards. In the z -domain that is

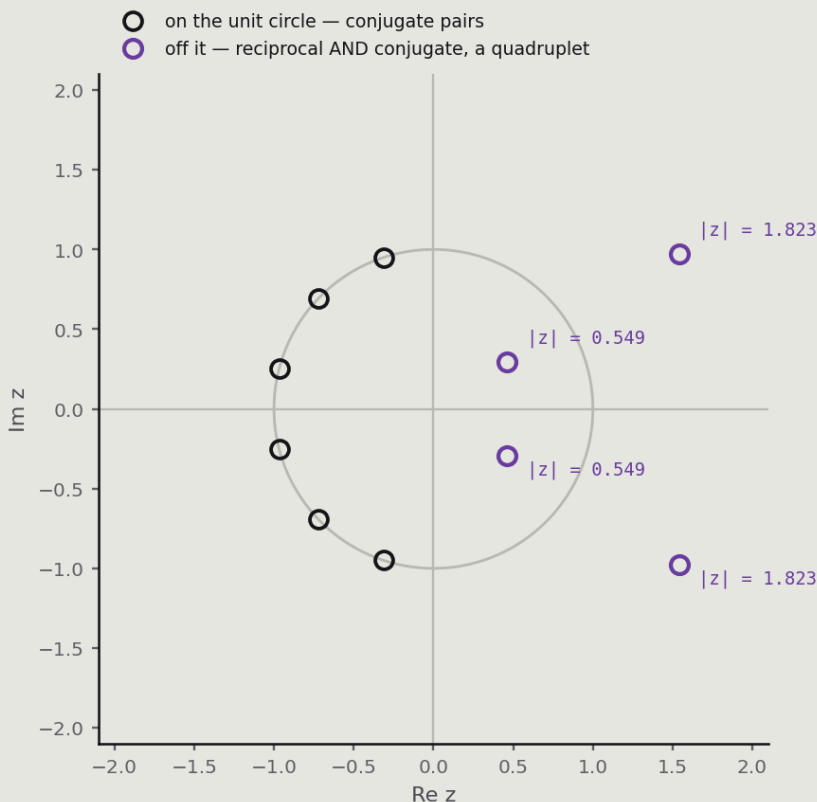
$$H(z) = z^{-(M-1)} H(z^{-1})$$

so if z_0 is a zero, so is $1/z_0$. Real coefficients also give the conjugate z_0^* of every zero. Put the two together:

- a complex zero off the unit circle comes with three partners — z_0 , $1/z_0$, z_0^* , $1/z_0^*$ — a group of four (a **quadruplet**);
- a complex zero *on* the unit circle is its own reciprocal, so it comes only as a **conjugate pair**;
- a real zero off the circle comes with its reciprocal: a pair at r and $1/r$;
- a zero at $z = 1$ or $z = -1$ can stand alone. (These are the forced zeros of Types 2, 3 and 4.)

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Zeros of the 11-tap low-pass, $\omega_c = \pi/2$ (rectangular window). Symmetric $h(n)$ means $H(z) = z^{-(M-1)} H(1/z)$: every zero at z comes with one at $1/z$, and real coefficients add the conjugate of each.



The zeros of the 11-tap low-pass, found with `np.roots`. Six zeros sit on the unit circle in the stopband, in conjugate pairs — they are what keeps the stopband small. Four more form a group at $|z| = 0.549$ and $|z| = 1/0.549 = 1.822$, at $\pm 0.18\pi$: each inside zero has its reciprocal outside. Every linear-phase FIR filter has a zero pattern like this.

WHAT THE EXAM WANTS HERE

“State the location of zeros of a linear-phase FIR filter” wants the two symmetries and what follows from them: zeros in reciprocal pairs (from the symmetric coefficients) and in conjugate pairs (from the real coefficients), so groups of four in general, pairs on the unit circle, and single zeros at ± 1 . Add this: a linear-phase FIR filter can never be minimum-phase unless all its zeros are on the unit circle, because every inside zero has a partner outside.

6.5

The window method, step by step

Write down the ideal response. Turn it back into $h_d(n)$, which is infinitely long. Keep M samples of it, shifted so they start at $n = 0$.

This is also called the **Fourier series method**: the $h_d(n)$ are the Fourier coefficients of the periodic function $H_d(\omega)$, and cutting the series short is what a window does.

THE IDEAL LOW-PASS, AS YOUR NOTEBOOK STARTS IT

$H_d(\omega) = 1$ for $|\omega| \leq \omega_c$ and 0 for $\omega_c < |\omega| \leq \pi$. Its impulse response is the inverse DTFT:

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \begin{cases} \frac{\omega_c}{\pi} & n = 0 \\ \frac{\omega_c}{\pi} \cdot \frac{\sin \omega_c n}{\omega_c n} & n \neq 0 \end{cases}$$

a sinc that runs from $-\infty$ to $+\infty$. Two problems: it never ends, and it is not causal. The window fixes the first. Shifting it right by τ samples fixes the second — and the shift is exactly the linear phase $e^{-j\omega\tau}$. (Your notebook: “due to the linear phase, $h(n)$ is shifted by n_d delay”.)

- 1 **Write the desired response** $H_d(e^{j\omega})$ with the delay term $e^{-j\omega\tau}$, $\tau = (M-1)/2$, so that the answer comes out causal.

- 2 **Invert it** with the DTFT integral to get $h_d(n)$. For a low-pass with cutoff ω_c :

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\tau} e^{j\omega n} d\omega = \frac{\sin \omega_c(n - \tau)}{\pi(n - \tau)}, \quad h_d(\tau) = \frac{\omega_c}{\pi}$$

At $n = \tau$ the formula gives 0/0. L'Hôpital's rule (or $\sin x/x \rightarrow 1$) gives ω_c/π .

- 3 **Cut it to M samples** by multiplying with a window $w(n)$ that is non-zero only for $0 \leq n \leq M-1$: $h(n) = h_d(n) w(n)$. With the rectangular window this just means keeping $h_d(0) \dots h_d(M-1)$.
- 4 **Write** $H(z) = \sum h(n) z^{-n}$, pairing the mirror taps as in §6.3.
- 5 **Find the magnitude** $|H(e^{j\omega})| = |H_r(\omega)|$ and the phase $-\omega\tau$. Make a table at $\omega = 0, 0.1\pi, \dots, \pi$, or plot it — the exam usually asks for the table.

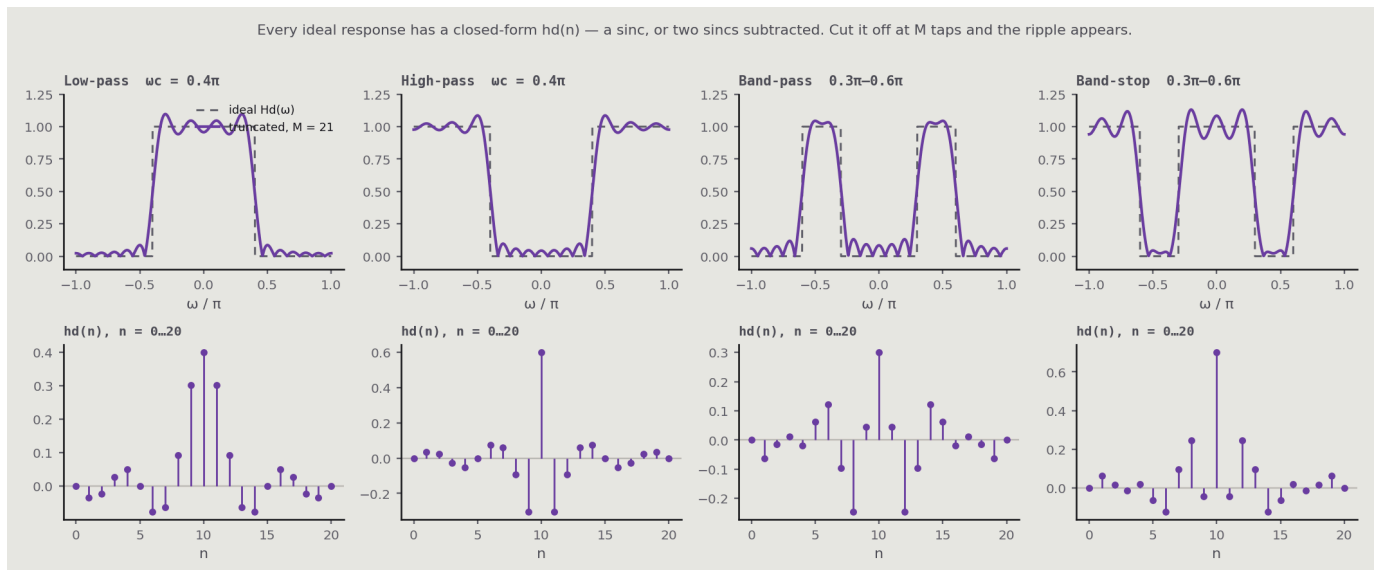
The ideal responses and their $h_d(n)$

Each one is the integral of step 2 over a different band. Each comes out as a sinc, or one sinc minus another. With $\tau = (M-1)/2$ and $n \neq \tau$:

FILTER	$H_D(\Omega) = E^{-j\Omega T}$ ON	$H_D(N), N \neq T$	$H_D(T)$
Low-pass	$ \omega \leq \omega_c$	$\sin \omega_c(n-\tau) / \pi(n-\tau)$	ω_c/π
High-pass	$\omega_c \leq \omega \leq \pi$	$[\sin \pi(n-\tau) - \sin \omega_c(n-\tau)] / \pi(n-\tau)$	$1 - \omega_c/\pi$
Band-pass	$\omega_1 \leq \omega \leq \omega_2$	$[\sin \omega_2(n-\tau) - \sin \omega_1(n-\tau)] / \pi(n-\tau)$	$(\omega_2 - \omega_1)/\pi$
Band-stop	everything except $\omega_1 \leq \omega \leq \omega_2$	$[\sin \omega_1(n-\tau) - \sin \omega_2(n-\tau) + \sin \pi(n-\tau)] / \pi(n-\tau)$	$1 - (\omega_2 - \omega_1)/\pi$

Note that $\sin \pi(n-\tau) = 0$ whenever $n-\tau$ is a whole number, which is whenever M is odd. So for odd M the high-pass is simply $\delta(n-\tau)$ minus the low-pass, and the band-stop is $\delta(n-\tau)$ minus the band-pass. That is the quick way to remember them.

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The four ideal responses, and what 21 taps of each actually give. Top: the ideal (dashed) and the 21-tap version (ink). Bottom: $h_d(n)$ for $n = 0 \dots 20$, each a sinc centred on $\tau = 10$. The ripple next to every edge and the sloping transition are the price of cutting the sinc short. §6.6 says why, §6.7 what to do about it.

6.6

Gibbs, and why the window must taper

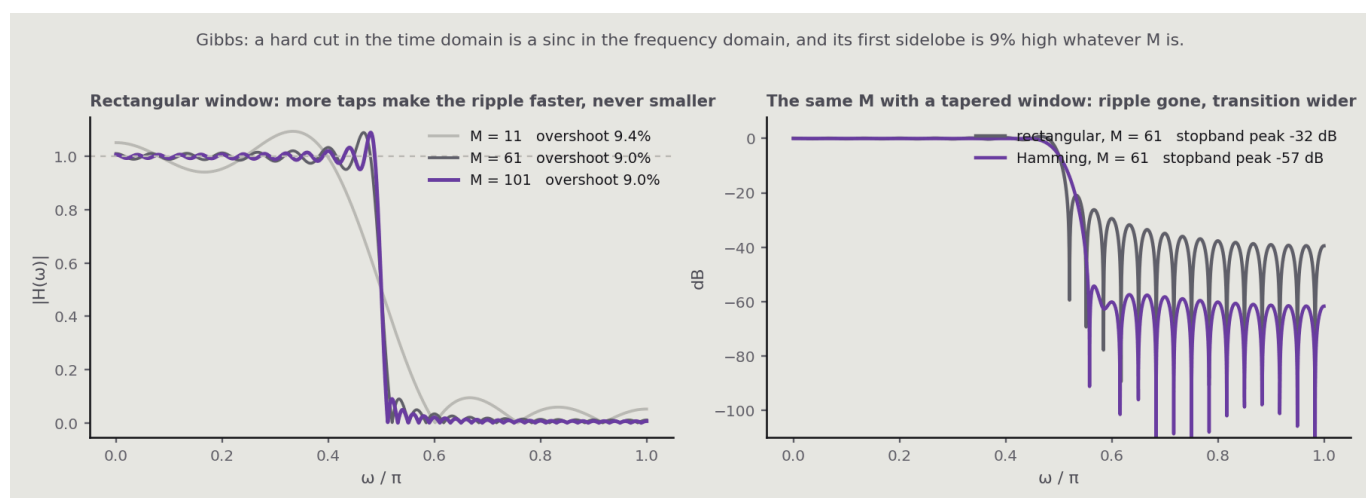
Cutting the series short is multiplying by a box in time. In frequency that is convolving with a sinc. The sinc's side lobes are the ripple, and they do not get smaller when M grows.

Multiplying in time is convolving in frequency (§4.4). So windowing does this to the ideal response:

$$H(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\theta) W(\omega - \theta) d\theta \quad \text{with, for the rectangular window,} \quad |W_R(\omega)| = \left| \frac{\sin(\omega M/2)}{\sin(\omega/2)} \right|$$

$W_R(\omega)$ has a main lobe of width $4\pi/M$ and a first side lobe only **13 dB** down. Sliding this across the sharp edge of the ideal response smears the edge (the main lobe sets the transition width) and pushes the side lobes into the passband and stopband as ripple. A bigger M makes the main lobe narrower, so the transition gets sharper. But the side-lobe *height* is fixed by the shape of the box. So the overshoot at the edge stays at about 9% however long the filter is. This is the **Gibbs phenomenon**.

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Gibbs, measured. Left: the same ideal cutoff cut to 11, 61 and 101 taps with a rectangular window. The ripples get faster and crowd towards the edge, but the overshoot stays at 9% — the legend shows it measured on each curve. Right: at $M = 61$ the Hamming window gives a stopband floor near -53 dB instead of -21, and pays for it with a transition band twice as wide.

THE RULE

A window that **tapers smoothly to zero** has smaller side lobes and a wider main lobe. So every tapered window trades transition width for less ripple. Choose the window from the stopband attenuation you need. Then choose M from the transition width you can afford (§6.9).

6.7

The window family

Five fixed shapes and one adjustable one. The one number to know about each is how far down its side lobes are.

All of them are defined for $0 \leq n \leq M-1$ and are zero outside. The raised-cosine ones are $a - b \cos(2\pi n/(M-1))$ with different a , b :

Rectangular: $w(n) = 1$

Bartlett (triangular): $w(n) = 1 - \frac{2|n - \frac{M-1}{2}|}{M-1}$

Hann (Hanning): $w(n) = 0.5 - 0.5 \cos \frac{2\pi n}{M-1}$

Hamming: $w(n) = 0.54 - 0.46 \cos \frac{2\pi n}{M-1}$

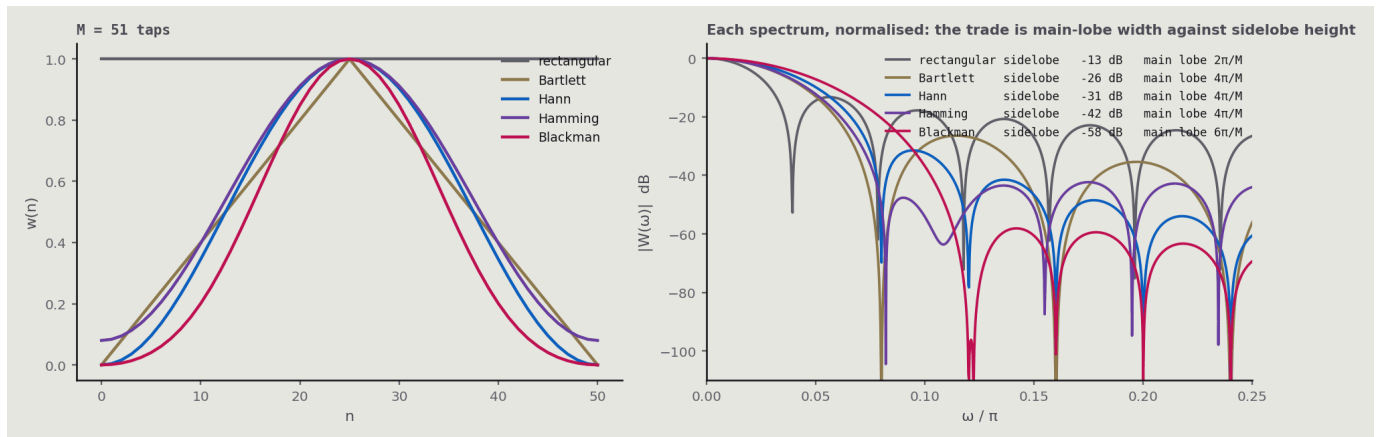
Blackman: $w(n) = 0.42 - 0.5 \cos \frac{2\pi n}{M-1} + 0.08 \cos \frac{4\pi n}{M-1}$

Kaiser: $w(n) = \frac{I_0\left(\beta \left(1 - \left(\frac{2n}{M-1} - 1\right)^2\right)\right)}{I_0(\beta)}$

WINDOW	MAIN-LOBE WIDTH	PEAK SIDE LOBE	MIN. STOPBAND ATTENUATION OF THE FILTER
Rectangular	$4\pi/M$	−13 dB	−21 dB
Bartlett	$8\pi/M$	−25 dB	−25 dB
Hann	$8\pi/M$	−31 dB	−44 dB
Hamming	$8\pi/M$	−41 dB	−53 dB
Blackman	$12\pi/M$	−57 dB	−74 dB
Kaiser	set by β	set by β	set by β

The two dB columns are different things, and the exam asks for both. **Peak side lobe** is about the window's own spectrum. **Minimum stopband attenuation** is what the *filter* made with that window reaches. It is better than the side lobe, because the side lobes partly cancel when they are added up across the edge. (The slides' table prints −21 dB in both columns for the rectangular window. Its side lobe is −13 dB, as the figure below measures.)

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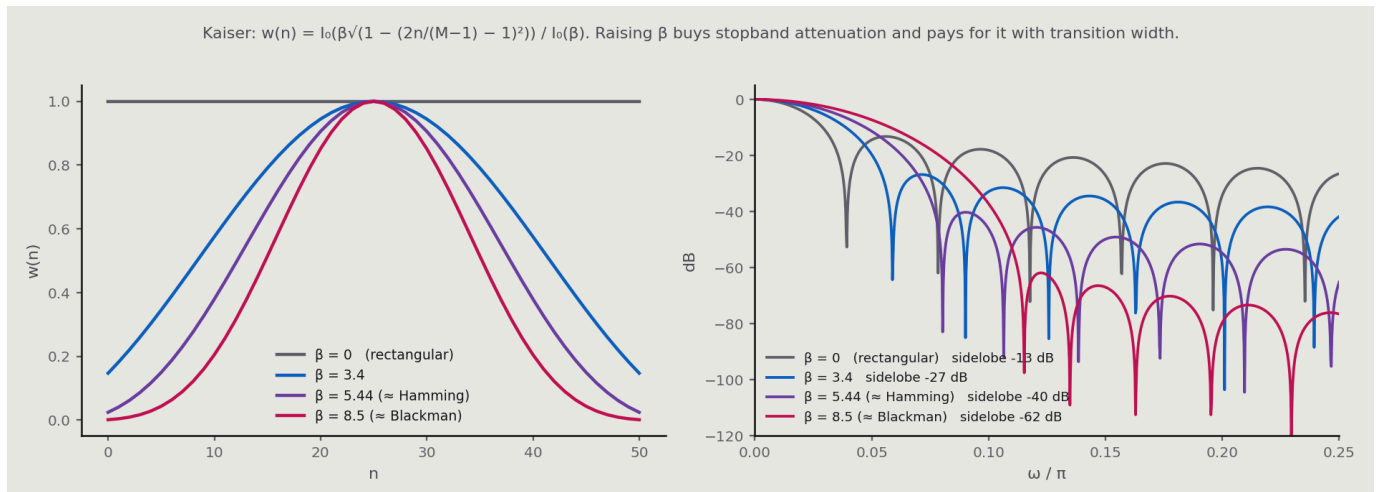


The family, computed at $M = 51$. Left, the shapes. Right, each window's spectrum with its highest side lobe measured: -13, -26, -32, -42 and -58 dB. The main lobe gets wider in the same order — you cannot have both.

Kaiser: one window, one knob

I_0 is the modified Bessel function of order zero, $I_0(x) = 1 + \sum_{k \geq 1} [(x/2)^k / k!]^2$. The parameter β (the slides call it α) sets the trade-off smoothly: $\beta = 0$ is the rectangular window, $\beta \approx 5.44$ behaves like Hamming, $\beta \approx 8.5$ like Blackman. Because β and M can be chosen separately, the Kaiser window is the only one in the family with a proper **recipe** from a specification — §6.9.

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The Kaiser window for four values of β . The side-lobe level falls smoothly as β rises, and the main lobe widens with it. The values 5.44 and 8.5 land on the Hamming and Blackman side-lobe levels, which is why those two are used as reference points.

6.8

Worked designs

Four filters, done the way the exam wants: the ideal response, the coefficients, the window, the product, and the magnitude at a row of frequencies.

WORKED**(a) Low-pass, rectangular window,** $\omega_c = \pi/2$, $M = 11$ (slides p. 27–30) – $\tau = (11-1)/2 = 5$. From the table in §6.5,

$$h_d(n) = \frac{\sin \frac{\pi}{2}(n-5)}{\pi(n-5)}, \quad h_d(5) = \frac{\omega_c}{\pi} = \frac{1}{2}$$

Every even distance from the centre gives $\sin(k\pi) = 0$, so half the taps are zero:

N	0	1	2	3	4	5	6	7	8	9	10
$h(n)$	0.0637	0	-0.1061	0	0.3183	0.5	0.3183	0	-0.1061	0	0.0637

The rectangular window changes nothing, so $h(n) = h_d(n)$. By §6.3, $|H(\omega)| = |0.5 + 0.636 \cos \omega - 0.212 \cos 3\omega + 0.127 \cos 5\omega|$, and in dB:

Ω	0	0.1π	0.2π	0.3π	0.4π	0.5π	0.6π	0.7π	0.8π	0.9π	π
$ H $	1.052	0.981	0.953	1.076	0.996	0.500	0.004	0.076	0.047	0.019	0.052
dB	0.44	-0.17	-0.42	0.64	-0.04	-6.02	-47.4	-22.4	-26.6	-34.3	-25.7

Read the row: a passband ripple of about ± 0.6 dB, exactly half at the cutoff, and a stopband that never gets below about -22 dB except at the accidental null near 0.6π . That -21-ish dB floor is the mark of the rectangular window.

WORKED**(b) Low-pass,
Hamming window,**

$$\omega_c = 3\pi/4, \quad M = 7$$

**(June 2015, Dec 2014,
June 2012)**

$\tau = 3$. $h_d(n) = \sin[3\pi(n-3)/4] / \pi(n-3)$, $h_d(3) = 3/4$. The Hamming window is $w(n) = 0.54 - 0.46 \cos(2\pi n/6)$:

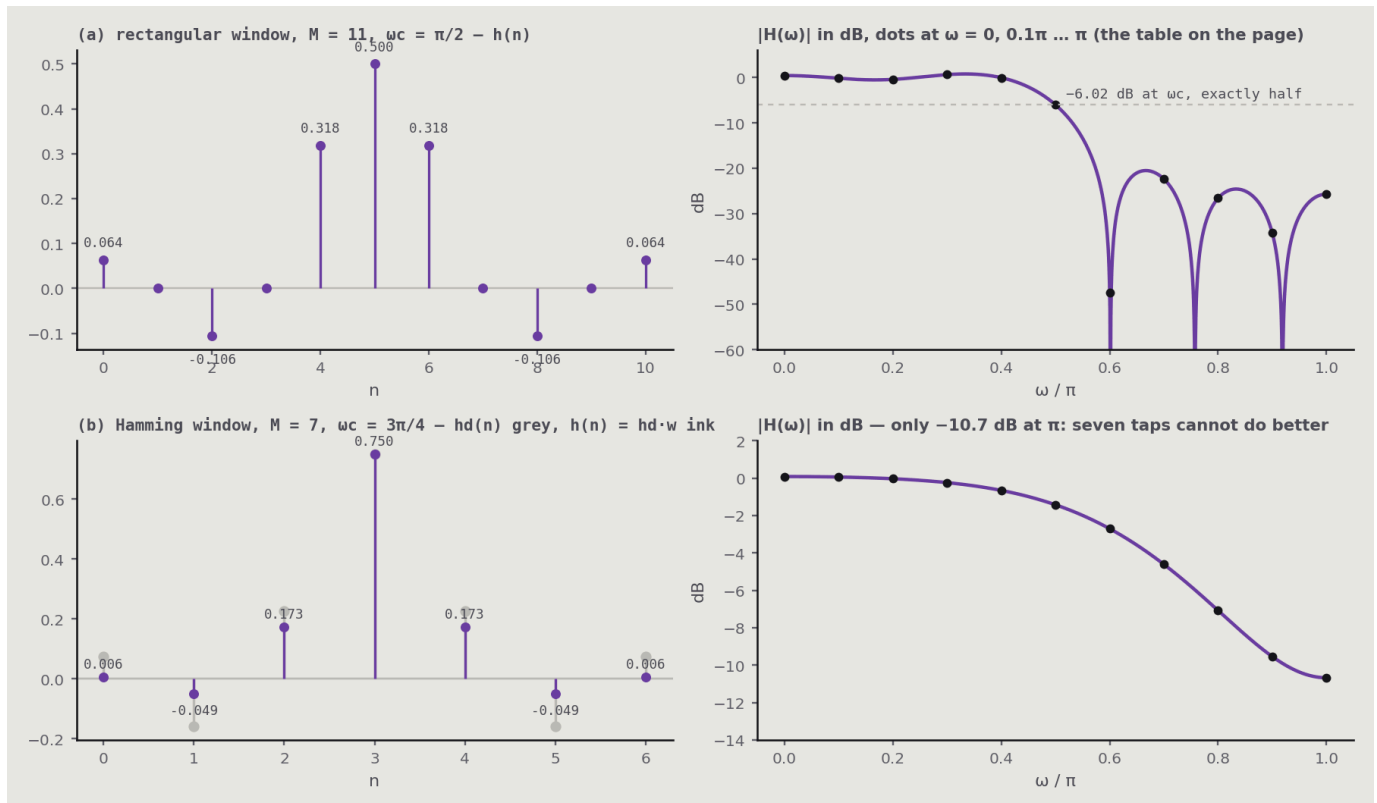
N	0	1	2	3	4	5	6
$h_d(n)$	0.0750	-0.1592	0.2251	0.7500	0.2251	-0.1592	0.0750
$w(n)$	0.08	0.31	0.77	1.00	0.77	0.31	0.08
$h(n) = h_d w$	0.006	-0.049	0.173	0.750	0.173	-0.049	0.006

$$|H(\omega)| = |0.75 + 0.346 \cos \omega - 0.098 \cos 2\omega + 0.012 \cos 3\omega|$$

Ω	0	0.1π	0.2π	0.3π	0.4π	0.5π	0.6π	0.7π	0.8π	0.9π	π
$ H $	1.010	1.007	0.996	0.973	0.927	0.849	0.732	0.588	0.443	0.334	0.293
dB	0.09	0.06	-0.03	-0.24	-0.66	-1.43	-2.70	-4.61	-7.08	-9.54	-10.67

Seven taps with a Hamming window give a transition band about $8\pi/7 \approx 1.14\pi$ wide — wider than the whole axis. So this “filter” only reaches -10.7 dB at π . The question is testing the method, not making a useful filter. Say that in one line and you have shown you understand the table in §6.7. (The slide prints 1.9722 at 0.3π ; it is 0.9722.)

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Designs (a) and (b), computed. Left, the coefficients. Right, $|H(\omega)|$ in dB with a dot at each of the eleven frequencies in the tables. In (a) the response passes through -6.02 dB at ω_c exactly and the stopband floor sits near -22 dB. In (b) the grey stems are $h_d(n)$ before the window and the ink stems after it; the window has pulled the outer taps almost to zero.

WORKED (c) High-pass, Hamming window, $\omega_c = 1$ rad/sample, $M = 7$ (slides p. 55–57) -

$\tau = 3$. From the high-pass row of the table:

$$h_d(n) = \frac{\sin \pi(n-3) - \sin(n-3)}{\pi(n-3)} = -\frac{\sin(n-3)}{\pi(n-3)} \quad (n \neq 3), \quad h_d(3) = 1 - \frac{1}{\pi} = 0.6817$$

($\sin \pi(n-3)$ is zero for every whole number n — the “delta minus low-pass” form.)

Multiply by the same Hamming window as in (b):

N	0	1	2	3	4	5	6
$h_d(n)$	-0.0150	-0.1447	-0.2679	0.6817	-0.2679	-0.1447	-0.0150
$h(n)$	-0.0012	-0.0449	-0.2062	0.6817	-0.2062	-0.0449	-0.0012

$$|H(\omega)| = |0.6817 - 0.4124 \cos \omega - 0.0898 \cos 2\omega - 0.0024 \cos 3\omega|$$

Check the two ends. At $\omega = 0$ all the cosines are 1 and the sum is

$0.6817 - 0.5046 = 0.177$ (small, as a high-pass should be). At $\omega = \pi$ the signs alternate and it is $0.6817 + 0.4124 - 0.0898 + 0.0024 = 1.007$. A symmetric odd-length filter (Type 1) is allowed to have gain at π ; an even-length one would not have been.

WORKED (d) Band-pass, rectangular window, $\omega_{c1} = 1$, $\omega_{c2} = 2$ rad/sample, $M = 7$ (slides p. 58) -

$$h_d(n) = \frac{\sin 2(n-3) - \sin(n-3)}{\pi(n-3)} \quad (n \neq 3), \quad h_d(3) = \frac{2-1}{\pi} = 0.3183$$

N	0	1	2	3	4	5	6
$h(n)$	-0.0446	-0.2652	0.0216	0.3183	0.0216	-0.2652	-0.0446

Rectangular window, so these are the final coefficients. The band-stop with the same edges is $\delta(n-3)$ minus this:

$\{0.0446, 0.2652, -0.0216, 0.6817, -0.0216, 0.2652, 0.0446\}$. Every tap flips sign and the centre becomes $1 - 0.3183$.

WHERE MARKS ARE LOST

Three slips cause most of the lost marks on a window-design question. Forgetting the delay τ in H_d (then the answer is non-causal and not symmetric). Getting $h_d(\tau)$ wrong — it is ω_c/π , not 1 and not 0. And writing the Hamming window with M instead of $M-1$ in the denominator, which makes $w(0) \neq w(M-1)$ and breaks the symmetry.

6.9

Choosing the length, and the Kaiser recipe

The window fixes the stopband attenuation. The length fixes the transition width. Kaiser gives both from two formulas.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** §6, 70–74 · class notebook

For the fixed windows, first pick the window from the attenuation you need (the table in §6.7). Then set the length from the transition band. Your notebook's rule: the transition band is about as wide as the window's **main lobe**, and the main lobe is $k \cdot 2\pi/M$. So

$$\omega_s - \omega_p \approx \frac{k \cdot 2\pi}{M} \implies M \approx \frac{k \cdot 2\pi}{\omega_s - \omega_p}, \quad k = 2 \text{ (rectangular)}, 4 \text{ (Bartlett, Hann, Hamming)}, 6 \text{ (Blackman)}$$

Round up, and make it odd if you need Type 1. (A finer version uses the measured transition widths, about $1.8\pi/M$ for rectangular, $6 \cdot 2\pi/M$ Hann, $6 \cdot 6\pi/M$ Hamming, $11\pi/M$ Blackman — a bit smaller than the main lobe, because the edge sits in the middle of the lobe. In the exam, use the main-lobe rule your teacher gave.) The notebook also has a rough window-free estimate, $N \approx [-10 \log_{10}(\delta_1 \delta_2) - 15] / (14 \Delta f)$ with $\Delta f = (\omega_s - \omega_p)/2\pi = f_s - f_p$. It is only a starting point.

WORKED

Order of the filter for

$$\omega_p = 0.3\pi, \quad \omega_s = 0.45\pi$$

, rectangular window (class notebook)

$$\Delta\omega = 0.45\pi - 0.3\pi = 0.15\pi, \quad M \approx \frac{2 \cdot 2\pi}{0.15\pi} = \frac{4}{0.15} = 26.7 \Rightarrow M = 27 \text{ taps (order 26)}$$

Rectangular means $k = 2$: the main lobe is $4\pi/M$. Twenty-seven taps for a transition band of 0.15π . With a Hamming window ($k = 4$) the same edges would need about 54 taps, but the stopband would be -53 dB instead of -21 .

THE KAISER DESIGN EQUATIONS

- 1 Put the ideal cutoff in the middle of the transition band: $\omega_c = (\omega_p + \omega_s)/2$.
- 2 One ripple for both bands: $\delta = \min(\delta_p, \delta_s)$, where $\delta_p = (10^{0.05A_p} - 1)/(10^{0.05A_p} + 1)$ and $\delta_s = 10^{-0.05A_s}$. The attenuation you actually design for is $A = -20 \log_{10} \delta$.

3 The shape parameter from A:

$$\beta = \begin{cases} 0 & A \leq 21 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 < A \leq 50 \\ 0.1102(A - 8.7) & A > 50 \end{cases}$$

4 The length from the transition width $\Delta f = (\omega_s - \omega_p)/2\pi$:

$$M \geq \frac{A - 7.95}{14.36 \Delta f}$$

5 $h(n) = h_d(n) w_K(n)$ with $\tau = (M-1)/2$, then $H(z)$ as usual.

WORKED

Kaiser low-pass:

$\omega_c = \pi/4$, $\Delta\omega = 0.02\pi$, $\delta_s = 0.01$ (slides p. 74)

$$A = -20 \log_{10}(0.01) = 40 \text{ dB}$$

$$\beta = 0.5842(19)^{0.4} + 0.07886(19) = 1.896 + 1.498 = 3.39$$

$$\Delta f = 0.02\pi/2\pi = 0.01$$

$$M \geq \frac{40 - 7.95}{14.36 \times 0.01} = 223.2 \Rightarrow M = 225 \text{ (odd)}, \quad \tau = 112$$

$$h(n) = \frac{\sin \frac{\pi}{4}(n - 112)}{\pi(n - 112)} \cdot \frac{I_0\left(3.39 \sqrt{1 - \left(\frac{n}{112} - 1\right)^2}\right)}{I_0(3.39)}$$

Two hundred and twenty-five taps for a 1% transition band. That is what a sharp FIR filter costs, and it is the plain answer to “what does FIR cost” in §6.1.

6.10

Design by frequency sampling

Instead of windowing the ideal impulse response, take M samples of the ideal frequency response and do an inverse DFT. The result is exact at those samples and free to ripple between them.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** §82-93 · Proakis 10.2.3

Choose M equally spaced frequencies $\omega_k = 2\pi k/M$, $k = 0, \dots, M-1$, and set

$$H(k) = H_d\left(\frac{2\pi k}{M}\right), \quad h(n) = \frac{1}{M} \sum_{k=0}^{M-1} H(k) e^{j2\pi kn/M}, \quad n = 0, \dots, M-1$$

That is just the M-point IDFT from Unit 5. Since $h(n)$ is then a sequence of length M whose DFT is $H(k)$, the frequency response of the filter passes through the samples **exactly**: $H(e^{j2\pi k/M}) = H(k)$. Between the samples it is whatever the interpolation makes it.

MAKING $h(n)$ REAL, AND THE PHASE LINEAR

For real coefficients the samples must be conjugate-symmetric: $H(M-k) = H^*(k)$, the same DFT property as §5.6. For linear phase, give each sample the phase of the delay $\tau = (M-1)/2$: $H(k) = |H(k)| e^{-j\pi k(M-1)/M}$ for $k = 0, \dots, (M-1)/2$. Then the sum turns into a real cosine series:

$$h(n) = \frac{1}{M} \left[H(0) + 2 \sum_{k=1}^P \operatorname{Re} \left\{ H(k) e^{j2\pi kn/M} \right\} \right], \quad P = \begin{cases} \frac{M-1}{2} & M \text{ odd} \\ \frac{M}{2} - 1 & M \text{ even} \end{cases}$$

and for an ideal magnitude of 1 in the passband the term inside is $|H(k)| \cos[2\pi k(n-\tau)/M]$.

WORKED

Low-pass by frequency sampling,

$\omega_c = \pi/2$, $M = 17$ (slides p. 86–87)

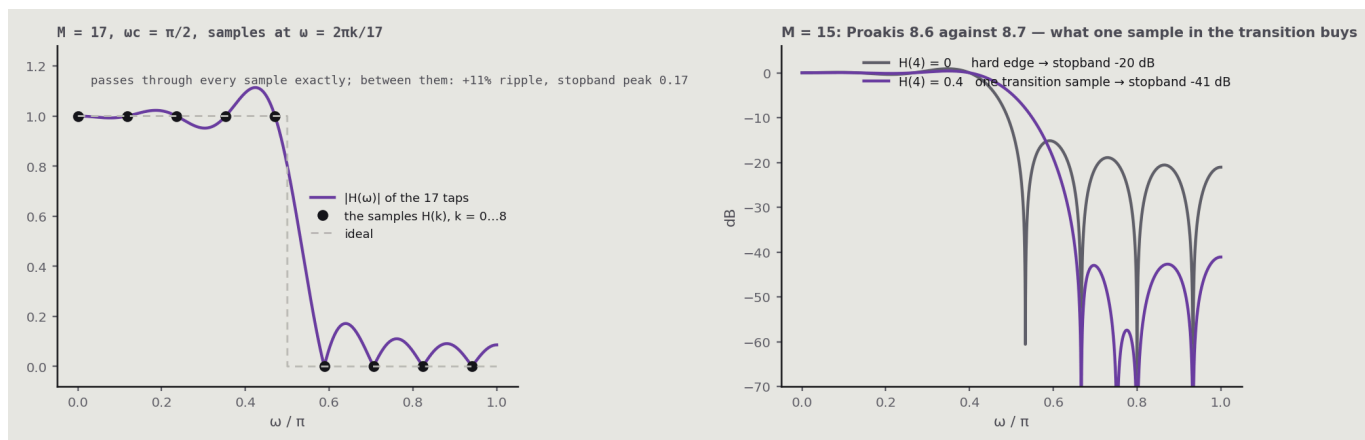
$\tau = 8$. The sample spacing is $2\pi/17$, so the cutoff falls at $k = 17/4 = 4.25$: samples $k = 0 \dots 4$ are in the passband ($|H(k)| = 1$), $k = 5 \dots 8$ in the stopband (0), and $k = 9 \dots 16$ mirror them. So

$$h(n) = \frac{1}{17} \left[1 + 2 \sum_{k=1}^4 \cos \frac{2\pi k(n-8)}{17} \right], \quad n = 0, \dots, 16$$

N	0	1	2	3	4	5	6	7	8
$h(n) = h(16-n)$	0.0398	-0.0488	-0.0346	0.0660	0.0315	-0.1075	-0.0299	0.3188	0.5294

Check: $h(8) = (1 + 2 \cdot 4)/17 = 9/17 = 0.5294$, since every cosine is 1 at $n = \tau$. And the DFT of these 17 numbers really is $\{1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1\}$ — exact at the samples. Between them the passband rises to 1.11 and the stopband peaks at 0.17 (-15 dB). That is the price of a hard edge in the samples.

COMPUTED · TOOLS/MAKE-FIGURES.PY



Exact at the samples, free in between. Left: the $M = 17$ design passes through every one of its samples (dots) and ripples between them — measured: +11% in the passband, 0.17 in the stopband. Right: Proakis's $M = 15$ pair. Leaving the edge sample at 0 gives a stopband of -20 dB; setting that one sample to 0.4 gives -41 dB. One transition sample, twenty decibels.

WORKED

Proakis 8.6 and 8.7: $M = 15$, with and without a transition sample

$H(2\pi k/15) = 1$ for $k = 0, 1, 2, 3$ and 0 for $k = 4, \dots, 7$, symmetric impulse response. $\tau = 7$, $P = 7$, and only $k = 1, 2, 3$ contribute:

$$h(n) = \frac{1}{15} \left[1 + 2 \sum_{k=1}^3 \cos \frac{2\pi k(n-7)}{15} \right]$$

N	0	1	2	3	4	5	6	7
8.6 $h(n) = h(14-n)$	-0.0498	0.0412	0.0667	-0.0365	-0.1079	0.0341	0.3189	0.4667
8.7 with $H(4) = 0.4$	-0.0141	-0.0019	0.0400	0.0122	-0.0914	-0.0181	0.3133	0.5200

For 8.7 the extra term is $(2/15)(0.4) \cos[8\pi(n-7)/15]$. The coefficients barely change, but the stopband improves from -20 dB to -41 dB (measured). This is why real frequency-sampling designs always put one or two samples in the transition band, with values found by optimisation (Proakis has tables of them).

WINDOW METHOD OR FREQUENCY SAMPLING?

Both give linear phase and both are closed-form. Windowing controls the *ripple* (through the window) and lets the exact edges move a little. Frequency sampling fixes the *values at M frequencies* exactly and lets the ripple move. Frequency sampling is the natural choice when the specification is given as a list of samples. It can also be built very cheaply as a comb filter followed by M resonators, but that belongs to the realisation unit.

6.11

Two special filters: the differentiator and the Hilbert transformer

Both have a 90-degree phase shift built in, so both need an antisymmetric impulse response. Their $h_d(n)$ are one line each.

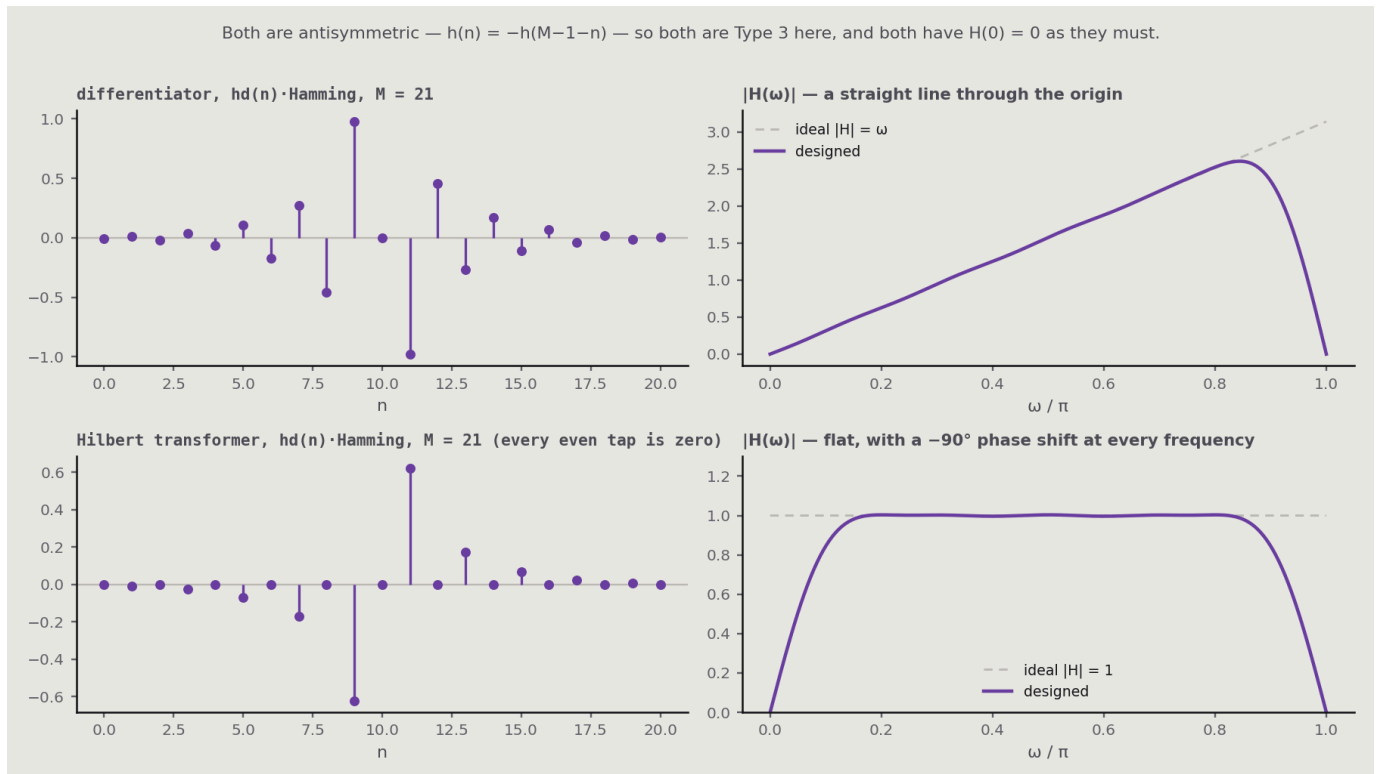
Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** · Proakis 10.2.4

An ideal **digital differentiator** multiplies each frequency by $j\omega$. An ideal **Hilbert transformer** shifts every positive frequency by -90° and every negative one by $+90^\circ$ without changing the magnitude (it is what makes the analytic signal and single-sideband modulation possible):

$$\begin{aligned} \text{differentiator: } H_d(\omega) &= j\omega e^{-j\omega\tau}, & h_d(n) &= \frac{\cos \pi(n - \tau)}{n - \tau} \quad (n \neq \tau), & h_d(\tau) &= 0 \\ \text{Hilbert transformer: } H_d(\omega) &= -j \operatorname{sgn}(\omega) e^{-j\omega\tau}, & h_d(n) &= \frac{1 - \cos \pi(n - \tau)}{\pi(n - \tau)} = \frac{2 \sin^2 \frac{\pi(n - \tau)}{2}}{\pi(n - \tau)} \end{aligned}$$

Both $h_d(n)$ are odd about τ — **antisymmetric** — which is why Types 3 and 4 exist. For odd M (Type 3) the Hilbert transformer has $h(n) = 0$ at every even distance from the centre and $2/\pi(n - \tau)$ at every odd one, so half the multiplications are free. Both filters have $H(0) = 0$, as they should. The differentiator prefers Type 4 (even M), because Type 3 also forces $H(\pi) = 0$, which cuts off the top of the $j\omega$ ramp.

COMPUTED · TOOLS/MAKE-FIGURES.PY



Both special filters at $M = 21$ with a Hamming window. Top: the differentiator's antisymmetric $h(n)$ and its magnitude, a straight line $|H| = \omega$ that only falls off near π because the odd length forces $H(\pi) = 0$. Bottom: the Hilbert transformer, every even tap zero, and a magnitude that is flat except at the two ends, where the same symmetry forces zeros.

6.12

Optimum equiripple design, in one page

Windowing spends its error unevenly: big near the edge, tiny far from it. Spread it evenly and the same specification needs fewer taps.

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The window and frequency-sampling methods are *closed-form*: you write the answer down. Their weak point is that the error is largest next to the band edge and wasted everywhere else. The **Chebyshev** or **minimax** approach chooses the coefficients to make the largest weighted error as small as possible:

$$\min_{h(n)} \max_{\omega \in S} |E(\omega)|, \quad E(\omega) = W(\omega) [H_{dr}(\omega) - H_r(\omega)]$$

over the passband and stopband S (the transition band is left free), with a weight $W(\omega)$ that lets you ask for a smaller ripple in one band than in the other. The **alternation theorem** says the best filter is the one whose error ripples with *equal* height and alternating sign at least $L+2$ times, where L is the number of cosine terms in $H_r(\omega)$ — hence the name **equiripple**. The **Parks–**

McClellan algorithm finds it by the Remez exchange: guess the frequencies where the error peaks, solve for the coefficients, find where the error really peaks, move the guesses there, repeat.

	WINDOW	FREQUENCY SAMPLING	EQUIRIPPLE (PARKS-MCCLELLAN)
How	cut $h_d(n)$ with $w(n)$	IDFT of M samples of H_d	iterative minimax fit
Ripple	equal in both bands, largest at the edge	not controlled between samples	equal height, set per band
Band edges	approximate	exact only at the sample points	exact
Length for a given spec	longest	middle	shortest — the optimum
By hand?	yes	yes	no — software (remez)

A useful estimate of the length an equiripple design needs (Kaiser):

$M \approx [-20 \log_{10} \sqrt{(\delta_1 \delta_2)} - 13] / (14.6 \Delta f) + 1$. For the Kaiser-window example of §6.9 ($\delta = 0.01, \Delta f = 0.01$) it gives about 185 taps against 225 — the saving from spreading the error evenly.

6.13

Building an FIR filter

The direct form is the equation drawn as a tapped delay line. Symmetry lets you add the mirror taps before multiplying, and halve the multipliers.

Syllabus unit 3 · **FIR filter design** · class notebook · slides **fir-filter-design.pdf** · syllabus unit 5, in brief

The **direct (transversal) form** is $y(n) = \sum h(k) x(n-k)$ drawn as it is: $M-1$ delays in a line, M multipliers, $M-1$ adders. With linear phase, first add the two samples that share a coefficient:

$$y(n) = \sum_{k=0}^{\tau-1} h(k) [x(n-k) + x(n-M+1+k)] + h(\tau) x(n-\tau) \quad (M \text{ odd})$$

which needs only $(M+1)/2$ multipliers instead of M — the “operations of the order of $N/2$ ” in your notebook. The full set of structures (cascade, lattice, polyphase, and the frequency-sampling structure) is the realisation unit of the syllabus.

UNIT 6 IN SIX LINES

- FIR = no feedback: always stable, and **linear phase whenever** $h(n) = \pm h(M-1-n)$, with delay $(M-1)/2$.

- Four symmetric families. Type 2 cannot pass π ; Types 3 and 4 cannot pass 0; Type 3 cannot pass π either. Zeros come in reciprocal and conjugate sets.
- **Window method:** $h_d(n) = \sin \omega_c(n-\tau)/\pi(n-\tau)$, $h_d(\tau) = \omega_c/\pi$, times $w(n)$. The window sets the stopband floor (rect -21, Hann -44, Hamming -53, Blackman -74 dB); $M \approx k \cdot 2\pi/\Delta\omega$ with $k = 2, 4, 4, 6$.
- Gibbs: a rectangular cut leaves 9% overshoot whatever M is. Taper to remove it, and pay in transition width.
- **Kaiser:** $A = -20 \log_{10} \delta$, β from $A, M \geq (A - 7.95)/(14.36 \Delta f)$.
- **Frequency sampling:** $h(n) = (1/M) [H(0) + 2 \sum \text{Re}\{H(k)e^{j2\pi kn/M}\}]$, exact at the samples; one transition sample buys about 20 dB. Equiripple is the optimum, done by software.

APPENDIX A · YOUR FIVE TUTORIALS

Your tutorials, solved

Every question from Tutorials 1–5, and where the worked answer lives.

Tutorial 1 — Review of Signals and Systems

Q	ASKS YOU TO	GO TO
1	Say whether each $h(n)$ is causal (and stable)	§2.13
2	Sketch $x(n-2)$, $x(4-n)$, $x(2n)$, $x(n)u(2-n)$, $x(n-1)\delta(n-3)$	§1.13
3	Which sinusoids are periodic, and their period	§1.6
4	Periodicity of sums and products of sinusoids	§1.6
5	More transformations; even and odd parts	§1.13, §1.14
6	Energy signal or power signal	§1.12

Tutorial 2 — Review of Signals and Systems

Q	ASKS YOU TO	GO TO
1	Which systems are linear	§2.3
2	Time-variant or time-invariant	§2.3
3	Static or dynamic	§2.3
4	Causal / memory / time-invariant	§2.3
5	Convolution, numeric and analytic	§2.12
6	Impulse response of a cascade	§2.4

Tutorial 3 — Response of System

Q	ASKS YOU TO	GO TO
1	Structure; impulse, step and full response with initial conditions	§2.8 (all four parts); structure §2.1; faster route §3.5
2, 3	Convolution sums by algebra	§2.12
4, 5	Convolution from stem plots	§2.12
6	Circular convolution of two 4-point sequences	§2.14
7	Eight-point circular convolution	§2.14; DFT shortcut §5.3

Tutorial 4 — z-transform

Q	ASKS YOU TO	GO TO
1	z-transform of seven finite-duration signals	§3.1
2	z-transform of $a^n u(n)$	§3.1
3	z-transform of an anti-causal signal	§3.1
4	z-transform of a two-sided signal	§3.1
5	Inverse by the complex inversion integral	§3.3
6	Inverse for two different ROCs	§3.3
7	Causal signal from a second-order $X(z)$ with complex poles	§3.3

Tutorial 5 — System Analysis and Its Response

Q	ASKS YOU TO	GO TO
1	ROC and $h(n)$ for stable / causal / anti-causal	§3.4
2	$H(z)$, poles, unit-sample response, stable version	§3.4
3	Causal or not, from the degrees of $H(z)$	§3.4
4	Block diagram $\rightarrow H(z)$; stability vs k ; output for a given input	§3.4
5	Show $\alpha = 1 - \beta/2$ from $\sum h(n) = 2$	hint below; §3.2
6	ROC and $h(n)$, stable-causal and anti-causal	§3.4
7	Values of α , β for stability	§3.4
8	Step response of the accumulator; stability	§3.6
9	Transient and steady-state response to a cosine	§3.6, §4.7

TUTORIAL 5, Q5 IN THREE LINES

$y(n) = \alpha y(n-1) + \beta x(n)$ with $\sum_{n=0}^{\infty} h(n) = 2$. You never need $h(n)$:

$$\sum_{n=0}^{\infty} h(n) = \sum_n h(n) \cdot 1^{-n} = H(z) \Big|_{z=1} = \frac{\beta}{1 - \alpha} = 2 \Rightarrow \alpha = 1 - \beta/2$$

APPENDIX B · REVISION

Formula sheet

Everything from the six units worth having on one page.

Frequency and sampling (Unit 1)

$$\Omega = 2\pi F \quad \cdot \quad \omega = 2\pi f = \Omega T_s \quad \cdot \quad f = F/F_s \quad \cdot \quad -\pi \leq \omega \leq \pi, \quad -\frac{1}{2} \leq f \leq \frac{1}{2}$$

DT sinusoid periodic $\Leftrightarrow f = k/N$ is a fraction; period N = denominator;
combination $\rightarrow$ LCM

ω_0 and $\omega_0 + 2\pi k$ are the same signal $\cdot$ fastest at $\omega = \pi$ $\cdot$ only N distinct harmonics $e^{j2\pi kn/N}$

$F_s > 2F_{\max}$ $\cdot$ Nyquist rate $2F_{\max}$ $\cdot$ folding frequency $F_s/2$ $\cdot$ aliases $F_0 + kF_s$

$$\Delta = (x_{\max} - x_{\min})/(L - 1) \quad \cdot \quad |e_q| \leq \Delta/2 \quad \cdot \quad \text{SQNR}_{\text{dB}} = 1.76 + 6.02b$$

Signals (Unit 1)

$$E = \sum |x(n)|^2$$

$$P = \left(\frac{1}{N}\right) \sum_0^{N-1} |x(n)|^2 (\text{periodic})$$

energy signal $\Rightarrow P = 0$; power signal $\Rightarrow E = \infty$; periodic $\Rightarrow$ power

$$x_e = \left(\frac{1}{2}\right) [x(n) + x(-n)] \cdot x_o = \left(\frac{1}{2}\right) [x(n) - x(-n)] \cdot \text{fold and delay } \textit{do not commute} \cdot \text{interpolate} \rightarrow \text{shift} \rightarrow \text{decimate}$$

$$\delta(n) = u(n) - u(n-1)$$

$$x(n)\delta(n-k) = x(k)\delta(n-k)$$

$$\sum x(n)\delta(n-k) = x(k)$$

$$x(n) = \sum x(k)\delta(n-k)$$

Systems and convolution (Unit 2)

TI test: $y(n,k) = T[x(n-k)]$ vs $y(n-k)$ · non-linear: products, powers, constants · time-varying: coefficient in n , $x(2n)$, $x(-n)$

$y(n) = \sum_k x(k)h(n-k)$ · length n_1+n_2-1 · start = start + start · $\Sigma y = (\Sigma x)(\Sigma h)$

causal $\Leftrightarrow h(n) = 0, n < 0$ · BIBO stable $\Leftrightarrow \sum |h| < \infty \Leftrightarrow$ all roots $|\lambda| < 1$
· FIR always stable

cascade $h_1 * h_2$ · parallel $h_1 + h_2$ · step response = running sum of h ·

$u * u = (n+1)u(n)$ · $a^n u * b^n u = (a^{n+1} - b^{n+1})/(a - b)$

circular: $y_p(n) = \sum_0^{N-1} x_p(k)h_p[(n-k) \bmod N]$ · linear from circular needs $L \geq N+M-1$ · wraparound: split $2N-1$ (+0) into halves and add

deconvolution = polynomial division · $r_{xh}(n) = x(n) * h(-n)$ · $r_{xh}(n) = r_{hx}(-n)$ · r_{xx} even, max at $0 = E_x$

Difference equations (Unit 2)

total = natural + forced = zero-input + zero-state · first order: $y(n) = a^{n+1}y(-1) + \sum_0^n a^k x(n-k)$

$\lambda^N + a_1 \lambda^{N-1} + \dots + a_N = 0$ · distinct: $\sum C_i \lambda_i^n$ · pair $re^{\pm j\theta}$: $r^n [C_1 \cos \theta n + C_2 \sin \theta n]$ · repeated: $r^n [C_0 + C_1 n + \dots]$

particular: $A \rightarrow K$; $AM^n \rightarrow KM^n$; $An^M \rightarrow$ polynomial; $A \cos \omega_0 n \rightarrow K_1 \cos + K_2 \sin$ · fit C 's from $y(0)$, $y(1)$ computed by the equation

impulse response: homogeneous solution with $h(0) = 1$ (then $h = b_0 h_0(n) + b_1 h_0(n-1) + \dots$) · inverse system: swap x and y

z-transform (Unit 3)

$X(z) = \sum x(n)z^{-n}$ · $x(n-k) \leftrightarrow z^{-k}X(z)$ · $x * h \leftrightarrow X(z)H(z)$ · $a^n x(n) \leftrightarrow X(z/a)$
 · $nx(n) \leftrightarrow -z \frac{dX}{dz}$

$a^n u(n) \leftrightarrow 1/(1 - az^{-1}), |z| > |a|$ · $-a^n u(-n-1) \leftrightarrow \text{same}, |z| < |a|$ · $na^n u(n) \leftrightarrow az^{-1}/(1 - az^{-1})^2$

ROC: right-sided $\rightarrow$ outside outermost pole; left-sided $\rightarrow$ inside innermost;
 never contains a pole · causal AND stable $\Leftrightarrow$ all poles inside $|z| = 1$
 inverse: expand $X(z)/z$, or treat z^{-1} as the variable; ROC outside pole $\rightarrow$
 causal term, inside $\rightarrow$ anti-causal (minus sign)

complex pair $re^{\pm j\omega_0} \rightarrow A r^n \cos(\omega_0 n + \varphi)u(n)$ · long division: descending z for
 causal, ascending for anti-causal

$x(n) = (1/2\pi j) \oint X(z)z^{n-1}dz = \sum \text{residues inside } C$ · initial value $x(0) = \lim_{z \rightarrow \infty} X(z)$

$Z^+\{y(n-1)\} = z^{-1}Y^+(z) + y(-1)$ · second-order stable $\Leftrightarrow |a_2| < 1$ and $|a_1| < 1 + a_2$
 · transient from poles of H , steady state from poles of X

Frequency analysis (Unit 4)

DTFS: $x(n) = \sum_0^{N-1} c_k e^{j2\pi kn/N}$, $c_k = (1/N) \sum_0^{N-1} x(n) e^{-j2\pi kn/N}$, $c_{k+N} = c_k$

DTFT: $X(\omega) = \sum x(n) e^{-j\omega n} = X(z)|_{z=e^{j\omega}}$, period 2π · $x(n) = (1/2\pi) \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$

$x(n-k) \leftrightarrow e^{-j\omega k} X$ · $x * h \leftrightarrow XH$ · $x \cos \omega_0 n \leftrightarrow \frac{1}{2}X(\omega + \omega_0) + \frac{1}{2}X(\omega - \omega_0)$ · real $x \Rightarrow$
 $|X|$ even, $\angle X$ odd

$a^{|n|} \leftrightarrow (1 - a^2)/(1 - 2a \cos \omega + a^2)$ · $a^n u(n) \leftrightarrow 1/(1 - ae^{-j\omega})$ · pulse of N
 ones $\leftrightarrow e^{-j(N-1)\omega/2} \sin(N\omega/2)/\sin(\omega/2)$

$H_p(F) = H(z)|_{z=e^{j2\pi F}}$ · $H_p(0) = H(1)$, $H_p(0.5) = H(-1)$ · $A \cos(2\pi F_0 n + \theta) \rightarrow$
 $A|H_0| \cos(2\pi F_0 n + \theta + \varphi_0)$

DFT and FFT (Unit 5)

$X(k) = \sum_0^{N-1} x(n) W_N^{kn}$ · $x(n) = (1/N) \sum_0^{N-1} X(k) W_N^{-kn}$ · $W_N = e^{-j2\pi/N}$ · $X(0) = \sum x$
 · real $x \Rightarrow X(N-k) = X^*(k)$

$W_N^{k+N} = W_N^k$ · $W_N^{k+N/2} = -W_N^k$ · $W_N^2 = W_{N/2}$ · $W_4^k = 1, -j, -1, j$ · $X = W_N x$,
 $x = (1/N) W_N^* X$

finite signal: $X(k) = X(\omega)|_{\omega=2\pi k/N}$ exactly · zero padding: denser dots, same curve · $F_k = kF_s/N$

infinite signal: $x_N = x \cdot P$, $X_N(\omega) = (1/2\pi) X * P \rightarrow$ leakage; $P(\omega) = \sin(N\omega/2)/\sin(\omega/2) \cdot e^{-j(N-1)\omega/2}$

three errors: aliasing (F_s) · smearing (more data, window) · grid (zero padding)

direct: N^2 mults, $N(N-1)$ adds; real: $4N^2$, $2N(N-1)$, $2N^2$ trig · radix-2: $(N/2)\log_2 N$ mults, $N \log_2 N$ adds, $\log_2 N$ stages, $N/2$ butterflies/stage

DIT: $A = a + W^r b$, $B = a - W^r b$; input bit-reversed, output natural · DIF: $A = a + b$, $B = (a - b)W^r$; input natural, output bit-reversed

IFFT: conjugate twiddles, divide by N · memory $2N + N/2 = 5N/2$ · radix-4: $(3N/8)\log_2 N$

composite $N = m_1 N_1$: $X(k) = \sum_r W_N^{rk} \sum_n x(m_1 n + r) W_{N_1}^{nk}$ · divide & conquer $N = LM$:
 row DFTs $\rightarrow W_N^{lq} \rightarrow$ column DFTs, cost $N(M + L + 1)$

FIR filter design (Unit 6)

linear phase: $h(n) = \pm h(M-1-n)$, delay $\tau = (M-1)/2$ · Type 1 sym odd (any) · Type 2 sym even ($H(\pi)=0$) · Type 3 anti odd ($H(0)=H(\pi)=0$) · Type 4 anti even ($H(0)=0$)

sym, M odd: $H(e^{j\omega}) = e^{-j\omega\tau} [h(\tau) + 2\sum_{n=0}^{\tau-1} h(n)\cos\omega(\tau-n)]$ · M even:

$e^{-j\omega\tau} \cdot 2\sum_{n=0}^{M/2-1} h(n)\cos\omega(\tau-n)$ · antisym: $j e^{-j\omega\tau} \cdot 2\sum_{n=0}^{\tau-1} h(n)\sin\omega(\tau-n)$

zeros: $H(z) = z^{-(M-1)}H(1/z) \Rightarrow z_0, 1/z_0, z_0^*, 1/z_0^*$ · causal: $h(n)=0$ for $n<0$, finite zeros $\leq$ finite poles · Paley-Wiener: $\int |\ln|H||d\omega < \infty \Rightarrow$ no zero band, no jump

ideal LP: $h_d(n) = \sin\omega_c(n-\tau)/\pi(n-\tau)$, $h_d(\tau) = \omega_c/\pi$ · HP = $\delta(n-\tau)$ - LP ·

BP: $[\sin\omega_2(n-\tau) - \sin\omega_1(n-\tau)]/\pi(n-\tau)$, $h_d(\tau) = (\omega_2 - \omega_1)/\pi$ · BS = $\delta(n-\tau)$ - BP

window: $h(n) = h_d(n)w(n)$ · Hann $0.5-0.5\cos(2\pi n/(M-1))$ · Hamming

$0.54-0.46\cos$ · Blackman $0.42-0.5\cos+0.08\cos(4\pi n/(M-1))$ · Bartlett

$1-2|n-\tau|/(M-1)$

main lobe $k \cdot 2\pi/M$: $k = 2$ rect, 4 Bartlett/Hann/Hamming, 6 Blackman $\Rightarrow M \approx$

$k \cdot 2\pi/(\omega_s - \omega_p)$ · side lobe -13, -25, -31, -41, -57 dB · stopband -21, -25, -44, -53, -74 dB

$|W_R(\omega)| = |\sin(\omega M/2)/\sin(\omega/2)|$ · Gibbs: 9% overshoot, independent of M ·

$|H(\omega_c)| = \frac{1}{2}$ (-6.02 dB) for a windowed LP · rough N =

$[-10\log_{10}(\delta_1\delta_2)-15]/(14\Delta f)$

Kaiser: $w(n) = I_0(\beta\sqrt{1-(2n/(M-1)-1)^2})/I_0(\beta)$ · $A = -20\log_{10}\delta$ · $\beta = 0$

($A \leq 21$); $0.5842(A-21)^{0.4} + 0.07886(A-21)$ ($21 < A \leq 50$); $0.1102(A-8.7)$ ($A > 50$) · $M \geq (A-7.95)/(14.36\Delta f)$, $\Delta f = (\omega_s - \omega_p)/2\pi$

frequency sampling: $H(k) = H_d(2\pi k/M)$, $h(n) = (1/M)\sum H(k)e^{j2\pi kn/M}$ · real h:

$H(M-k) = H^*(k)$ · linear phase: $h(n) = (1/M)[H(0) +$

$2\sum_{k=1}^P |H(k)|\cos(2\pi k(n-\tau)/M)]$, $P = (M-1)/2$ or $M/2-1$

differentiator: $H_d = j\omega$, $h_d(n) = \cos\pi(n-\tau)/(n-\tau)$ · Hilbert: $H_d = -j \operatorname{sgn}\omega$,

$h_d(n) = [1 - \cos\pi(n-\tau)]/\pi(n-\tau)$ · both antisymmetric

equiripple: $\min \max |W(\omega)[H_{dr} - H_r]|$, alternation theorem, Parks-McClellan

(Remez) · $M \approx [-20\log_{10}\sqrt{\delta_1\delta_2} - 13]/(14.6\Delta f) + 1$ · linear-phase

structure: $(M+1)/2$ multipliers

Mistakes that cost the most marks

MISTAKE	THE CORRECT POSITION
Assuming a DT sinusoid is always periodic	Only if f is a ratio of whole numbers; $\sin(3n)$ is not periodic.
Losing the arrow (the $n = 0$ position)	It sets where the whole answer sits — track it through every shift, fold and convolution.
Folding and shifting in the wrong order	$x(-n-k) \neq x(-n+k)$. Rewrite $x(4-n)$ as $x(-(n-4))$ first.
Skipping the sum check on a convolution	$\Sigma y = (\Sigma x)(\Sigma h)$ — five seconds, catches nearly every slip.
Mixing up natural / zero-input, forced / zero-state	Zero-input = natural form <i>after</i> the ICs; zero-state contains a transient too.
Giving $X(z)$ without the ROC	The ROC decides which signal you mean.
Forgetting the minus sign on the anti-causal term	$-a^n u(-n-1)$, always.
Thinking zeros affect stability	Only poles decide stability.
Not transforming back to check a DFT/IDFT	One sign slip rotates the whole answer; $X(0) = \Sigma x$ and conjugate pairs are free checks.
Believing zero padding improves resolution	It only adds dots. Resolution needs a longer record.
Bit-reversing the wrong end of an FFT	DIT takes bit-reversed <i>input</i> ; DIF gives bit-reversed <i>output</i> .

APPENDIX C · SOURCES

Where each part comes from

This version is built only from the lecture modules in your teacher folder and the five tutorials. The grey badge under every section heading names its source.

TEACHER MATERIAL	COVERS	SECTIONS HERE
Module 01 — Review of Signal & System (ModuleCombineLectures)	DSP, classification, frequency, sampling, quantization	1.1–1.10
M01L003 — Concept of Frequency (AnalogToDigitalFrequencyConcepts)	Analog vs digital frequency, DT sinusoids, harmonics	1.5–1.8
M01L001 — Discrete Time Signal	Representations, sums, operations, symmetry, elementary signals	1.11–1.15
M02L002 — Discrete Time System	Blocks, I/O description, superposition, classification, interconnection	2.1–2.4
M02L003 — Analysis of LTI Systems	FIR/IIR, recursion, formal solution	2.5–2.7
M02L004 — Analysis of LTI Systems	Impulse response, anti-causal response, stability, inverse system	2.8–2.10
M02L005 — Convolution Operations	Derivation, computation, properties, causality and stability	2.11–2.13
M02L005b — Periodic Convolution	Periodic inputs, circular convolution, transforms, deconvolution, correlation	2.14–2.17
Tutorials 4 and 5	z-transform, inverse, system function, stability, responses	3.1–3.6
M04L001 — Frequency Analysis of Signals & Systems	DTFS, DTFT, properties, examples, frequency response, steady state	4.1–4.8
M04L002 — Introduction to DFT (via DTFT)	Frequency sampling, DFT, matrix, examples, finite and infinite cases	5.1–5.9
M04L004 — Fast Fourier Transform	Complexity, DIT, DIF, memory, IFFT, composite radix, divide and conquer	5.10–5.17

TEACHER MATERIAL	COVERS	SECTIONS HERE
Syllabus unit 3 — FIR Filter Design (class notebook; fir-filter-design.pdf, which follows Proakis ch. 10)	Causality, linear phase, the four types, zero locations, window method, windows, order, Kaiser, frequency sampling, differentiator, Hilbert, equiripple	6.1–6.13

REMOVED IN THIS VERSION, ON PURPOSE

Topics that were in the earlier course but are not in the teacher's modules or the tutorials: continuous-time Fourier series and transform details, special filters (resonators, notch, comb, all-pass, oscillators), ideal filters and pole–zero filter design, minimum-phase systems, chirp-z and split-radix (the last two are only named in the MO4L004 tree). The DFT property tables, overlap-add / overlap-save and Goertzel were in that list too; they are now §5.6, §5.7 and §5.11, because the syllabus lists them and the class has reached them. If the teacher issues a module on any of these, it can be added back.

WHERE THE SLIDES HAVE SLIPS

A few printed answers in the modules do not pass their own checks; each place says so and gives the corrected value: the constant C_1 in the second-order example (§2.7), the difference equation from $2(-0.5)^n + (0.5)^n$ (§2.10), the steady-state period $\{-2, 3, -1\}$ and the padded circular convolution row (§2.14, §2.15), the 8-point DFT of four ones and the IDFT $\{4, 1, 2, 3\}$ (§5.5), and the speed-up for $N = 512$ (§5.11).